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Dedicated to my parents Eleni and Konstantinos

1 Introduction

1.1 Pulsars

In the year 1934 Walter Baade and Fritz Zwicky (Baade and Zwicky, 1934), presented an idea that when a star reaches the end of its life and explodes as a supernova, the remnants of its core, consisting mainly of iron, will collapse into an incredibly dense object known as a neutron star. The key requirement for this process to occur is that the mass of the star's core must exceed a critical value, known as the Chandrasekhar limit (Chandrasekhar, 1931) which is ≈ 1.4 times the mass of the Sun ($M_{\odot}$) and is the maximum mass that a white dwarf star (the end state of a star with mass less than 8 solar $M_{\odot}$) can have. At this stage, the core holds against the collapse due to the electron degeneracy pressure, but as it approaches the $1.4 M_{\odot}$ limit, gravity wins over as the dominant force. If the mass of the dying star's core exceeds this limit, the intense pressure and gravity will cause the core to collapse and form a neutron star, which can have a radius of just a few kilometers but a mass comparable to that of the sun.

Pulsars are fast rotating neutron stars whose strong magnetic field which can be approximated by a magnetic dipole (Gunn and Ostriker, 1969). They were first discovered in 1967 (Hewish et al., 1968) and although they are being observed for more than 50 years some aspects regarding their emission mechanism, still remain unknown. Ever since their discovery, they are being used to study a variety of astronomical phenomena, including the behavior of highly magnetized plasma, the nature of the dense matter in neutron stars, and the properties of gravity in strong-field regimes. Furthermore, observations of their pulses with a high time resolution can be used to study the distribution of matter in the interstellar medium, the properties of binary star systems, and the effects of general relativity in strong-field regimes. In recent years, extremely precise pulsar timing is being used for studying the gravitational waves background. Pulsars continue to be an important tool for astronomers and physicists, providing insights into a wide range of astrophysical phenomena. Pulsars are also interesting from a theoretical perspective, as they provide a unique laboratory for studying the behavior of matter in extreme conditions. The extreme densities and pressures in the interior of a neutron star cause the electrons and protons in its matter to combine, forming a uniform sea of neutrons. This state of matter is called neutron-degenerate matter and is not found in any other object in the universe. The behavior of neutron-degenerate matter is still not well understood, and the study of pulsars has the potential to provide new insights into this area of physics.

Another possible state of matter that can be present in pulsar cores is the strange matter (Witten, 1984). This is a hypothetical form of matter that consists of a mix of up, down, and strange quarks (Gell-Mann, 1964; Zweig, 1964). Strange matter is believed to be more stable than conventional nuclear matter because of strangeness saturation (Gell-Mann, 1964). This

occurs when the number of strange quarks in a system becomes large enough that they begin to interact more strongly with each other, forming a bound state that is more stable than the equivalent system made up of only up and down quarks.

The possibility of strange matter in the cores of neutron stars and pulsars is based on the idea that the extreme conditions in these objects, such as high densities, pressures, and temperatures, could cause a phase transition to occur, converting conventional nuclear matter to strange matter. This transition could be triggered by the conversion of one or more nucleons (protons or neutrons) into strange quarks, a process known as strangeness production.

There exist even more theories concerning the state of matter inside a pulsar core (Lattimer, 2012) and all of them are taking into consideration the extreme physical conditions inside them.

1.1.1 The birth of Pulsars

At the beginning of their existence, stars combine hydrogen (protons) to form helium through fusion. As the amount of hydrogen in the star's core diminishes, it becomes denser and hotter, which facilitates the fusion of progressively heavier elements. If a star's initial mass is higher than $\approx 8M_{\odot}$, it continues fusing heavier elements, up to iron (more specifically, Fe_{56}). Beyond Fe_{56} , which is a very stable nucleus, any fusion processes that attempt to create heavier nuclei than Fe_{56} would actually require an input of energy rather than releasing energy. This is because such fusion processes would involve overcoming the strong nuclear force, which resists the merging of heavy nuclei. The end result of the ongoing fusion, at the star's core, is a buildup of heavier elements. When the fusion ceases, the core will still grow, as it will be supplied with the heavy nuclei synthesized in the outer shells and eventually the gravitational force overcomes the radiation pressure. This leads to the star collapsing and becoming even denser. For a time this collapse is held by the electron degeneracy pressure, just like in the case of white dwarf stars. The force causing electron degeneracy pressure is a quantum phenomenon resulting from the Pauli principle, which states that two identical fermions (with half-integer spin) cannot occupy the same quantum state simultaneously. However, once a body reaches the Chandrasekhar limit the gravity overcomes the pressure. If a collapsing star does not reach this limit, it transforms into a white dwarf, sustained by electron degeneracy pressure. If it exceeds the Chandrasekhar mass, it continues resulting in an incredibly high density that leads to electron capture (Wick, 1933; Alvarez, 1937) which can be expressed by:

$$p^+ + e^- \rightarrow n + \nu_e, \quad (1.1)$$

where p^+ is a proton, e^- an electron, n is a neutron and ν_e is an electron neutrino. The high density of neutrons resulting from the electron capture leads to neutron degeneracy pressure which has the potential to stop the collapse. This means that neutron degeneracy pressure acts as a counter force to gravitational forces that might otherwise cause the object to continue collapsing.

1.1.2 Pulsar Magnetic Fields

Pulsars are characterized not only by their strong magnetic fields but also by the fast rotation speeds around their own axis.

As the core of the star collapses and a pulsar is being formed, the magnetic field of the star is amplified due to magnetic flux conservation. As the volume of the core decreases, the magnetic field lines become more tightly packed, resulting in a stronger magnetic field. The rotation of the neutron star also contributes to the amplification of the magnetic field through a process called dynamo action (Thompson and Duncan, 1993). Their magnetic fields lie in the range between 10^8 and 10^{14} G¹. At the neutron star's surface the Lorentz forces acting are stronger ($\approx 10^{12}$ times) than the gravitational forces and with a sufficiently high temperature ($\approx 10^6$ K) the charged particles are forced out from the polar regions of the pulsar. The positively and negatively charged particles that escaped from the surface will co-rotate with the pulsar, forming the pulsar's magnetosphere.

The magnetosphere extends far from the pulsar, up to a distance where the charged particles' orbital speed would be equal to the speed of light (c). This distance is known as the light cylinder radius (r_{LC}) and if we denote the pulsar's period as P then r_{LC} can be estimated by the relation: $r_{LC} = Pc/2\pi$. The magnetic field lines that cross the light cylinder must remain open. The particles moving along these lines can escape the magnetosphere and traveling along the curved lines they emit radiation that forms a highly collimated beam, meaning that it is focused into a narrow cone rather than being emitted in all directions. In a way the strong magnetic field constrains the emission of radiation to a narrow cone around the magnetic poles. If the emission beam is directed towards Earth and if there is a misalignment between the magnetic field and rotational spin axes the intensity of pulsar's observed signal varies periodically, leading to a "lighthouse-like" effect. Since pulsars are rotating their emitted radiation leads to loss of energy.

Since the emitted electrons carry with them the angular momentum and as a result the pulsar's spin slows down. The amount of the emitted radiation is affected by the pulsar's spin period and its magnetic field. The energy loss rate, $\dot{E}$ (also referred to as spin down luminosity) is:

$$\dot{E} = -\frac{d(I\Omega^2/2)}{dt} = -I\Omega\dot{\Omega} = 4\pi I \frac{P}{P^3}, \quad (1.2)$$

where $\Omega = 2\pi/P$ is the rotational angular frequency and I is the moment of inertia. The strength of the pulsar's magnetic field and its period (P) affect the amount of radiated emission, thus also the slowing down rate, $\dot{P}$.

In the case when the angle between the rotational and magnetic axis is equal to 90° the rotating dipole emission model predicts that we can estimate the strength of the field as:

$$B = 3.2 \times 10^{19} (G) \sqrt{P\dot{P}}. \quad (1.3)$$

Fig. 1.1 portrays the anatomy of a pulsar magnetosphere. The finer details of the structure of the pulsar magnetosphere are still not well understood and there are several models that attempt to describe it. One of the most widely used models is the "outer gap" model (Cheng, Ho,

¹ 1 Gauss denoted as 1 G is a cgs unit of magnetic fields strength, and is equal to $G = 10^{-4}$ T

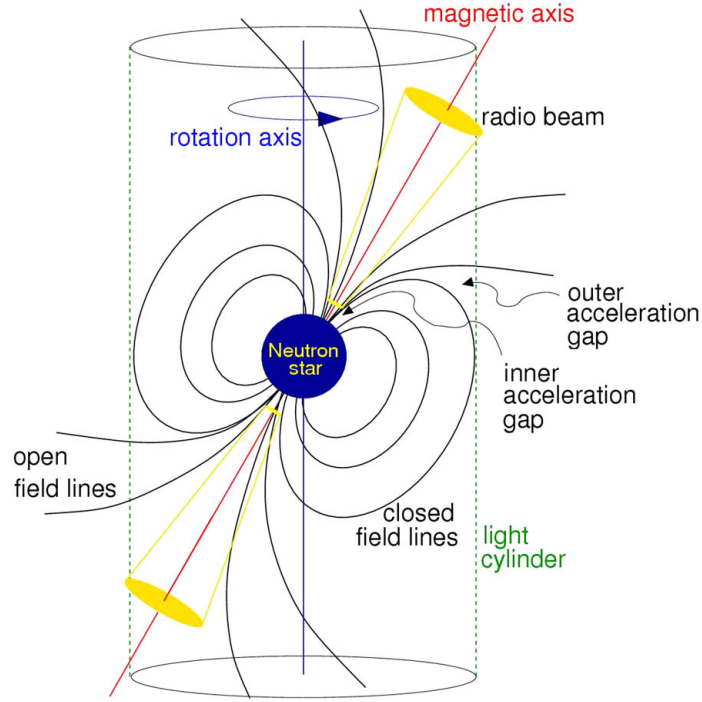


FIGURE 1.1: The anatomy of a pulsar magnetosphere. When a neutron star rotates on its axis, the misalignment between its spin and magnetic axes, along with a favorable positioning of the pulsar in the sky, results in the radio beam it emits intersecting our line of sight. This regular occurrence of the beam crossing our viewpoint causes us to observe radio pulses over time. Credit: Lorimer and Kramer (2004)

and Ruderman, 1986; Romani, 1996), which suggests that the magnetosphere contains regions where electrons and positrons are accelerated to high energies, producing the observed beams of radiation.

From pulsar timing observations we can estimate the pulsar's rotational period P , and the slow-down rate $\dot{P}$ which is defined as the first derivative of the pulsar period. Since the rotation of a pulsar is slowing down due to the emitted radiation and the angular momentum carried away by the escaping electrons, the amount of emitted radiation will decrease over time (see Eq. 1.2) and that will also cause the decrease of the slow-down rate. This can be expressed as:

$$\dot{\nu} = -C\nu^n, \quad (1.4)$$

where n is a dimensionless variable called the braking index and C is a constant. For a magnetic dipole $n = 3$ (Lorimer and Kramer, 2004). However, deviations from this value have been mentioned (Johnston and Galloway, 1999; Kaspi and Helfand, 2002). The value of the braking index provides information about the underlying mechanisms responsible for slowing down the rotation of the pulsar, such as the loss of rotational energy through emission of electromagnetic radiation or gravitational waves.

Eq. 1.4 can be re-written in terms the rotational period as:

$$\dot{P} = -CP^{2-n}. \quad (1.5)$$

If we assume that at their origin, pulsars have a rotational period P_o such that $P_o \ll P$ and that the magnetic dipole radiation mechanism is responsible for the pulsar's slowing down, then from Eq.1.5 we can estimate the characteristic age (τ_c) of the pulsar according to:

$$\tau_c = \frac{P}{2\dot{P}}. \quad (1.6)$$

The characteristic age is only a rough estimate of the true age of the pulsar and not a direct measurement of the pulsar's age. For example, some pulsars may have been spun up or slowed down by the emission of gravitational waves or other processes, which can cause their spin rate to change over time. Furthermore, Eq. 1.6 assumes that when a pulsar is born, its period is equal to zero seconds which is a large simplification. In reality, the observations that attempted to estimate the initial spin periods of a pulsar indicate the values in the 10 to 20 millisecond range.

1.1.3 Pulsar Population

Currently, there are ≈ 3300 known pulsars² (Manchester et al., 2005). Their population is quite diverse, with pulsars having different spin periods, magnetic field strengths, and emission properties.

The study of the pulsar population provides important information about the physical conditions in these objects and their evolution. For example, the distribution of spin periods and magnetic field strengths can provide information about the birth and evolution of neutron stars. The diversity of emission properties can offer important information about the physical processes responsible for producing the emission and the conditions in the pulsar magnetosphere.

The pulsar spin period P and its time derivative $\dot{P}$ are very useful parameters in the study of these properties and variations. When plotted against each other the, so called, P - $\dot{P}$ diagram helps in classifying pulsars and understanding their evolutionary history. For example, young pulsars are characterized by large values of $\dot{P}$, indicating that they are rapidly slowing down due to the loss of rotational energy.

The P - $\dot{P}$ diagram also provides important information about the internal structure of pulsars and the physics of neutron stars. For example, some models predict that the $\dot{P}$ of a pulsar depends on its magnetic field strength (Xie and Zhang, 2019), which can be determined from its position on the P - $\dot{P}$ diagram. Such a plot is shown in Fig. 1.2, where we can see the two main pulsar population groups. Millisecond pulsars, most of which are members of binary systems, mostly occupy the lower left corner of the plot, whereas its right part is occupied by the typical pulsars.

Young pulsars are located in the top left region of the Fig. 1.2. As they mature, they lose energy and transition into a group referred to as normal or canonical pulsars. These pulsars have a slower rotation rate and make up the majority of the pulsar population, with spin periods ranging from 0.1 seconds to a few seconds and surface magnetic fields of approximately 10^{10} to 10^{13} Gauss. They have a long lifetime, with characteristic ages spanning from 300,000 years

²<http://atnf.csiro.au/research/pulsar/psrcat/>

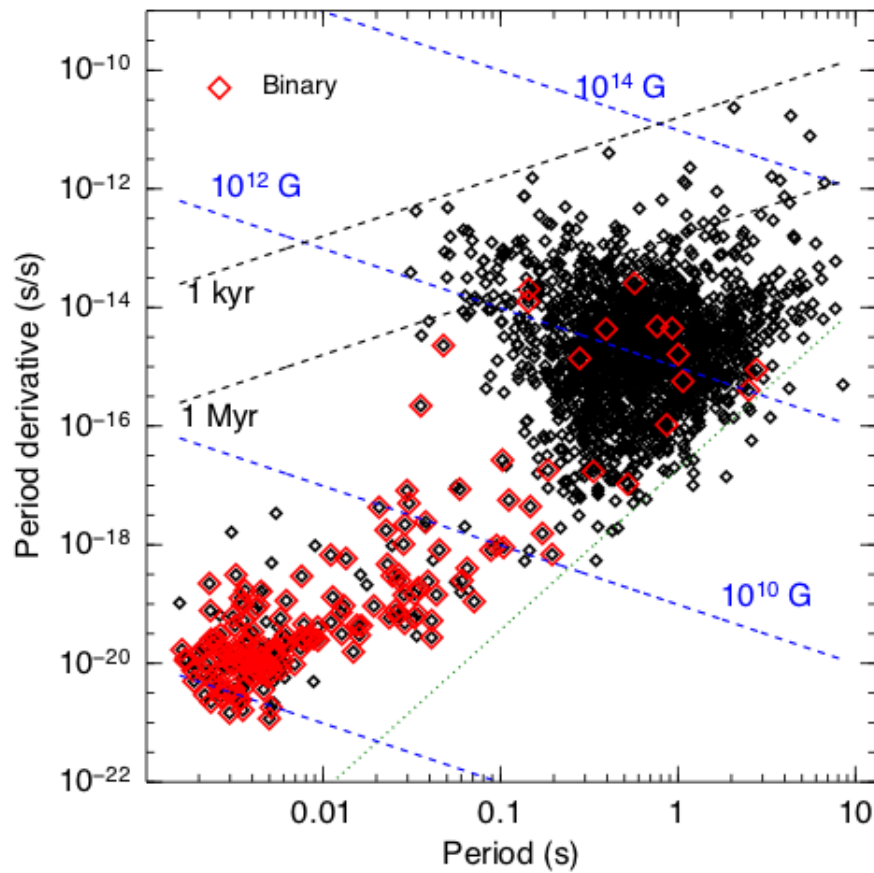


FIGURE 1.2: This diagram is a $P - \dot{P}$ diagram, showing the two main groups of pulsars based on their period and its derivative. The dashed lines represent the constant magnetic dipole field strength (B) and characteristic age (τ). Credit: Wikimedia Commons.

to 1 billion years. If these pulsars don't interact with other celestial bodies, they will eventually decelerate to the point where the slow rotation causes a lower electric field, which, in turn, prevents the electrons from being accelerated to sufficiently high energies. Consequently, the gamma-ray photons lack the necessary energy for pair creation, leading to the absence of electron cascades or "sparks," and as a result, there are no electrons available to emit radio waves. As a result pulsars become invisible at radio wavelengths and cross the "death line." The exact location of this line strongly depends on the details of the underlying theories which slightly differ in their predictions, and some pulsars have even been discovered beyond this line.

Binary pulsars occupy a range of positions on the $P - \dot{P}$ diagram, depending on their binary parameters and the stage of their evolution. They are very useful for studies of the nature of gravity and the properties of dense matter, because of their precise periods and compactness.

Most of the millisecond pulsars belong to the binary group. There also exist isolated millisecond pulsars, but it is believed that they were also members of a binary system, that at some point got disrupted, for example, by a supernova explosion. They have extremely fast spin periods, ranging from a few milliseconds to a few tens of milliseconds. These pulsars are thought to

be "recycled" pulsars that have been spun up by the accretion of matter and angular momentum from a companion star. They are the leftovers of a binary system that had two main sequence stars. The more massive component evolves faster and undergoes a supernova (SN) explosion that leads to a pulsar formation. If the binary system does not get disrupted by the explosion, it now consists of a pulsar and a main sequence star. The later will evolve into a red giant and fill its Roche lobe. The Roche lobe is the theoretical boundary around a celestial body in a binary star system within which the gravitational force of that body dominates over the gravitational force of its companion. As the outer plasma layers of a star become unbound by gravity, matter will begin to escape through the inner Lagrange point (L1) of the binary system. This escaping matter is then gravitationally pulled towards the companion pulsar, resulting in the pulsar gaining angular momentum from its companion star.

1.2 Pulsar Profile Morphology

Pulsars emit radio waves in the form of periodic pulses, each with a unique shape. Each recorded pulse corresponds to a single pulse profile. A single pulse profile is a graph that shows the shape and intensity of a single pulse of radiation as a function of time.

When these single pulse profiles are added together over time to create an integrated (or averaged) profile, the resulting profile is stable and characteristic of the pulsar at a particular frequency. This integrated profile, which represents the average of many individual profiles, maintains a consistent structure over time and is indicative of both the pulsar's emission process and its geometry. On the other hand, the individual profiles bring us information about the emission's momentary state.

A widely accepted model of the profile morphology has been established in Rankin (1983), which suggests that the emission beam is split into an inner (core) and an outer (conal) region. In Rankin (1993) this model was modified. This version suggests that the cone region may be surrounded by more than one conal regions. Furthermore, the emission from each region is not uniform but originates from discrete regions inside them (Fig. 1.3(a)). There exist more models such as the patchy beam model (Lyne and Manchester, 1988) which suggests that the radio emission from a pulsar is not uniform, but rather is composed of isolated regions, or "patches," of emission within the overall emission beam. However, the core and cone model's predictions, about the beam geometry, are the ones that agree the most to real data.

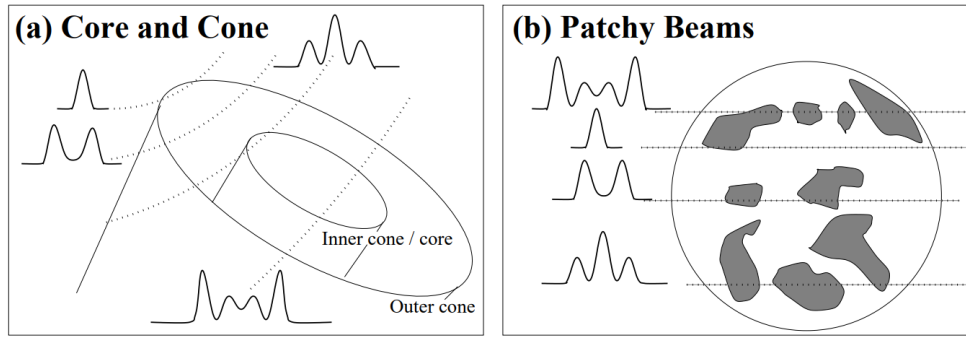


FIGURE 1.3: **Left:** The number of observed profile components depends on the number of nested emission cones, as well as on the cutting angle of the line of sight. Different lines of sight result to different profiles. **Right** The patchy beam model which suggests that the radio emission from a pulsar does not originate from a single, well-defined point on its surface but instead comes from multiple emission regions (or patches), distributed across its surface. Figure from Lorimer (2005).

1.3 Radius to Frequency Mapping

Regardless of the true emission mechanism, a common property of all pulsar profiles is the fact that they exhibit a radius to frequency mapping (RFM). A commonly observed phenomenon among pulsars is the tendency their profiles to broaden as the observing frequency decreases. This can be attributed to the properties of curvature radiation, which is the most likely mechanism of pulsar radiation. It involves the emission of photons from the electrons traveling along the magnetic field lines, and due to the curvature of these lines the radiation will be emitted in a tangent direction. Therefore, at lower frequencies, the radiation must be emitted from regions where the magnetic field is oriented at larger angles with respect to the magnetic axis (Manchester and Taylor, 1972a). This is in line with the principles of electrodynamics as it states that the frequency of a photon discharged by an electron in motion is linked to its acceleration. When an electron moves along a curved magnetic field line, it undergoes a form of acceleration, and the emitted frequency is determined by the radius of curvature. Therefore, high-frequency radiation can only originate from the magnetosphere region with a larger radius of curvature, which is closer to the star. Conversely, low-frequency photons are created at much higher altitudes, where the magnetic field's curvature reduces and the magnetic field lines are more open, leading to broader profiles.

1.4 The Interstellar Medium

While pulsars are fascinating and unique celestial objects, this thesis is not focused on their characteristics. Rather, we are exploiting their observational properties to examine phenomena associated with the propagation of their radiation through interstellar medium (ISM). The concept of using pulsars only as tools to study various effects is widely exercised, not only in

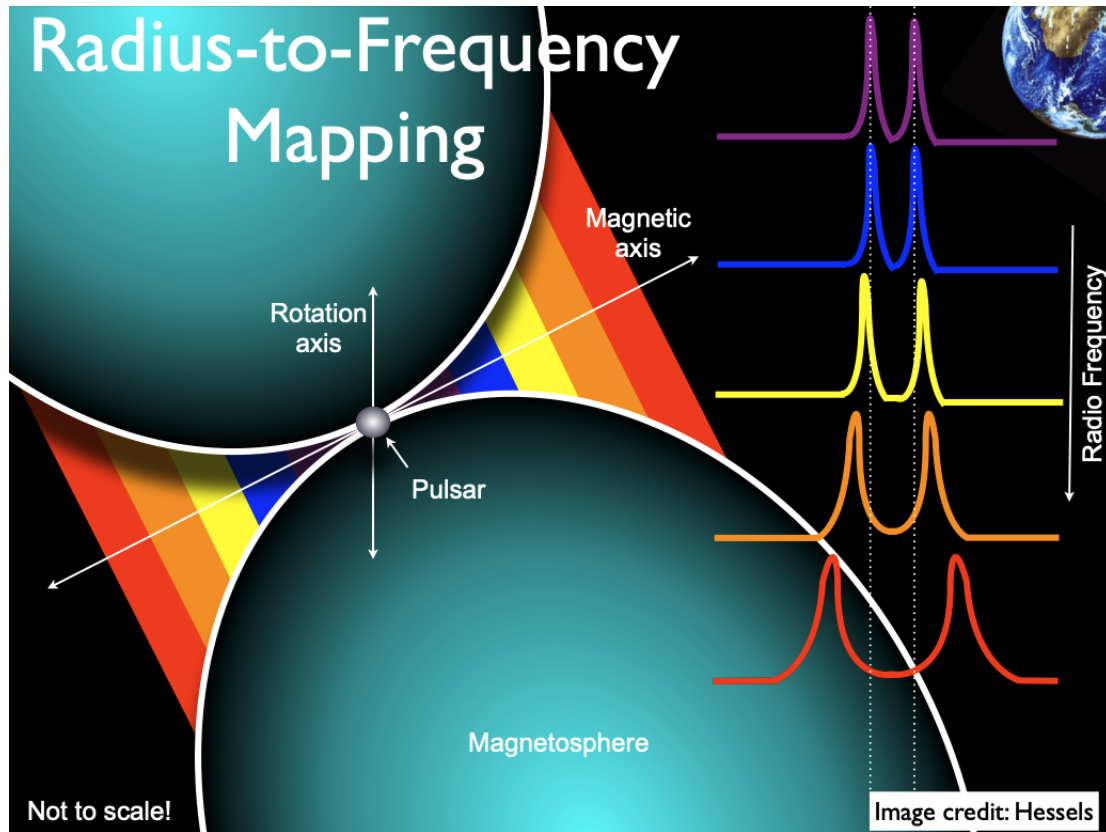


FIGURE 1.4: Radius to Frequency Mapping: According to the radius-to-frequency mapping (RFM) model, the emission of higher frequencies occurs nearer to the surface, resulting in narrower pulse shapes. Conversely, the emission heights for lower frequency radiation are higher, which leads to broader pulse shapes. Image by Jason Hessels.

ISM research but also in other areas such as timing studies, where they play a significant role in studying strong gravity in binary systems and exploring gravitational waves.

The ISM is the matter that fills the space between stars in our Galaxy. This space is not empty, but consists of low density dust and gas. The ISM's mass is irregularly distributed and accounts for 10 -15 % of our Galaxy's mass. Almost half it is located in clouds that cover 1-2 % of our Galaxy's volume (Ferrière, 2001).

1.4.1 Interstellar Medium Categories

The ISM can be classified into four categories.

- The Cold Neutral Medium. It consists of neutral atoms molecules and dust. Because of its low temperature ($\approx 10 - 10^2$ K) clouds that host protostars (a combination of gas and dust that is being turned into a star) can be formed.
- The Warm Neutral Medium. Its temperature is $\approx 10^4$ K and consists of neutral particles, mostly of neutral Hydrogen (HI), observed at 1.4 GHz.
- The Warm Ionised Medium, which also has a temperature $\approx 10^4$ K. It consists of ionised gas (mostly HII). The ISM close to massive hot OB stars gets ionised because of their UV

radiation. OB associations are large groups of massive OB stars that are typically very young. The gravitational attraction among the stars within an OB association can cause them to come closer together, leading to the formation of young open clusters. As time passes, the stars within the open cluster can continue to evolve and interact, ultimately resulting in their dispersal and integration into the surrounding population of stars.

- The Hot Ionised Medium, is a halo around our Galaxy, that is observed in X-rays. The general belief is this halo consists of the hot gas that supernova explosions eject far from the Galactic disk and gas that was heated by the supernova explosion shockwaves. Its temperature is $\approx 10^6$ K and because it is ionised, it also interferes with the pulsar signal.

The warm and hot ionised medium comprise the ionized interstellar medium (IISM) that is responsible for the propagation effects discussed in Sec. 1.5.

1.4.2 Turbulence in the Interstellar Medium

The ISM is turbulent. The kinetic energy that feeds the turbulence, possibly, originates from supernova explosions (Korpi et al., 1999; Kirchschrager, Mattsson, and Gent, 2022), expanding ionising shells (Krumholz, 2007), magnetorotational instability (Dobbs et al., 2008; Piontek and Ostriker, 2007), gravitational instability (Krumholz and Burkhardt, 2016) and stellar winds (Owociki, 1999; Gallegos-Garcia et al., 2020).

The presence of turbulence in the ISM has a number of important effects. For example, turbulence can support molecular clouds against gravitational collapse, leading to star formation. It can also create dense, high-pressure regions in the ISM that are able of triggering the formation of new stars. Additionally, turbulence can mix and redistribute elements and isotopes throughout the ISM, influencing the chemical composition of the universe.

The Gaussian and Kolmogorov models are widely used in describing turbulence. Gaussian turbulence is a statistical model that assumes that the velocity fluctuations in a turbulent flow follow a Gaussian distribution. This model is based on the central limit theorem, which states that the sum of many independent variables will tend to follow a Gaussian distribution. That is why the statistical properties of turbulence are described in terms of the mean and variance of the velocity fluctuations.

The Kolmogorov turbulence, on the other hand, is a scaling model that assumes that turbulence has a universal form, regardless of the length scale at which it is observed. The model is based on the work of the Soviet physicist Andrei Kolmogorov, who proposed that turbulence could be described by a series of cascading eddies that transfer energy from large to small scales. In the Kolmogorov model, the scaling exponents that describe the turbulence are independent of the fluid properties and are determined solely by the energy transfer between eddies.

The fluctuations of the electron density inside the ISM belong to a range of various scales, which are described by a three-dimensional wavenumber. The extended power law model (Rickett, 1990; Armstrong, Rickett, and Spangler, 1995) is used for describing the power spectrum of these fluctuations, according to which:

$$P_{n_e} = \frac{C_{n_e}^2(z)}{(\kappa^2 + k_o^2)} e^{-\frac{q^2}{4\kappa_i^2}}, \quad (1.7)$$

where κ is the magnitude of the three-dimensional wavenumber, κ_o and κ_i are the outer and inner turbulence scales and $C_{n_e}^2(z)$ is the strength of the fluctuations along a given line of sight (in units of $m^{-20/3}$). According to Rickett (1990) Eq. 1.7 can be simplified to:

$$P_{n_e} = \begin{cases} C_{n_e}^2 q^{-\beta} & , \kappa_o \ll \kappa \ll \kappa_i \\ 0 & , \text{otherwise} \end{cases} \quad (1.8)$$

in the case of a Kolmogorov power spectrum, $\beta = 11/3$ (Rickett, 1977; Bhat et al., 2004). According to Romani, Narayan, and Blandford (1986), $4/3 \leq \beta \leq 11/3$. In this case, the turbulence is propagating with a speed smaller than the speed of sound in the ISM.

1.5 Propagation Effects

The ISM is affecting the emitted pulsar signal in a few different ways. Ideally, we would like to be able to split the recorded pulsar profile into two parts. The first part would correspond to the intrinsic pulsar profile and carries information about the pulsar emission mechanism. The second part would be the signature the ISM leaves on the intrinsic profile.

The radio emission from pulsars interacts with the free electrons in the ionised interstellar medium (IISM) as it travels, resulting in observable propagation effects. In this section, we will focus on three effects that can impact the pulsar signals in the IISM: scintillation, scattering, and dispersion.

1.5.1 Dispersion

One very interesting effect we can observe using pulsars is the dispersion effect. It is caused by the fact that the radiation at different wavelengths propagates through the IISM with different velocities. Even if the variation of the propagation velocity is very small the dispersion effect causes measurable differences of the pulse arrival times between different observing frequencies. For modelling purposes, we usually assume that the IISM is homogenous and isotropic and that the free electrons of the IISM cause the dispersion of the E/M waves. The optical index of a wave with a wavelength λ , is:

$$\mu_p = \sqrt{1 - \frac{n_e r_o \lambda^2}{\pi}} < 1, \quad (1.9)$$

where $r_o = \frac{e^2}{m_e c^2}$ The wave's propagation speed, u_p that corresponds to μ_p is

$$u_p = \frac{c}{\mu_o} > c, \quad (1.10)$$

which may seem contrary to the Special Theory of Relativity. However, u_p is the phase velocity of a wave with a specific frequency and a signal does not consist only of just one monochromatic

wave. That means that there is a wave group with slightly different frequencies from each other. This group propagates with a velocity $u_g \neq u_p$. Because $u_g u_p = c^2$, we have:

$$u_g = c\mu_p = c\sqrt{1 - \frac{nr_o\lambda^2}{\pi}} < c. \quad (1.11)$$

If d is the distance the wave group will travel in time τ :

$$\tau = \frac{d}{u_g}, t \propto \lambda. \quad (1.12)$$

The time τ is different than the time required for the wave group to travel the same distance, d , in vacuum. So the wave group will seem delayed compared to its propagation in vacuum. The delay will be

$$\Delta\tau = \frac{d}{u_g} - \frac{d}{c} = \mu_p^{-1} - 1. \quad (1.13)$$

In the case of ISM, $n_e \ll 1$ the delay time may be shown to be:

$$\Delta\tau = \frac{e^2}{2\pi m_e c} \frac{DM}{f^2} \quad (1.14)$$

$$DM = \int_0^d n_e dl,$$

where DM is the so called Dispersion Measure (DM). This quantity describes the column density of electrons along the line of sight inside a cylinder in the IISM, with an area equal to unity and height equal to d . The existence of the dispersive delay, which is inversly proportional to the square of the observing frequency means that the pulse will arrive at the higher frequencies first, and only later at lower observing frequencies. If the pulsar is observed in frequencies that are Δf apart, then according to Manchester and Taylor (1972b) the difference in the pulse arrival times between these frequencies will be:

$$\Delta\tau = \frac{e^2}{\pi m_e c} DM \frac{\Delta f}{f^3} = 1.25 \times 10^4 DM \frac{\Delta f}{f^3}, \quad (1.15)$$

where f is measured in MHz and DM in pc cm^{-3} .

1.5.2 Interstellar Scattering

Shortly after the first pulsar discovery (Hewish et al., 1968), it was noticed that such sources flux is in some cases highly variable and that one possible reason for this behavior, is the scattering and scintillation of their signal (Lyne and Rickett, 1968; Scheuer, 1968) as it propagates through the IISM. The pulsar rays travel different lengths as they are being scattered inside the ISM. As a result, some of the rays travel longer distances arrive later to the observer and the observed pulsar profiles become broader. More analytical work was conducted a few years later by Williamson (1972) and Williamson (1973), where the thin, thick and extended ISM screen models were introduced, each one with different assumptions.

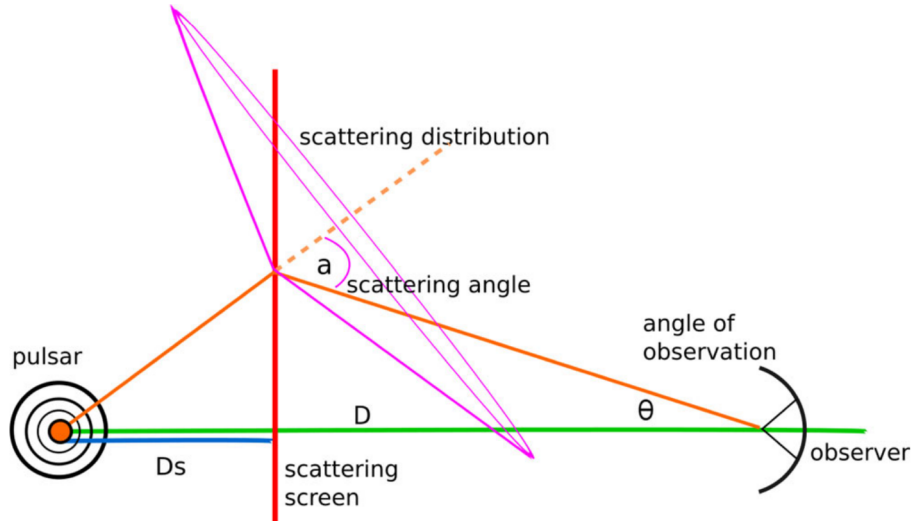


FIGURE 1.5: Thin screen model. Top: A schematic that describes how the emitted pulsar signal is modified by the ISM. The scattering screen is assumed to be located at a distance D from the Earth and D_s from the pulsar. α is the angle by which the signal is scattered and θ the angle at which the signal is observed. Schematic from Geyer and Karastergiou (2016).

The thin screen model assumes that scattering happens in a medium which is thin along the propagation axis and placed in a single location between the observed pulsar and Earth. This is in contrast to the thick screen model according to which, irregularities in the ISM occupy a larger area and the screen can be located close to the pulsar or the observer. Another model, called the extended screen model assumes that the ISM irregularities are spread throughout the whole propagation path, between the pulsar and the observer. Some other, more complicated models of scattering were also proposed, like multiple scattering screens, however they are not, yet, fully developed (Kirsten et al., 2019). The thin screen model has gained popularity among other models, because of its simplicity and its ability to describe the observational data very well. In Geyer and Karastergiou (2016) the isotropic and anisotropic versions of the thin screen model are suggested, depending on whether the distribution of the angles of the rays exiting the screen, is symmetrical or not.

Scattering happens when the pulsar rays propagate through regions of the warm ionised medium (1.4.1). Compared to the rest of the ISM, these regions are small and relatively rare because our Galaxy is not very active in terms of supernovae explosions and star formation. Hence, the probabilities of the rays getting scattered are small and the effect of scattering is not much severe. This is why the thin screen model seems more suitable, compared to the thick screen model, for modeling the scattering medium.

Scattered Profiles

As the pulsar signal propagates through the ISM it becomes scattered because of inhomogeneities. As a result, the apparent pulse profile will acquire a scattering tail, because a part of the pulsar's emission travels along longer paths thus they arrive later.

The pulse profile we observe is the result of a convolution between the intrinsic pulsar profile and the "response function" of the scattering screen, which is commonly referred to as the PBF (Pulse Broadening Function). Typically, we simplify the intrinsic pulse profile using a Gaussian function (Kramer et al., 1994) or a combination of Gaussian functions that represent distinct profile components. In the context of the thin screen model, the simplest PBF employed is an exponential decay function. When a Gaussian signal is convolved with this particular PBF, it gives rise to an exponentially modified Gaussian (EMG). Therefore, in such scenarios, the observed pulse profile can be interpreted as an EMG or a sum of EMGs in the case of multiple component profiles.

Since the PBF is the one describing the scattering effect, a variety of forms have been proposed. If we focus on the thin screen model, $PBF \sim e^{t/\tau_d}$. Still within the thin screen model framework, some more PBF variations exist. For example: $PBF \sim \tau^{-1} e^{-\frac{t}{\tau}}$ or $PBF \sim \frac{e^{-\frac{t}{\tau}}}{\sqrt{\pi\tau t}}$ for an isotropic/anisotropic IISM screen respectively (Geyer and Karastergiou, 2016). According to the thick screen model $PBF \sim \sqrt{\frac{\pi\tau}{4t^3}} \times e^{-\frac{\pi^2\tau}{16t}}$. The extended screen model uses the assumption that $PBF \sim \sqrt{\frac{\pi^5\tau^3}{8t^5}} e^{-\frac{\pi\tau}{4t}}$.

The scattering effects are highly frequency dependent, since (as it was mentioned in Sec. 1.5.1) different frequencies (wavelengths) will undergo refraction in the IISM to a different degree (see eq. 1.9) with longer wavelengths experiencing higher refractive index. This leads to a situation where lower frequency radio waves may be scattered at higher angles (Fig. 1.5) than the higher frequencies, and that in turns means that they may travel much longer paths. The result of this is that at lower frequencies the pulse broadening effect is much stronger, and the scatter time which measures this effect varies with frequency as:

$$\tau \propto \nu^{-\alpha} \propto \nu^{2\beta/(\beta-2)}, \quad (1.16)$$

where α is the so called frequency scaling index and indicates that τ_d is frequency dependent.

The spectral index (β) of the fluctuation spectrum can be used to derive the theoretical value of α , by the relation $\alpha = 2\beta/(\beta - 2)$. If we assume a Kolmogorov spectrum for the density functions then $\beta = 11/3$ and $\alpha = 4.4$. If the inhomogeneities inside the ISM follow a Gaussian distribution, then $\alpha = 4.0$.

Recent studies of the scattering properties, performed by Lewandowski et al. (2015), Geyer and Karastergiou (2016), Lewandowski et al. (2018), Kirsten et al. (2019), and Bansal et al. (2019a) show that in reality α can have different values. Some proposed explanations are the existence of multiple screens and non Kolmogorov turbulence.

Regardless of the α value, because of Eq. 1.9, at low frequencies the profile components seem broader compared to higher frequencies. An example of the scattering frequency evolution is shown in Fig. 1.6.

Fig. 1.7 illustrates the dependence of scattering time on observing frequency for a sample of pulsars observed using the PL611 LOFAR station. These plots provide critical insights into the interstellar medium's influence on pulsar signals, particularly how scattering varies with frequency. The data gathered from the PL611 station enables a detailed analysis of the frequency-dependent scattering that can shed light on the distribution and properties of turbulent plasma

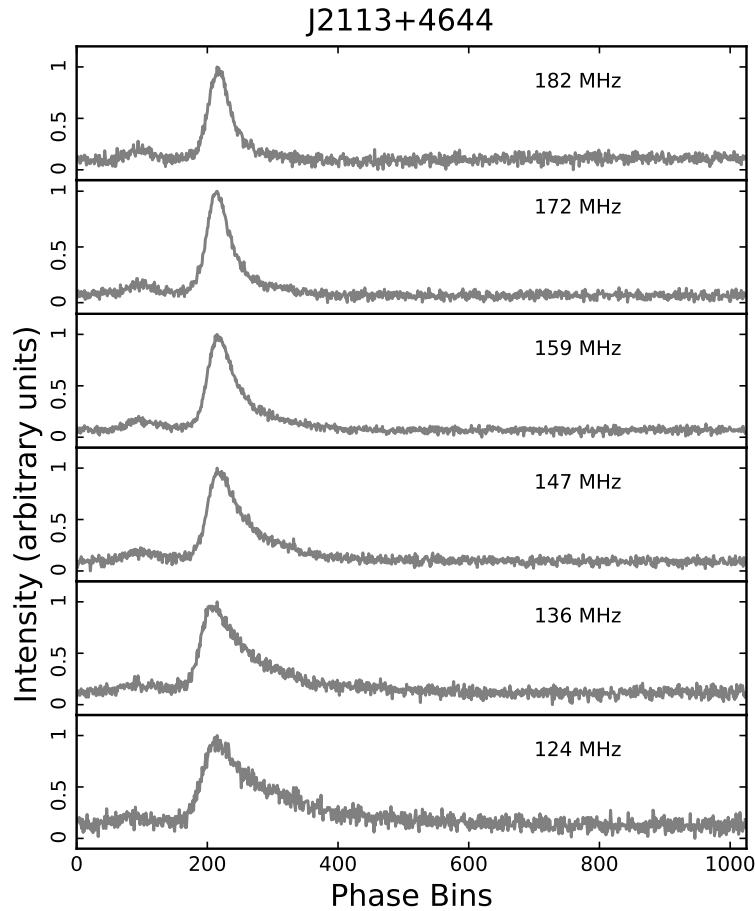


FIGURE 1.6: An example of the scattering's frequency dependence. The data was collected using the PL611 station located in Łazy, Poland. The 72 MHz bandwidth was split into 6 sub-bands. We can see how the scattering tail becomes shorter as the observing frequency increases.

structures along the line of sight to the pulsars. This can lead to a better understanding of dispersion and scattering phenomena in pulsar observations at low frequencies.

1.5.3 Scintillation

Scintillation is the effect of short-time intensity variations of the received signal. It is caused by electron density variations between the observed pulsar and Earth. The thin screen model (Fig. 1.8) is also used for the explanation of scintillation. These variations lead to multi-path propagation of the signal, and a disturbed wavefront of the pulsar radiation will interfere with itself, leading to amplification or dampening of the pulsar signal.

Because of scattering the received signal is showing phase changes ($\delta\Phi$), over a time interval τ_s , which are observed as interference patterns. The phases changes follow the relation:

$$\delta\Phi \sim 2\pi f \tau_s, \quad (1.17)$$

where f is the observational frequency.

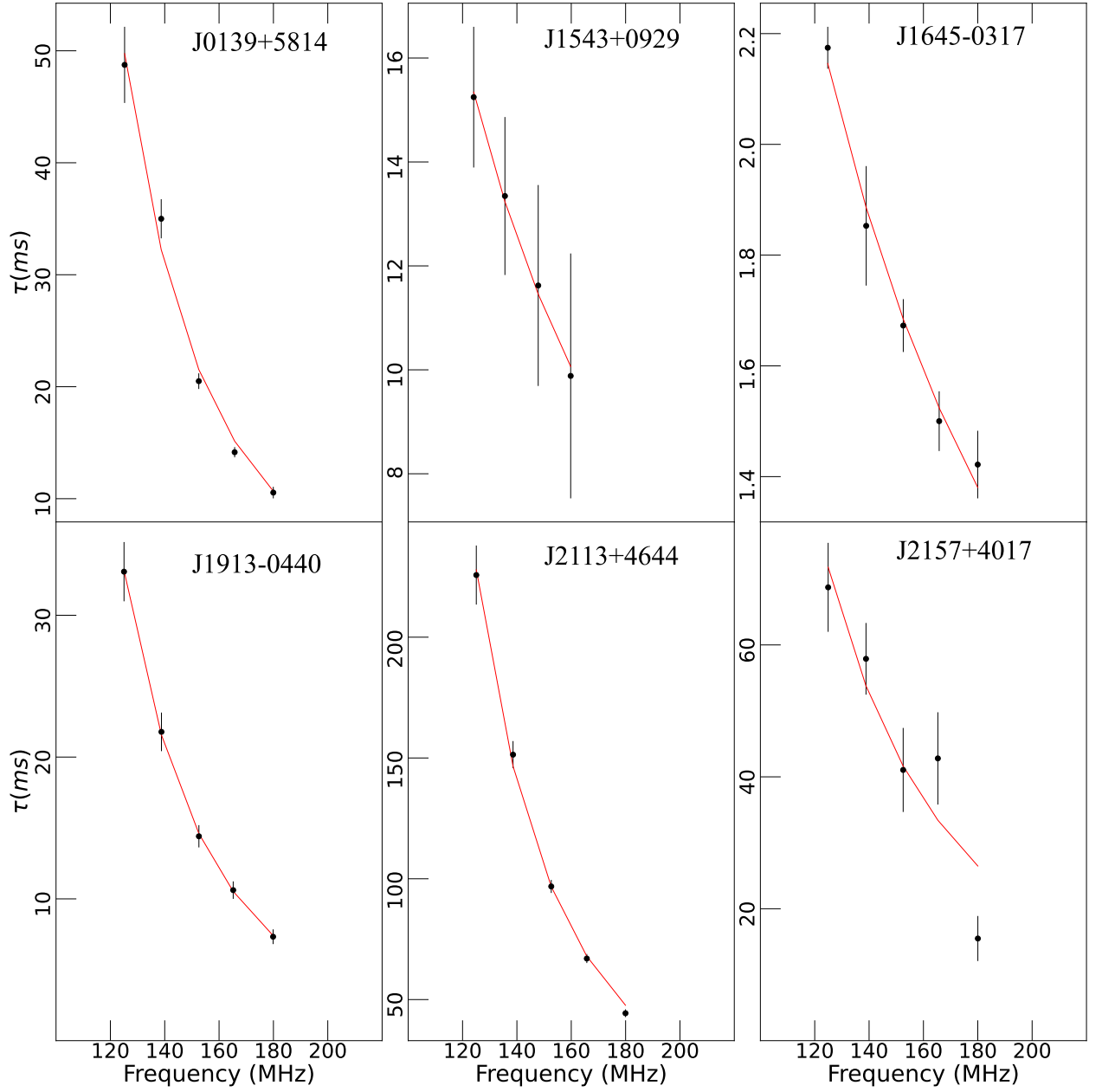


FIGURE 1.7: Plots showing the scatter time frequency dependence for a sample of pulsars observed with the PL611 LOFAR station.

The interference patterns are observed if :

$$2\pi\delta f\tau_s \sim 1 \text{ radian}, \quad (1.18)$$

where δf is the decorrelation bandwidth (Lyne and Rickett, 1968; Cordes, 1986)

The interference patterns can be seen in the dynamic spectra of the pulsar flux as regions of constructive and destructive interference (Fig. 1.8). Dynamic spectra are two dimensional observations where the intensity of a pulsar measurement is a function of frequency and time.

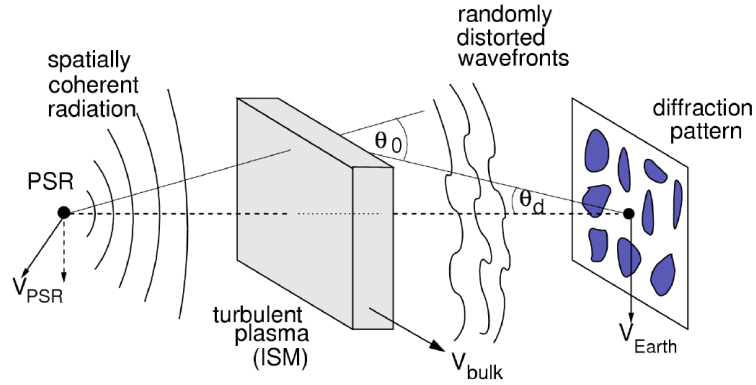


FIGURE 1.8: The thin screen model, used to explain the scintillation effect, due to inhomogeneities in the ISM. Figure from Lorimer and Kramer (2004).

1.5.4 Faraday Rotation

The polarization of the radiation varies inside the ISM, the radiation emitted by pulsars is usually highly polarized and when travelling through the ISM it encounters the Galactic magnetic field. In such a case we will see the effect of a phenomenon known as Faraday's rotation, which changes the position angle of the polarization, according to the formula:

$$\Delta\phi = \frac{e^3}{2\epsilon_0 m_e^2} c \omega^2 \int_0^d n_e(z) B_o(z) dz, \quad (1.19)$$

where n_e is the electron density, $B_o(z)$ is the longitudinal magnetic field and ω is the wave frequency.

To be able to measure the RM, the source needs to be (partly) linearly polarised. Many pulsars are polarised and thus the RM can be measured for many different directions and distances. From this, it is possible to map the Galactic magnetic field (Han et al., 2018). Typical magnetic field strengths are of the order of a few μG (Sobey et al., 2019).

1.6 Pulsars as tools to understanding the Interstellar Medium

By studying the properties of pulsars, such as their pulse periods and pulse profiles we can learn about the fundamental properties of neutron stars, such as their masses, sizes, magnetic fields, and spin rates. By calculating the DM values of many pulsars, located in different distances and locations, we can estimate our Galaxy's electron density distribution (Cordes and Lazio, 2003).

The variability of dispersion provides information about the structure and properties of ionized material along the line of sight from the radio source to the observer. The DM values may vary with time due to changes in the ionized material or changes in the position of the observer and/or the pulsar with respect to the ionized material. By analyzing the variability of DM, we can gain insights into the dynamics of the ionized material in the ISM.

Scintillation is a result of the interference between different paths that radio waves take as they propagate through the ISM. The variations in the refractive index of the medium cause the

phase of the wave to change, leading to constructive and destructive interference. This leads to fluctuations in the amplitude and phase of the radio wave, which can be observed as variations in the brightness of the radio source. Scintillation provides information about the scale size of the inhomogeneities in the ISM, and can be used to probe the small-scale structure of the ionized material.

Scattering is a result of the inhomogeneities in the ISM that cause the radio wave to be redirected in different directions. The scattered wavefront is distorted and broadened, leading to a reduction in the apparent brightness and spatial resolution of the radio source. By analyzing the scattering of radio waves, we can derive information about the size and distribution of inhomogeneities in the ISM, and gain insights into the turbulence and evolution of the ISM.

In this thesis I focus on the study of pulsar scattering using a single LOFAR station, located in Poland. The data preparation and analysis pipeline are described in the next section. During the analysis we implemented two different models of the turbulence distribution inside the ISM and its variability of the course of 4 years.

2 Observations

In this section I will describe the telescope used for conducting observations and a basic operational pipeline that was followed for recording and manipulating the data. Then I will proceed with the description of the data analysis workflow created for studying the effects of interstellar scattering on the observed pulsar profiles.

2.1 LOFAR

The Low Frequency Array (LOFAR) is a network of single radio observing stations each capable of observing at low radio frequencies. LOFAR is unique in its ability to observe the universe at frequencies as low as 10 MHz, which is much lower than what is possible with most other radio telescopes. LOFAR is made up of a network of thousands of small antennas which are grouped in the so-called "stations" spread across 8 countries in Europe, with 2 more countries (Italy and Bulgaria) being in the process of planning the commissioning of their LOFAR stations (Fig. 2.1).

The signals from these stations are combined to form a highly-sensitive radio telescope that can detect and study a wide range of astrophysical phenomena, including pulsars, radio galaxies, and the cosmic microwave background.

A large number of LOFAR stations exist in the Netherlands. As shown in Fig. 2.1 the rest of the stations are spread throughout Europe and are considered part of the International LOFAR Telescope (ILT). Regardless of their location, the stations can be also used in combination, as one radio telescope (with a baseline of ≈ 1980 km) or they can also be used as separate telescopes (van Haarlem et al., 2013).

LOFAR has a large field of view (FOV), which allows it to observe a large area of the sky simultaneously. In contrast, the dish radio telescopes (both the individual ones as well as the interferometers made of single dish telescopes) typically have a smaller FOV and must be pointed precisely at the target in the sky. LOFAR's resolution is lower compared to dish radio telescopes, but on the other hand its large field of view allows us to observe a large number of sources, simultaneously.

2.2 Phase array radio telescopes

Individual LOFAR stations are interferometer radio telescopes working as phased arrays made of a large number of individual dipole antennas. If a radio wave arrives at the station with an angle not perpendicular to the surface, it will arrive slightly later at the furthest antenna. Using this time delay, it is possible to adjust the signal from each antenna in such a way that, simplifying, only the signal originating from a chosen direction is not cancelled out. This process

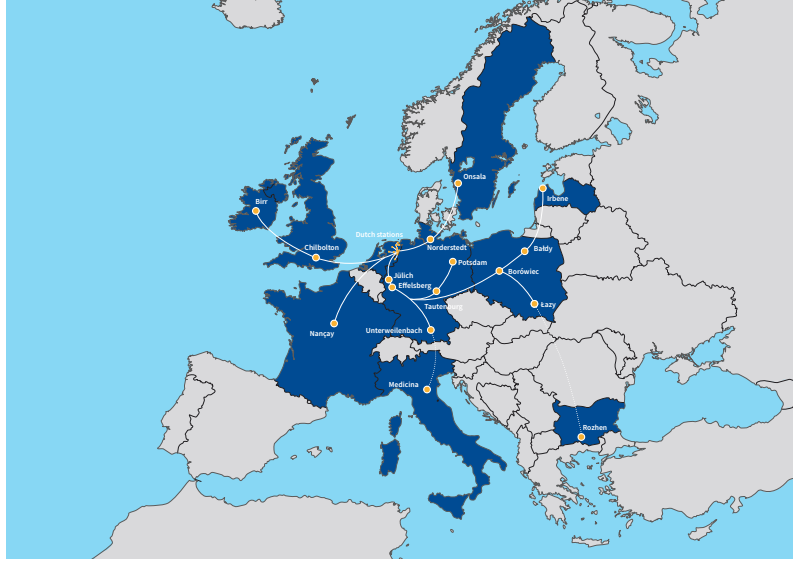


FIGURE 2.1: The most current distribution (as of October 2023) of LOFAR stations including the future Medicina (Italy) and Rozhen (Bulgaria) stations. Each dot represents a single LOFAR station and each solid white line the fiber optic cable that connects each single station to a supercomputer located at the LOFAR headquarters. The image is property of ASTRON.¹

is called beamforming and allows the user to digitally point the telescope in the direction of any observed source.

The geometric delay of a radio front wave reaching an antenna at a location x_n , compared to its arrival to a phase center x_o is:

$$\tau_n = t_n - t_o, \quad (2.1)$$

where t_n is the time at which the wave arrives at x_n .

The way the time delay is applied, depends on the hardware used to observe at a set frequency range. When the station is observing in the higher frequency range between 110-250 MHz, these delays, with respect to the centre of the tile, are applied by physically between cables with different lengths. However, when the station is observing as low frequencies between 10-90 MHz the time delay is performed inside the electronics cabin, where each subband signal is multiplied with the complex phase factor:

$$e^{\pm 2\pi i \nu_m \tau_n}, \quad (2.2)$$

before adding the signals. ν_m is the central frequency of a sub band m .

Thanks to beamforming, despite the antennas being stationary the pointing or directivity of the phase array can be altered by incorporating distinct phase delays into each one of them.

2.3 Anatomy of a LOFAR station

Each single LOFAR station consists of two groups. Both parts of LOFAR, the Low Band Antennas (LBA, 2.3.1)) and the High Band Antennas (HBA 2.3.2) provide a wide bandwidth of

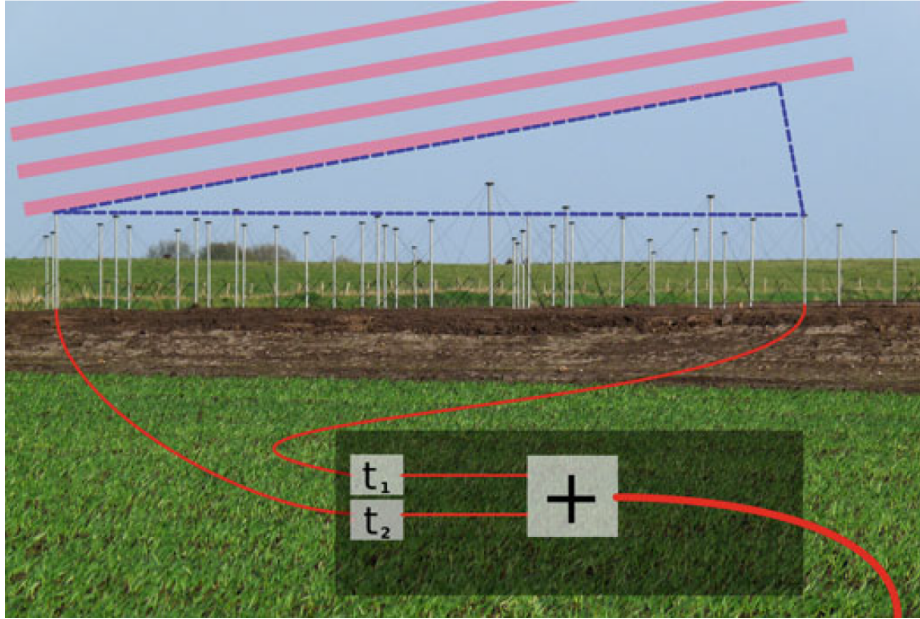


FIGURE 2.2: The LOFAR antenna stations operate as phased arrays, where their aiming is achieved by manipulating the signals in a way that ensures waves arriving from the desired direction are perfectly synchronized and combined. In the arrangement seen in this figure, the wave fronts reach the leftmost antenna first and the rightmost antenna last. To compensate for this delay, the signal from the leftmost antenna is intentionally delayed by the exact same amount before being combined with the signal from the rightmost antenna. Figure from Heald, McKean, and Pizzo (2018)

~ 80 MHz and employ non moving antennas. LOFAR is a versatile radio telescope. Its high sensitivity allows it to detect very faint radio signals from distant objects. This makes it ideal for studying the early universe and other cosmological phenomena. LOFAR's has the ability to produce high-resolution images of the sky, which is important for studying individual sources of radio emission, such as galaxies and active galactic nuclei (AGN). Because no moving parts are employed, targeting and tracking are only limited by the target's visibility.

2.3.1 The Low-Band Antennas

The LBAs are designed to operate from 10 MHz up to 90 MHz, but in practice because of low frequency ionospheric effects their band pass is restricted between 30-80 MHz. The LBA elements are inverted-V dipoles, sensitive to two orthogonal linear polarizations. Each polarization is detected using two copper wires that are connected to a low-noise amplifier. Each polarization has its own output hence two coaxial cables per LBA are connected to the receiver unit in the electronics cabinet which is placed at the station.

2.3.2 The High-Band Antennas

The LOFAR HBA operate in the 110-250 MHz range. In contrast to the LBAs that are simple passive elements a single HBA tile consists of a square, 4x4 element (dual polarized) phased array with built-in amplifiers and an analog beam-former consisting of delay units and summaters.

Like with the LBA, the resulting signals are transferred through coaxial cables to the receiver unit in the electronics cabin.

2.3.3 Electornics Cabinet

The electronics cabinet of a LOFAR station is a critical component of the LOFAR radio telescope system. It contains the electronics that are used to control the antenna and process the radio signals that are collected by the antenna.

The electronics cabinet of a LOFAR station typically includes the following components:

- **Antenna control system:** This system is responsible for controlling the position and orientation of the virtual beam of the antenna, and for performing any necessary adjustments to ensure that the beam is "pointed" correctly at the target in the sky.
- **Radio frequency (RF) front-end:** This component amplifies and filters the radio signals that are collected by the antenna before they are digitized and sent to the central processing system.
- **Analog-to-digital converters (ADCs):** These converters are responsible for converting the analog radio signals into digital signals that can be processed and analyzed by the central processing system.
- **Timing and synchronization system:** This system is responsible for synchronizing the signals from the different antenna stations to ensure that they can be combined to form a high-resolution image of the sky.
- **Power supply:** The electronics cabinet must be equipped with a reliable power supply or to perform beamforming.

In addition to these components, the electronics cabinet of a LOFAR station may also include additional hardware, such as amplifiers, filters, and signal processing electronics, to support the specific scientific goals of the LOFAR telescope.

2.4 Pulsar Observations with LOFAR stations

LOFAR's low frequency coverage makes it particularly valuable for observing pulsars, as the radio signals emitted by them are strongest at low frequencies (Bilous et al., 2016). This allows for the detection of faint or weak pulsars that might not be observable with other instruments.

LOFAR's high time resolution is also important for studying pulsars. Pulsars emit pulses of radio emission at very regular intervals, and the accuracy with which LOFAR can sample the pulse period is critical for accurately determining the properties of the pulsar.

In addition, LOFAR's wide field of view and its ability to observe many pulsars simultaneously is particularly valuable for conducting surveys of the sky to search for new pulsars and to study the properties of large samples of pulsars. This allows for a more complete understanding of the pulsar population and the relationships between different pulsar properties.

Finally, LOFAR's digital signal processing capabilities allow for the formation of high-precision pulse profiles and the detection of weak or faint pulsars. This is particularly valuable for studying pulsars, as it allows for the measurement of subtle changes in the pulse profile over time, which can provide important information about the properties of the pulsar and its environment.

2.5 POLFAR

Three operating single LOFAR stations (PL610, PL611, PL612) are located in Poland. They have been functioning since 2016 and they belong to the POLFAR Consortium, established in 2007, which consists of a number of Polish educational and research institutions (Białkowski et al., 2018; Błaszczewicz et al., 2020). Pulsars have been a target for POLFAR stations ever since they became functional. The POLFAR stations are integrated into the larger LOFAR network, which allows astronomers to use the combined capabilities of the stations to study a wide range of astronomical phenomena, as discussed in Sec. 2.1.

2.6 Observations and data treatment

2.6.1 Observations

From Eq.1.9 it is obvious that scattering is a frequency dependent phenomenon. This relationship justifies the need to conduct observations at multiple than one frequencies and this is possible to do with LOFAR stations. From the same equation it is also obvious that the lower the frequency, the larger the time delay because of scattering. Since LOFAR operates in low frequencies, this is one more reason why it is ideal for pulsar scattering studies. Radio pulsars are known to have an average spectral index with a value equal to either -1.8 ± 0.2 (Maron et al., 2000) or -1.60 ± 0.02 (Jankowski et al., 2018). Most pulsars' flux density spectra show a turnover (which means that they are brightest) at frequencies of ≈ 100 MHz (Maron et al., 2000; Bilous et al., 2016). These three facts, make LOFAR an ideal option for pulsar scattering studies.

For this work, we used the HBA part of the PL611 International LOFAR station, close to Łazy near Krakow, Poland. The pulsar observations are conducted usually once a week, during the station's stand alone mode. The pulsar observation pipeline is similar to the one developed by the German LOFAR Consortium (GLOW)².

The observing frequency range is set between 110–190 MHz and the signal is sampled at 200 Hz and is later fed to the Polyphase Filter Bank where it is split into 512, 195.3125 kHz wide, subbands.

Then the process of beamforming takes place, during which the system is capable of creating up to 488 beamlets (for 8-bit mode), each acting as a single beam for a single subband. Later, the 488 beamlets are usually divided into four groups (lanes) of 122 beamlets each.

²www.mpifr-bonn.mpg.de/en/lofar/glow

2.6.2 Data Treatment

Our observations are centered at of 154.466 MHz with a bandwidth of 71.4844 MHz. We disregard the signal from the edges of the observing radio frequency band, because the instrumental response is very poor in these ranges and the data contain very little valuable information. As a result, we typically use only three of these lanes, which are positioned towards the center of the band.

The data transfer, from the station to the Computer Center of the University of Zielona Góra, of these three lanes is facilitated by the PIONIER company, using a dedicated 10Gb/s connection. The LuMP³ software package is used to save the beamformed data in 2 computers with two Xeon3, 10-core processors and 128 GB of RAM each.

Once the data is stored, DSPSR software package (van Straten and Bailes, 2011) is used to dedisperse the signal and folds it per epoch. The dedispersion is performed coherently, by applying Fourier transformation to the recorded voltages and shifting the Fourier series coefficients until all the signals of each subband are aligned together. Incoherent dedispersion works by applying an appropriate time delay for each frequency channel, so that all the signals are aligned. Coherent dedispersion is a much more computationally demanding method, compared to the incoherent method, but it provides superior results. Its biggest advantage is the fact that it can completely compensate for dispersion inside each frequency channel, which coherent dedispersion cannot do.

The data is stored as a time series of full Stokes parameters, sampled at 1024 bins per pulsar period, and separately stored in each of the 366 spectral channels. This storage method facilitates straightforward pulse folding to obtain the average profile and also enables bandwidth, or partial bandwidth, integration of the signal. The end-result files of each observation, are **PSRCHIVE** (Hotan, van Straten, and Manchester, 2004) compatible files.

Radio frequency interference (RFI) cleaning is performed with the **PAZ** subroutine of the **PSRCHIVE** package. Initial Radio frequency Interference processing is performed automatically, using moving median filters and spike detection algorithms⁴. The final cleaning is performed manually using an interactive PAZ overlay.

As a result we obtained a *cleaned* output file that contains the measured voltage time series in each of 366 spectral channels (122 per lane), with a sampling rate of 1024 phase bins per pulse period. This gives us a lot of flexibility when handling the cleaned data, because we can integrate the signal in any desired frequency range between 100-200 MHz.

For this work, we used the The PL611 station, located close to Krakow, Poland. This station consists of 96 Low Band Antennae (LBA) and 48 (instead of the typical number of 96) High Band Antenna tiles.

Most of our pulsar scattering observations we have been using the HBA part of PL611. 17 pulsars are being monitored with LBA. 15 pulsars are being monitored with HBA and LBA. Each pulsar is observed for at least 3 hours every month.

³LuMP stands for LOFAR und MPiFR Pulsare is a software package developed for observing pulsars with the german LOFAR stations

⁴<https://psrchive.sourceforge.net/manuals/paz/>

2.6.3 Splitting of the signal into narrower sub-bands

The next part of our data treatment is targeted to the data preparation required for studying pulsar scattering. Since scattering is a frequency dependent phenomenon we need to split the HBA 72 MHz (in 366 spectral channels) bandwidth signal into several sub-bands. Determining the number of sub-bands (N_{bands}) is a compromise between having a large enough number of separate scattering estimates (each one for a different frequency) and a high enough signal-to-noise ratio, within each subband, in order to study the shapes of the profiles' components.

We used the code provided by Geyer and Karastergiou (2016)⁵ in order to simulate single component isotropically scattered profiles, with various Signal to Noise (SNR) ratios. In the original code, the SNR remains constant for each sub-band, but we introduced modifications so that the SNR of each sub-band drops as a function of their number ($SNR \propto \sqrt{N_{bands}}$). Then, we fitted the simulated profiles in order to obtain a τ estimation for each sub-band and then we compared the expected (simulated) values with the fitted ones in order to examine how the combination of N_{bands} and SNR affect the estimated α and τ values. This procedure was repeated many times with various combinations of SNR and N_{bands} .

In Fig. 2.3 we see examples of simulated split profiles, fitted to the isotropic thin screen model. This is a procedure which is performed automatically and was created for the needs of our work. The only necessary parameters, predefined by a user, are the number of subbands and the model to be fitted to the data.

We conducted simulations and processed the data using fitting algorithms to verify if the scatter time measurements and the scattering frequency dependence closely matched the simulated values. Our objective was to determine the optimal point, representing the highest number of subbands we could use. This choice would provide the best frequency coverage and most accurate α estimates while minimizing the reduction of signal-to-noise ratio (SNR) in individual subbands. Preserving SNR is crucial for accurate scatter time τ measurements, which, in turn, can impact the accuracy of the estimation of scatter time frequency scaling index α .

Given the SNRs of the data recorded with PL611, we decided to split the signal into 6-8 sub-bands, depending on the SNR.

2.7 Modelling of the pulse profile scattering tails

Now that our signal is split into a few subbands, our next goal is to estimate the amount of the scattering present in the data which is usually related to the broadening of the pulse profile.

As the scattering theory states (See Section 1.5), the profile that we obtain is a combination (convolution) of the intrinsic pulse profile with the scattering effect caused both during the propagation of the signal through the interstellar space and the instrumental effect related to the detection of the signal by the telescope (Kuz'min and Izvekova, 1993).

⁵https://github.com/marisageyer/LOFAR_scattering

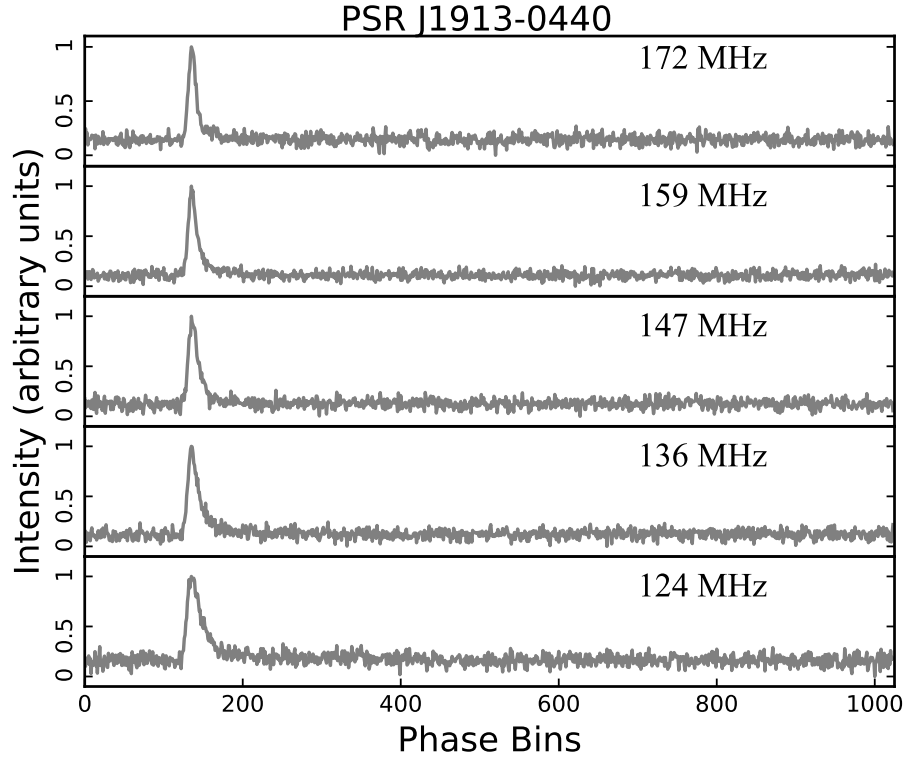


FIGURE 2.3: Signal splitting. An example of splitting a recorded signal into narrower band widths.

2.7.1 Deconvolution method

The observed pulse profile $P(t)$ can be described by Equation 2.3.

$$P(t) = P_i(t) * s(t) * D(t) * I(t) ; \quad (2.3)$$

where $*$ denotes convolution, $P(t)$ is the intrinsic pulse shape, $s(t)$ is the PBF, $D(t)$ the dispersion smear over a spectral channel range (if the data are not coherently dedispersed) and $I(t)$ the instrumental impulse response.

There is a variety of methods and approaches to estimate and model the intrinsic pulse shape. For example, $P(t)$ can be approximated either by using a high radio frequency profile of the observed pulsar in the place of $P_i(t)$ (Löhmer et al., 2001; Löhmer et al., 2004; Krishnakumar et al., 2015; Krishnakumar, Joshi, and Manoharan, 2017; Krishnakumar et al., 2019) or by a Gaussian curve. The idea behind this approach is that a high radio frequency profile is closer to the intrinsic pulsar profile, since it is much less affected by the ISM, compared to lower radio frequency profiles (Eq. 1.9). The χ^2 minimization is used for calculating $P(t)$ (Löhmer et al., 2001). Another way to calculate $P(t)$ is by means of deconvolution, via the CLEAN algorithm (Bhat, Cordes, and Chatterjee, 2003). This approach does not depend on pre-existing information about the pulse shape and can reconstruct intricate features of the pulse shape within time intervals shorter than the PBF's width.

In Demorest (2011) the intrinsic pulsar profile and PBF are constructed by means of cyclic

spectroscopy, applied to the observed profiles. In a cyclic spectroscopy analysis, the radio emission from the pulsar is observed at several different radio frequencies, and the data is used to create a "spectrogram," which shows the variation of the emission intensity as a function of time and frequency. Spectral estimation techniques typically rely on the assumption that the input signal is stationary, or nearly so, over the time frame used to calculate the spectrum. However, the pulsar radio signal is better characterized as cyclostationary, meaning it is a non-stationary random signal with statistical properties that change in a periodic manner over time (Rickett, 1975)

2.7.2 Forward Fitting (train + DC method)

The above methods try to decompose a scattered profile into two components. The first component corresponds to the profile of the emitted radiation as it leaves the pulsar's magnetosphere and in this method is approximated by a Gaussian curve (as mentioned in the previous section). The second part is the PBF that describes the ISM⁶. The first component is assumed to be Gaussian, whereas the second shape of the PBF is calculated by deconvolution.

In Geyer and Karastergiou (2016) and Oswald et al. (2021) the opposite approach is followed. This approach fits the scattered profile by assuming a Gaussian intrinsic pulse shape (or a sum of Gaussians in the case of multiple component profiles) and a baseline that is close to zero. The five parameters that are being fit are the amplitude, width, and center of the intrinsic pulse, as well as the scattering time caused by the ISM and a baseline fluctuation (of similar magnitude to the noise in the data) that deviates from zero. This method accounts for the strong scattering case where the pulses are so much scattered that part of the previous pulse of the sequence (i.e. its tail) is interfering with the next one. This results in a rise of the flux baseline level and as already mentioned, it is a parameter of the model.

2.7.3 Our approach

As already mentioned in Sec. 2.6 low radio frequency observations are ideal for pulsar scattering studies, because the steep spectral index in flux-density spectra (Maron et al., 2000; Bilous et al., 2020), makes pulsars brighter towards lower frequencies. Also, pulsar scattering is more evident in lower radio frequency observations, because of the relation shown in Eq. 1.9.

Thanks to the latest technological advances, low frequency radio telescopes such as LOFAR, the Murchison Widefield Array (MWA⁷) and the Long Wavelength Array (LWA⁸) came into existence, in the last decade. As a result, now it is possible to conduct good quality observations, in the low end of the radio spectrum.

We modelled pulsar scattering, following the methods described in Geyer and Karastergiou (2016) and Oswald et al. (2021). Each profile component is described by a Gaussian curve and each scattering tail by an exponentially decaying function. The form of this function depends on the model used to describe the ISM. The convolution of these two curves is used in order to model the observed profiles.

⁶the effects mentioned in Eq. 2.3 are usually negligibly small when using modern telescopes and receivers

⁷<https://www.mwatelescope.org>

⁸<https://lwa.unm.edu>

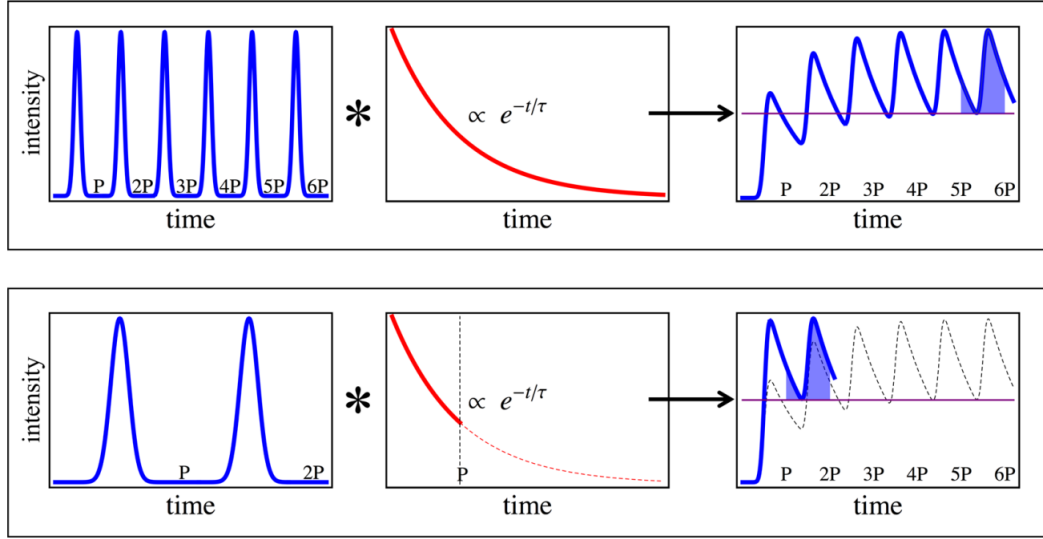


FIGURE 2.4: Forward fitting. The fitting method described in Geyer and Karastergiou (2016). A sequence of Gaussian pulses is convolved with a broadening function (PBF) in order to create a series of broadened pulse shapes. The last scattered profile (shaded) is used as a template for the fitting process. This method is known as **train+DC**. This figure shows two variations, where a long (upper row) and short (lower row) sequence of pulses are used. Image from Geyer and Karastergiou (2016)

We used the *broadfunc1D* (Eq. 2.4) and *broadfunc* (Eq. 2.5) temporal broadening functions (Geyer and Karastergiou, 2016). These are, respectively, the anisotropic and isotropic version of the thin screen model, respectively. As mentioned in Chap. 1, it has been reported that many pulsars have α values different than the expected theoretical ones. Anisotropic scattering mechanisms can lead to the case where $\alpha < 4.0$ (Stinebring et al., 2001). For this reason, we decided to model the thin screen ISM model with both the isotropic (Eq. 2.4) and anisotropic (Eq. 2.5 versions).

$$g(t) = \frac{1}{\sqrt{t\tau\pi}} \times e^{\frac{-t}{\tau}} \quad (2.4)$$

$$g(t) = \frac{1}{\tau} \times e^{\frac{-t}{\tau}} \quad (2.5)$$

2.7.4 Estimation of time delay and its frequency scaling

After the data is split (Sec. 2.6.3), the pulse profile in each sub-band is fitted according to the approach described in Sec. 2.7.3, with the **least squares** method (according to Levenberg (1944)). As a result of the fit, we obtained a τ value for each sub-band. Then, the power law function in Eq.2.6, given in Lewandowski, Kowalińska, and Kijak (2015) and Lewandowski et al. (2015):

$$\tau = 10^{-a \log(v) + b}, \quad (2.6)$$

where b is a free parameter that estimates the scattering's strength and α is the frequency scaling of the pulsar's scattering.

This model is fitted to the obtained τ values and their corresponding frequencies, in order to study how τ varies with frequency and estimate the observed value of the scattering frequency scaling index, α . Also, we use the results of fitting with Eq. 2.6 in order to interpolate the scattering time delay at 150 MHz. This, allows us to compare the time delay due to scattering, between different pulsars. In order to avoid reporting underestimated values of the error of α (α_{err}), due to overfitting, we used both **least square** (Levenberg, 1944) and Monte Carlo simulations (Goodman and Weare, 2010) methods and kept the largest α_{err} value (Lewandowski et al., 2015). Also, in this part of our analysis, the scatter time at 150 MHz was interpolated from Eq.2.6.

2.8 Trend analysis

The above sections describe the method of the estimation of the scattering parameters for a single observation. In our observing project we were, however, observing a number of pulsars, each of them was observed roughly once every month over a period of 3 to 4 years. This allowed us to gather a sizeable database of the scatter time τ_{150} and the frequency scaling indices estimates. The next natural step in our analysis was the study their variability in time, estimate their statistical significance and search for outliers. By having a model that describes the variability of a particular parameter, it becomes easier to understand its trend, look for any regular patterns, and also interpolate/extrapolate missing values. Another important aspect of trend analysis is the detection of seasonality. Seasonality refers to regular patterns that repeat over a specific time period. In our case data these patterns may be related to the Sun, and its influence on the propagation of the pulsar signal through the Solar system. Such patterns will be most likely related to the angular distance between the pulsar and the Sun for the exact time of observation. In our work, we focus on modelling the variability of α and τ_{150} with Deep Learning techniques (Sec. 2.10.2). In the next section some methods, helpful in these tasks, will be described in the following sections.

2.8.1 Gaussian Processes

Gaussian Processes (GPs) are already being used for studying pulsars. This method can be applied to modeling noise in pulsar timing observations (van Haasteren and Vallisneri, 2014; Ay et al., 2020).

GPs have been found to be very useful in non linear regression problems, especially when there is no prior knowledge about the form of the function that we are trying to fit.

Given the input points (regressors) and their outputs (targets), we need to find a function f that describes the connection between them. GPs are stochastic and don't any require prior knowledge about their form (e.g. linear, power law etc.). The only assumption is that each finite combination of GP function values f_{x_i} follows a multivariate joint Gaussian distribution (Rasmussen and Williams, 2006). A GP is of the form:

$$f \sim N(\mu(X), \Sigma(X)), \quad (2.7)$$

where μ is a mean function and $\Sigma (\Sigma(x_i, x_j) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R})$ a covariance function. The covariance matrix is also referred to, as a *kernel*. In matrix notation Eq. 2.7 is written as:

$$\begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix} \sim N \left(\begin{bmatrix} \mu(x_1) \\ \vdots \\ \mu(x_n) \end{bmatrix}, \begin{bmatrix} \Sigma(x_1, x_1) & \dots & \Sigma(x_1, x_n) \\ \vdots & \ddots & \vdots \\ \Sigma(x_n, x_1) & \dots & \Sigma(x_n, x_n) \end{bmatrix} \right) \quad (2.8)$$

When performing GP regression tasks, it is necessary to decide on the equations of the mean functions and the kernel. Usually, $\mu(x) = 0$ not only because this is a simple form, but also because the data can always be standardized to a zero mean. Additionally, setting $\mu(x)$ equal to zero, helps the fitting algorithm to predict a zero value for the cases where no input data exists.

Deciding on the kernel's form is not a tedious task. Usually in such tasks, a few different forms are used. All kernel matrices, as the one in Eq. 2.7 are positive definite, symmetric and invertible (Passerini, 2013).

A simple kernel is the white noise kernel (Duvenaud, 2014), defined as:

$$\Sigma(x, x) = \sigma^2 I_n, \quad (2.9)$$

where σ^2 is the variance of the noise and I_n the identity matrix.

Another one is the exponentiated quadratic kernel (Duvenaud, 2014), defined as:

$$\Sigma(x_i, x_j) = \sigma^2 e^{-\frac{\|x_i - x_j\|^2}{2l^2}}, \quad (2.10)$$

where σ is the variance amplitude and l the length scale, a parameter that affects the smoothness of the GP functions. Eq. 2.10 is also known as Gaussian RBF kernel.

More kernel functions exist and they can be combined with each other. The choice of the kernel and its parameters is not an easy task and easily leads in a case known as overfitting, where the model cannot separate noise from real data.

Some of the main disadvantages of GPs include:

- **Computational cost:** GPs are computationally expensive, especially when dealing with large datasets. The computational cost of GPs increases exponentially with the number of data points, making them unsuitable for very large datasets.
- **Hyperparameter tuning:** GPs have several hyperparameters that must be carefully tuned in order to obtain good results. The selection of appropriate hyperparameters can be difficult and time-consuming, and can greatly affect the performance of the model.
- **Model selection:** GPs are flexible models that can capture a wide range of underlying relationships between the features and the target. However, this flexibility can also make it difficult to determine the appropriate kernel function to use for a particular dataset.
- **Data requirements:** GPs assume that the data is generated from a Gaussian distribution. This assumption may not hold for all datasets, and if the data is not Gaussian, the results of a GP model may not be reliable.

- Overfitting: GPs have a tendency to overfit the data, especially when the noise in the data is high. This can result in poor generalization performance on new, unseen data.

2.9 Random Forest Regression

The Random Forest method (Breiman, 2001) is a machine learning method that uses a large number of decision trees in order to perform regression analysis. A decision tree is a graph method that predicts any possible outcome, given a specific input (Wu et al., 2008). Because this algorithm uses a number of decision trees (models) instead of just one, it belongs to the family of ensemble (Zhou, 2009) methods. The random forest algorithm gathers the outcomes of each decision tree and then determines the best (with respect to the loss function) outcome. In a random forest, multiple decision trees are trained on the data and their predictions are combined to make a final prediction. In regression tasks, each decision tree in a random forest makes predictions by finding the mean target value for each leaf node of the tree. The final prediction of the random forest is then the average of the predictions made by all the trees in the forest. This combination of individual predictions from multiple trees results in a more robust and stable prediction compared to using a single decision tree. On the subsequent sections we elaborate on the several advantages of employing Random Forests for regression tasks.

2.9.1 Non-linearity

Random forests can capture non-linear relationships in the data, making them well-suited for tasks where the relationship between the features and target is complex. In regression tasks, non-linear relationships between the features and the target can be difficult to model using linear methods, such as linear regression. Random forests, however, can capture non-linear relationships by using decision trees as the base estimators.

Each decision tree in a random forest splits the data into smaller subsets based on the values of the features, and the splits can be made on different features at each node. This allows decision trees to capture complex, non-linear relationships between the features and the target.

In addition, random forests can also handle interactions between features by splitting on combinations of features. This further increases the ability of random forests to capture non-linear relationships in the data.

The final prediction made by a random forest is an average of the predictions made by all the decision trees in the forest. This aggregation of predictions can further smooth out any non-linear relationships in the data, improving the overall performance of the model.

2.9.2 Feature importance

Feature importance can provide valuable information for understanding the relationship between the features and the target, as well as for feature selection and feature engineering. Random forests can provide information about the importance of each feature in the prediction, making them useful for feature selection and feature engineering.

Random forests provide a measure of feature importance through the use of decision trees. The data splitting by the decision trees not only find non-linear relationships (as mentioned in the previous section) but also estimate the feature importance by measuring how much the prediction accuracy improves with each split.

In a random forest, each decision tree in the forest provides a measure of feature importance. The final feature importance of a random forest is then calculated by averaging the feature importance of each tree.

There are several ways to calculate feature importance in random forests, including:

- Mean decrease impurity (MDI): MDI Li et al. (2019) and Scornet (2020) measures the decrease in impurity (e.g., variance for regression tasks) caused by each split in the decision tree. The feature with the largest decrease in impurity is considered the most important.
- Mean decrease accuracy (MDA): MDA (Han, Guo, and Yu, 2016; Altmann et al., 2010) measures the decrease in prediction accuracy caused by permuting the values of a feature. The feature with the largest decrease in accuracy is considered the most important.
- Permutation importance: Permutation importance (Altmann et al., 2010) is similar to MDA, but instead of permuting the values of a single feature, the values of the feature are randomly permuted for all the observations in the data.

2.9.3 Robustness to outliers

Random forests are robust to outliers in the data, as the individual decision trees in the forest can capture different patterns in the data. Outliers in the data are observations that are significantly different from the other data points and can have a major impact on the predictions made by machine learning models. In regression tasks, outliers can cause the model to fit to the noise in the data, leading to poor generalization and incorrect predictions.

Random forests are particularly robust to outliers in the data because each decision tree in a random forest makes its own predictions based on the data and these predictions are combined to form the final prediction. This means that even if one decision tree is affected by outliers in the data, the other decision trees will be affected to a less degree. Also, as the decision trees in a random forest are trained on different subsets of the data, they can capture different patterns and relationships in the data. This diversity of trees can help reduce the impact of outliers on the final prediction. The final prediction of a random forest is an average of the predictions made by all the decision trees in the forest. This aggregation of predictions helps to smooth out the impact of outliers, as their influence is diluted by the other predictions. Random forests can also be used for outlier detection, where the goal is to identify the data points that are significantly different from the others. This can be useful for identifying and removing outliers from the data, improving the performance of the model.

2.10 Neural Networks

In this section, we introduce the use of deep neural networks in studying the trend of the same dataset used in the previous section.

2.10.1 The universal approximation theorem

The universal approximation theorem (UAT) (Cybenko, 1989; Hornik, Stinchcombe, and White, 1989) states that any continuous function $f : [0, 1]^n \rightarrow [0, 1]$ can be approximated, up to any desired level of error, by a neural network that has at least one hidden layer and a finite number of weights. Practically, this means that for any dataset there exists a neural network that can compute, up to an indefinitely good approximation, a function that represents our data, regardless of the input/output dimensions. However, the UAT does not provide any ways of constructing the neural network that best describes our measurements. In the next section, we provide a short introduction to neural networks and then a description of the architecture of the neural network that best approximates our dataset. The aim of this initial study, is to show that only the UAT is a valuable tool for pulsar studies, although it is overlooked in favor of the GPs.

2.10.2 Neural Network Architecture

A neural network (NN) is a method that belongs to the field of artificial intelligence (AI). They are used as a way of mimicking the way a human brain processes any given information (Tsoukalas and Uhrig, 1996). The basic bricks of a NN are its neurons, that are grouped into layers. The first layer, is the one whose neurons accept the input data. The next group of neurons is called hidden layer, because it is placed between the input and output layers. The hidden layer, accepts the output of the input layer, processes this information and sends the new output to the next hidden layer or to the output layer. The output layer, accepts the data produced by the last hidden layer and creates the last output. In our case, we are interested in performing regression analysis. So, the output layer will consist of only one neuron that will output one number. This means that for a given input vector, we will get a single scalar value. If more than one hidden layers are used, then the neural networks is characterized as deep. Neurons are nodes that accept numerical data, perform some simple calculations and pass their results to the next layer, where the same process is being repeated until the last layer.

More specifically, each neuron receives an input and multiplies it with some value (weight). Then, all the weighted inputs are added together and with an additional bias value. This sum, becomes the argument of an activation function. If its value is larger than some threshold, then the data can be passed to each neuron of the next hidden layer (hence its name). Usually, the activation function is used so that it produces values in the range 0–1. Each neuron accepts input data and multiplies it with a weight, adds a bias value and sends this result to each neuron of the next layer.

A popular activation function is the sigmoid (logistic) function):

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad (2.11)$$

The hyperbolic tangent function:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad (2.12)$$

and the Rectified Linear Unit (ReLU):

$$\text{ReLU}(x) = \max(0, x). \quad (2.13)$$

The input data fed into the NN are part of the whole dataset. They are called training data, because the NN "learns" from them. The rest of the data are called test data, because they are used for checking the overall performance of the NN. This process, known as forward propagation, is taking place until the data reach the output layer. Then, the data travel backwards, in order to find the optimal values for the weight and bias values (backward propagation), by minimizing a predefined loss function.

A popular loss function, which we also use for our results, presented in the next sections, is the Mean Squared Error Loss Function (MSELF) defined as:

$$L(y, t) = \frac{1}{m} \sum_{i=1}^m (y_i - t_i)^2. \quad (2.14)$$

A NN's architecture (e.g. number of layers and neurons in each one, the chosen activation function, e.t.c.) is something that cannot be predetermined. Many different combinations, of all these parameters are tried and then the architecture with the best quality metrics is chosen.

2.10.3 Neural Networks Regression

Regression can be performed using a neural network architecture specifically designed for this purpose. The use of neural networks for regression offers several advantages, including:

- **Computational efficiency:** Neural networks are highly parallelizable, meaning that they can make use of multiple processors or GPUs to perform calculations in parallel. This allows for faster training and prediction times compared to traditional regression methods.
- **Good quality fitting results:** Neural networks are flexible and can fit complex, non-linear relationships between inputs and outputs. They can also learn from large amounts of data, which can result in high accuracy predictions.
- **Robustness to noise:** Neural networks can handle noise in the data, which makes them useful for problems where the data is noisy or incomplete.
- **Flexibility:** Neural networks can fit complex, non-linear relationships between inputs and outputs, making them a powerful tool for regression. They can capture non-linear patterns in the data that might not be easily captured by linear regression methods.
- **Large-scale learning:** Neural networks can handle large amounts of data, which can result in improved prediction accuracy. This is particularly useful in applications where the

relationships between inputs and outputs are complex and there is a lot of training data available.

- Large number of parameters and Generalization: Neural networks have a large number of parameters, which allows them to capture complex relationships in the data. This means that neural networks are designed to generalize from the training data. Thus, they are more robust to noise in the data as they can still learn the underlying patterns in the data and even make predictions, despite the presence of noise.
- Regularization: Regularization is a technique used to prevent overfitting in neural networks. It adds a penalty term to the loss function, which encourages the network to have small weights and to simplify the model. This can make the network more robust to noise in the data.
- Dropout: Dropout is a type of regularization technique that randomly drops out neurons from the network during training. This makes the network more insensitive to noise in the data as it reduces the dependence of the network on any single neuron.

Overall, the use of neural networks for regression can provide both fast and accurate results without any prior assumptions, making them a popular choice for many machine learning applications.

2.11 Pulsar sample

For this work, we used the The PL611 station, located close to Kraków, Poland. This station consists of 96 Low Band Antennae (LBA) and 48 High Band Antenna tiles (i.e. the station was build in a core-station configuration with only half of the normal station HBA antennas). The pulsar observations are conducted weekly within the so called stand alone time, which is currently about 30 hours a week. During this time, single station observations can be conducted independently from the LOFAR network station. In our case, this allows for observations of 10 pulsars per week with 3 hour integration times, and a total number of 40 separate observations per month. There are 30 targets observed regularly, for the purposes of our project. They were chosen such that their profiles have maximum three components and DM values below 150 pc/cm^3 .

From these pulsars, 18 have been observed at least 10 times, their observations' average SNR is higher than 10 and their integrated profiles show an obvious scattering tail. Furthermore, from these 18 pulsars we filtered out the ones whose split profiles don't have obvious scattering tails. This selection process left is with 10 pulsars, presented in Table 2.1. The use of a single LOFAR station, allows us to monitor our selected targets regularly over a long period of time. More particularly, hw have been able to observe our target group almost once per month for approximately 5 years in order to study the scattering effects in their integrated pulse profiles. This will allow us to study the variation of scattering parameters over time. Similar projects were until now performed only for selected individual sources, and more often to study their interstellar sctintillation properties and not the variations in their scattering - with one significant

TABLE 2.1: Our final target sample. The first part of the table refers to the pulsars that we study in our scattering analysis. The second part presents the pulsar whose DM variations are analyzed. The Dispersion Measure and period values have been obtained from the ATNF catalogue (Manchester et al., 2005).⁹

	$(pc\ cm^{-3})$	(ms)
J0139+5814	73.81	270
J0358+5413	57.14	160
J0525+1115	79.42	350
J0614+2229	96.91	330
J1543+0929	34.98	750
J1645-0317	35.76	390
J1913-0440	89.39	830
J1917+1353	94.54	190
J2113+4644	141.26	1000
J2139+2242	44.16	1100
J0335+4555	47.15	269
J0406+6138	65.41	595
J0454+5543	14.59	341
J0814+7429	5.75	1292
J1932+1059	3.18	227
J2157+4017	71.12	1525
J2219+4754	43.50	538

exception being the analysis of Long Wavelength Array(LWA) monitoring of scattering for a few pulsars at frequencies bellow 100 MHz (Bansal et al., 2019b).

⁹atnf.csiro.au/research/pulsar/psrcat

3 Scattering Analysis

The following section will present the data we acquired during our 5 year observing program, analysed using methods described in the previous chapter, with the aim to obtain the scattering parameters for individual observations, and also average values for the entire project.

3.1 Scattering Analysis results

The first step of our analysis was concerning the modeling of the scatter-broadened pulse profiles for a group of our pulsar sample (Tab. 2.1) that present an observable level of scattering. Both the isotropic (Eq. 2.4) and anisotropic models (Eq. 2.5) were fitted to the data, using the fitting process described in Sec. 2.7.3, in order to obtain the estimates of the scatter time at 150 MHz and the scattering frequency scaling index for individual observations. In this section, I focus on the average values of these results. In Figures 3.2 and 3.3, I present results of our scattering time estimations, on selected individual observations of 10 pulsars from our dataset. These figures also show the results of the fit for the frequency scaling model for scattering that I applied to calculate the frequency scaling index, denoted as α . Figure 3.2 presents the findings derived from the isotropic scattering model, while Figure 3.3 illustrates the results derived from the anisotropic scattering model. The particular epochs shown in these plots were carefully chosen to be with the individual observation's α close to the average value for a given pulsar. The result of the fitting of the model to the scatter time versus frequency data is the value of the frequency scaling index α . Also we can use the fitted model to calculate the representing "normalized" value of scattering within the observed bandwidth, which is the value obtained from the model at the frequency of 150 MHz (see Eq. 2.6). This frequency was chosen as it is relatively close to the central frequency of our observations (≈ 154 MHz) and was used before by other researchers for LOFAR pulsar data, which allows for an easy comparison with previously published values. Tables with the results of our α and τ_{150} estimations and their corresponding errors for each target and each separate epoch are located in Appendix A. I used these individual τ and α values to calculate their weighted mean values, using the inverse-square of their uncertainties as weights. These results are shown in Tab. 3.1.

Fig. 3.1 shows our the weighted average α values versus the weighted average τ_{150} estimations expressed as a fraction of the period. It is worth noting that, for about half of the pulsars, the isotropic time delay (τ_{iso}), occupies a large percentage of their period. In the case of the isotropic model, we observe that the ratio does not go beyond 0.1, whereas in the anisotropic model, its maximum value is almost doubled. We also notice that, for our sample, there is a linear trend of α increasing as τ_{150}/P increases. Such a trend was also noticed in Geyer et al. (2017a), but it should be noted that the pulsars in our sample shows significantly larger amount

TABLE 3.1: The weighted average values of α and τ_{150} . These values have been calculated for both the isotropic and anisotropic model (Sec. 2.7.2). The Dispersion Measure and period values have been obtained from Manchester et al. (2005).

pulsar	DM ($pc\ cm^{-3}$)	P(ms)	anisotropic		isotropic	
			α	$\tau_{150}(ms)$	α	$\tau_{150}(ms)$
J0139+5814	73.81	270	5.15 ± 0.25	40.28 ± 0.69	3.38 ± 0.20	14.62 ± 0.46
J0358+5413	57.14	160	1.70 ± 0.19	5.34 ± 0.14	1.54 ± 0.15	3.11 ± 0.39
J0525+1115	79.42	350	4.46 ± 0.31	69.29 ± 1.56	2.84 ± 0.28	20.51 ± 0.6
J0614+2229	96.91	330	3.29 ± 0.18	42.25 ± 0.39	2.19 ± 0.17	12.74 ± 0.2
J1543+0929	34.98	750	2.04 ± 0.38	6.39 ± 0.36	2.07 ± 0.38	9.11 ± 0.34
J1645-0317	35.76	390	1.20 ± 0.20	0.72 ± 0.08	1.19 ± 0.11	0.59 ± 0.19
J1913-0440	89.39	830	4.44 ± 0.3	15.97 ± 0.32	3.61 ± 0.23	9.13 ± 0.25
J1917+1353	94.54	190	3.94 ± 0.27	20.73 ± 0.56	3.27 ± 0.38	12.24 ± 0.35
J2113+4644	141.26	1000	4.40 ± 0.26	103.76 ± 0.51	3.59 ± 0.16	51.09 ± 1.20
J2139+2242	44.16	1100	2.74 ± 0.22	33.74 ± 0.45	1.55 ± 0.26	25.74 ± 0.48

of scattering (in terms of τ_{150}/P). Larger τ_{150}/P values indicate a higher possibility that the the scattering tail of the average pulse profile will wrap around the full rotational phase covering a wider range of rotational phases. When examining a particular τ_{150}/P value, a range of α values corresponds to it. This, suggests that there are no discernible systematic biases in the α values produced by the fitting algorithm (Geyer, 2017).

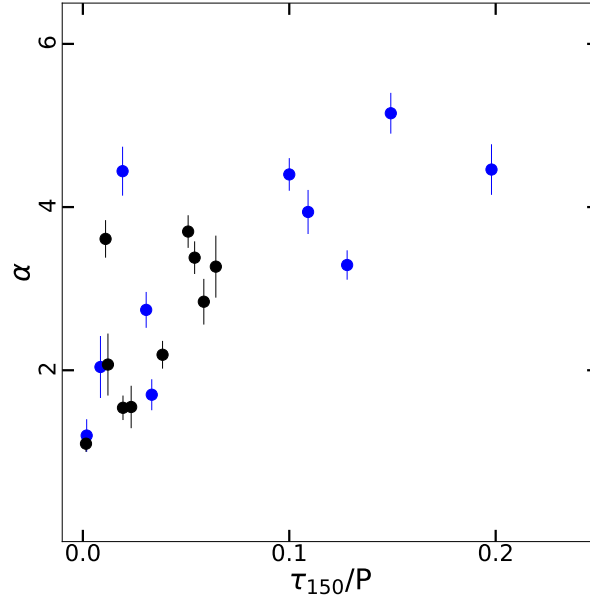


FIGURE 3.1: The mean α values of each pulsar, from both models, plotted versus their corresponding mean τ_{150} values, normalized over period. The isotropic values are plotted in black color and the anisotropic in blue. This plot shows the amount of scattering compared to the period of each pulsar. Also we see that to any range of τ_{150}/P correspond different values of α . This shows that our fitting results are unbiased.

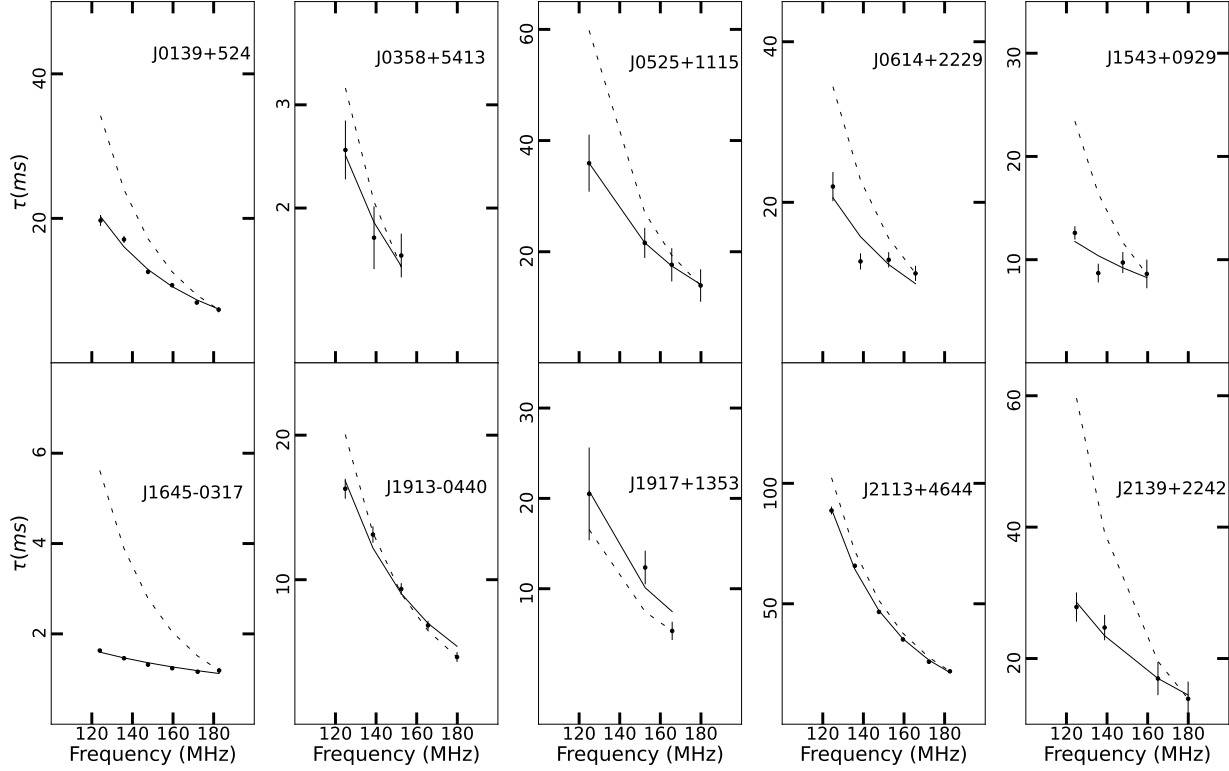


FIGURE 3.2: Isotropic scatter time versus observing frequency. The solid line in each plot represents our power-law fit to our estimations obtained for individual (exemplary) observations epoch for 10 pulsars from our sample. The dashed line shows the slope of the thin-screen Gaussian prediction with $\alpha = 4.0$ with the extra limitation that the model's prediction for the the highest frequency observation matches our estimation. This is done in order to graphically represent the disagreement between the model and our analysis results, especially towards the lower observing frequencies where scattering is more dominant.

Also, in Tab.3.1 we can notice that the α estimations do not necessarily agree to the theoretical values of $4.0 \leq \alpha \leq 4.4$. I have emphasized this disagreement in Figs. 3.2 and 3.3. In these Figs. we see the difference between the theoretical model for $\alpha = 4.0$ and our fitted power-law model. In order to further emphasize the disagreement between these models, during the fitting process of our model we added the extra restriction that both models must have the same value at the highest frequency.

3.2 Possible sources of errors in the τ_{150} estimates

In this section I consider the factors that may affect the reliability of our τ_{150} estimations.

3.2.1 Modelling of Profiles

Gaussian curves are commonly used to model pulsar profiles because they have a well-defined mathematical form that can be easily fit to the observed data. Additionally, Gaussian functions have a bell-shaped curve, which resembles the typical shape of pulsar profiles. This makes Gaussian curves a good initial approximation for the pulsar profile shape. However, it is worth

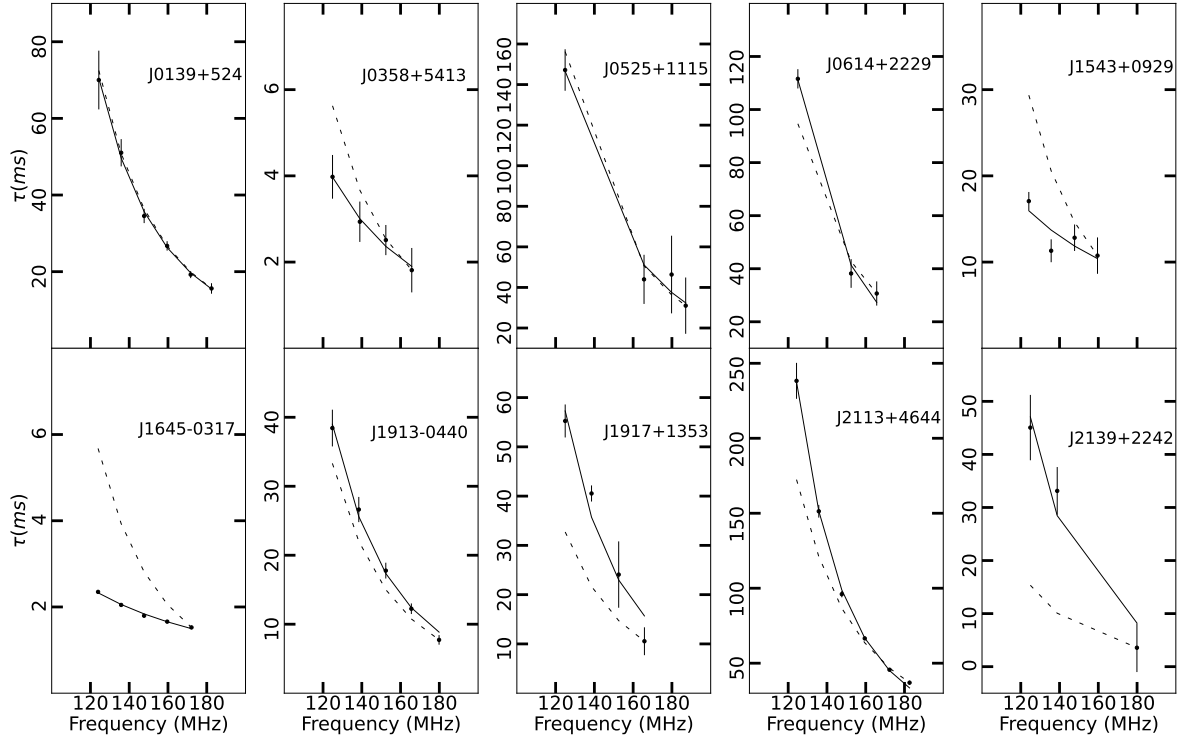


FIGURE 3.3: Anisotropic scatter time versus observing frequency. The solid line in each plot represents our power-law fit to our estimations. The dashed line shows the slope of the thin-screen Gaussian prediction with $\alpha = 4.0$ with the extra limitation that the model's prediction for the highest frequency observation matches our estimation. This is done in order to graphically represent the disagreement between the model and our analysis results, especially towards the lower observing frequencies where scattering is more dominant.

noting that Gaussian curves are not always an accurate representation of the true pulsar profile shape (as discussed, for example, in Skrzypczak et al. (2018)).

In low-frequency observations, pulsar scattering tails are long and Gaussian-approximation of the profile will have little effect on τ_{150} measurements. The points that contribute to the scattering tail shape are already smeared by the scattering effect and are better described by an exponentially decaying function (Sec. 1.5.2).

I did not use high-frequency profiles as templates of the profile components, a method that is sometimes used by other researchers, because they can also be dispersion-broadened, and there was a possibility of intrinsic profile changes with observing frequency, such as profile evolution and radius-to-frequency mapping. Such effects would make any attempts to use unscattered high-frequency profiles as templates for low-frequency scatter-broadening fit impractical, due to the time needed to.

3.2.2 Deviations from ISM models

The values quoted in Table 3.1 show that for the majority of our targets, the anisotropic α values are closer to the theoretically predicted range, than the values of α obtained for the isotropic model. This suggests that the presence of anisotropy appears to be a crucial factor in comprehending or explaining the α and τ values obtained when using the simple thin screen model.

Using an isotropic model for anisotropic scattering can result in significantly underestimated scattering values (Tab. 3.1).

The influence of this phenomenon becomes more pronounced at lower frequencies. As we move to lower frequencies, the impact of the same anisotropy becomes increasingly significant because the scattering angle becomes larger at lower frequencies. This is due to the fact that an anisotropy, usually represented by a truncated screen causes the scattering to become asymmetric. The light-rays of pulsar radiation traveling on one side of the break in the screen will get scattered towards the observer, and form the scattering tail, while the rays on the other side of the break will not encounter any screen at all and will not get scattered. This results in shortening/dampening of the observed scattering tail (with regard to an infinitely wide scattering screen model) because some of those highly scattered rays that were expected to reach the observer will be lost. Consequently, when attempting to simulate clearly anisotropic scattering using the isotropic model, you will tend to underestimate the extent of the relationship's slope.

Assuming a single thin screen model or a thick medium, and using Kolmogorov's distribution of spatial density irregularities with a negligibly small inner scale, the expected variation of the scatter time would follow a simple power-law function with a spectral index of $\alpha = 4.4$ over any range of observing frequencies. However, ISM distribution or larger fluctuation inner scales can result into lower α values of 4.0 or even lower (Cordes and Lazio, 2001). If the actual fluctuation density spectrum differs from Kolmogorov's (Gupta et al., 2000), the α values will lie outside the range 4.0–4.4. This can happen because of the configuration of the interstellar medium (ISM), which includes the potential for scattering on multiple screens (Löhmer et al., 2001; Löhmer et al., 2004). Another potential source of disparity might be the influence of inner scale effects, which can flatten the spectral slope below certain observing frequencies (Rickett et al., 2009).

However, the scenario that the multiple scattering case is responsible for the observed deviation is not very highly unlikely to be true taking into consideration the fact that the deviations are observed for all of the pulsars presented here and most of the objects studied in Geyer et al. (2017b), Bansal et al. (2019b), and Krishnakumar et al. (2019). The likelihood of the radio waves from all these sources experiencing multiple scattering events during their propagation is, essentially, negligible.

This is especially true when one takes into consideration, that the pulsars observed for scattering at LOFAR frequencies (or similar) are relatively close to Earth, which further limits the chance for multiple scatterings. In combination with our results (Tab. 3.1) this shows that anisotropy is the most probable reason for the disagreement between the theoretical expectations and our observations.

3.2.3 Correlation study between α and τ_{150}

I move on searching for any type of correlation between α and τ_{150} in the next steps of our studies. The number of good quality observations of each of our targets is quite small and this does not allow for a concrete correlation analysis between our α and τ_{150} estimations. Even so, in Table 3.2 I present the Pearson correlation factors (r), between α and τ and their corresponding

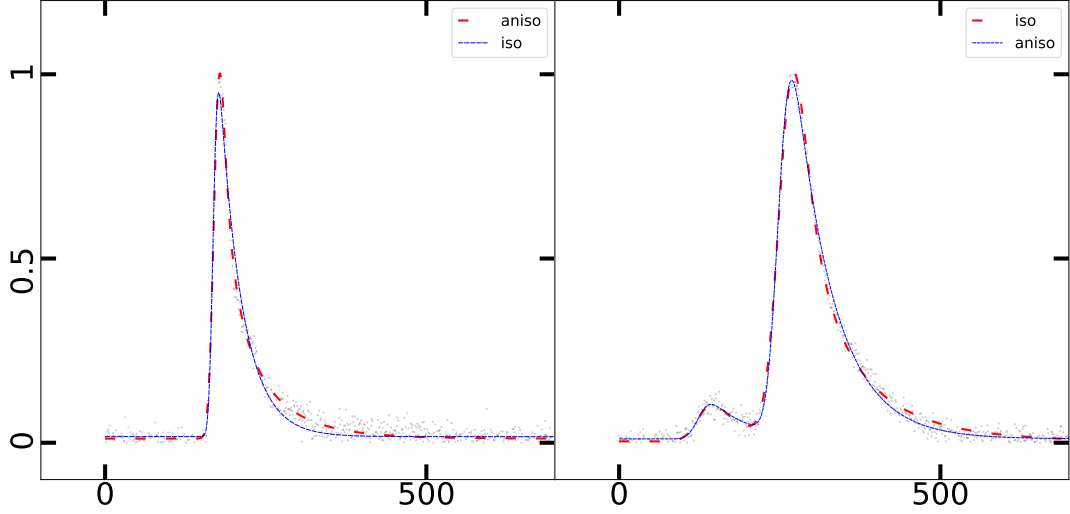


FIGURE 3.4: Left: An example of fitting both the isotropic and anisotropic models, to a single component scattered profile (shown partially). Right: An example of fitting both the isotropic (blue dashed lines) and anisotropic (red dashed lines) models, to a double component scattered profile.

p-values (p) for both scattering models. These values, collectively represent the relationship between α and τ_{150} as they vary over time for all the gathered data points. A weak anti-correlation exists between α and τ_{150} . The only exception is the anisotropic case of PSR J0358+5413. In the cases where there not enough good quality measurements, such us the anisotropic case of PSR J0358+5413, the correlation results are omitted.

TABLE 3.2: Correlation results. The Pearson correlation factor (r) between α and τ and their p-values, for both the isotropic and anisotropic models. The results are only for the pulsars from our sample that show scattering.

pulsar	r^{iso}	p^{iso}	r^{aniso}	p^{aniso}
J0139+5814	-0.22	0.17	0.32	0.07
J0358+5413	-0.49	0.05	0.85	0.03
J0525+1115	-0.54	0.27	-0.30	0.34
J0614+2229	0.11	0.56	-0.11	0.58
J1543+0929	-0.92	2.00e-08	-0.94	2.54e-7
J1645-0317	0.70	0.07	0.65	0.04
J1913-0440	-0.13	0.51	-0.05	0.81
J1917+1353	0.23	0.48	-0.11	0.67
J2113+4644	-0.68	0.02	-0.32	0.31
J2139+2242	-0.28	0.36	-0.72	0.04

Results Summary

Table 3.1 provides the average parameters for our pulsar targets for both scattering models, revealing a noteworthy observation: the frequency scaling index for all of them falls significantly below the range permitted by the simple scattering theory (i.e., $4.0 < \alpha < 4.4$). This discrepancy mirrors previous instances observed in pulsars with higher dispersion measure (DM)

and greater distance, as documented in studies such as Lewandowski, Kowalińska, and Kijak 2015 and Krishnakumar, Joshi, and Manoharan 2017, as well as in lower frequency observations (Geyer et al., 2017b; Bansal et al., 2019b; Krishnakumar et al., 2019).

In a prior work by Lewandowski et al. (2013), the authors explored various potential explanations for this deviation. These included factors such as non-Kolmogorov characteristics of interstellar medium (ISM) plasma turbulence, anisotropy in scattering, and the possibility of multiple scattering events along the line-of-sight. Given the results presented here, the likelihood of multiple scattering screens situated along the line of sight appears exceedingly improbable. This is true even for more distant objects, and for closer pulsars that exhibit significant scattering only at low observing frequencies within the LOFAR range, the probability is essentially negligible.

Further evidence against the multiple scattering model being responsible for the observed deviation lies in the fact that this deviation is evident in most of the pulsars discussed in this project, as well as in most of the objects examined in Geyer et al. 2017b, Bansal et al. 2019b, and Krishnakumar et al. 2019. The probability of radio waves from all these objects undergoing multiple scattering events along their paths is highly unlikely.

The average anisotropic α values presented in Table 3.1 are closer to the theoretical values of 4.0 to 4.4. Especially for three pulsars, *PSR J0515+1115*, *PSR J1913-0440* and *PSR J2113+4644*, their anisotropic α estimations are extremely close to the value 4.4. *PSR J1917+1353* is more special in the sense that its anisotropic α value is extremely close to 4.0. Also, within error bars its α estimations from both models, agree. This is also true for *PSR J0358+5413*, *PSR J1543+0929*, *PSR J1645-0317*, but as already discussed in this section, what we observe from these pulsars is their RFM instead of scattering. *PSR J0139+5814* is also unique in our target sample, because its mean anisotropic α value is much larger than 4.0 or 4.4. Figure 4.4 shows that there are no any large error estimations that could casue this high value. Also, the same figure shows that there is a wide range of our anisotropic α estimations, for this pulsar, much wider than the isotropic version of the scattering model.

Generally, as in Geyer (2017), we also report that the anisotropic model provides larger, than the isotropic model, α and τ values. For the anisotropic model, the τ values appear greater due to the observed appearance of the anisotropy, which effectively reduces the tail length. Employing an isotropic profile shape for an anisotropically scattered profile naturally leads to a reduced value, resulting in an underestimate of the scattering time. On the other hand, α increases because when erroneously applying an isotropic model to an anisotropically scattered pulsar, this effect and the associated underestimation become more pronounced as you move to lower frequencies, resulting in a considerably shallower τ slope.

4 Analysis of Temporal Variations in Pulsar Scattering

In this chapter, I investigate the variability exhibited by the scattering parameters and dispersion measure (DM) associated with the pulsars observed in the course of our research project. The examination of the evolution of these parameters over the duration of our project, serves as a crucial means by which we can conduct a comprehensive analysis of the inhomogeneities and anisotropy inherent in the Interstellar Medium (ISM). The structural and geometrical properties of the ISM, characterized by intricate complexities, may subtly manifest themselves through fluctuations in the observed scattering properties. These variations are a consequence of a multitude of factors, with the perpetual motion of the observer's line-of-sight being a primary contributing factor. This incessant motion is influenced by several determinants, including the dynamic motion of the Solar System, the pulsar itself, and the presence of possibly dense regions within the ISM responsible for the scattering phenomenon. Thus, observations conducted at distinct time points may be influenced by varying configurations of the line-of-sight concerning a non-uniform, non-homogeneous scattering screen, leading to discernible alterations in the scattering parameters. Additionally, our analysis extends to the critical study of DM variability. While DM quantifies solely the cumulative electron density along the line-of-sight, its variation provides a valuable avenue through which we can gain insights into the intricate structure of the ionized ISM through which the pulsar's signal traverses. The meticulous documentation of DM fluctuations will be presented in a dedicated section later in this chapter, offering a more in-depth understanding of this phenomenon.

4.1 Time variability Analysis

In this section I will focus on study of the variability of the scatter time τ_{150} and the frequency scaling of scattering, for both the isotropic and anisotropic scattering models (Eq. 2.4, 2.5). Figures 4.1, 4.3, 4.2, 4.4 show the variability of α and τ_{150} using both models, over a period of ~ 4 years. The variability of these parameters is obvious as well as the deviation of α from the theoretical values of 4.4 or 4.0. In the next section, we move on with confirming the significance of our estimation's variability. That means we determine whether the series exhibit a pattern or trend that is unlikely to occur by chance.

TABLE 4.1: The table presents each dataset's temporal coverage, indicating the duration of time for which the data was collected, and the quantity of good quality observations used for our analysis.

isotropic				anisotropic		
pulsar	no of observations	MJD range	time span (years)	no of observations	MJD range	time span (years)
J0139+5814	43	58074 - 59830	4.81	34	58407 - 59830	3.90
J0358+5413	16	58505 - 59831	3.63	6	58547 - 59831	3.52
J0525+1115	6	58505 - 59850	3.68	11	58491 - 59660	3.20
J0614+2229	29	58457 - 59779	3.62	27	58305 - 59688	3.79
J1543+0929	16	58596 - 59619	2.80	15	58596 - 59619	2.80
J1645-0317	9	58548 - 59331	2.15	10	58778 - 59331	1.52
J1913-0440	28	58368 - 59773	3.85	28	58368 - 59773	3.85
J1917+1353	12	58491 - 59709	3.34	17	58463 - 59807	3.68
J2113+4644	17	58491 - 59506	2.78	17	58491 - 59506	2.78
J2139+2242	12	58526 - 59640	3.05	8	58463 - 59584	3.07

4.1.1 White noise analysis

The first step of our analysis is to make sure that our α and τ_{150} value estimations, for both models, are not random white noise. The objective of this examination is to ascertain that these estimations are not manifestations of random white noise, which has no statistical significance. In the following sections, we present a few different white noise statistical tests.

Autocorrelation

Autocorrelation is a statistical technique used to analyze the relationship between a variable and its lagged values. In the context of checking for white noise, autocorrelation is used to assess the presence of any systematic patterns or dependencies in the data. The autocorrelation function (ACF) of a signal is a metric that quantifies the level of similarity or coherence between the signal and its time-delayed version.

$$R(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)x^*(\tau - t)dt, \quad (4.1)$$

where $R(\tau)$ is the real part of the ACF, τ is the delay parameter. For random data, autocorrelations should be near zero for all lags. To check for white noise using autocorrelation, we calculated the autocorrelation function of the data. The 95% confidence interval of the ACF is given by:

$$CI(k) = \pm z_{\alpha/2} \frac{\hat{\sigma}}{\sqrt{n}}, \quad (4.2)$$

where $CI(k)$ is the confidence interval at lag k , $z_{\alpha/2}$ is the critical value of the standard normal distribution for a given significance level α , $\hat{\sigma}$ is the estimated standard deviation of the sample mean using Bartlett's formula, and n is the sample size. The standard deviation of the sample mean is estimated using Bartlett's formula, we first calculate the sample autocorrelation function at each lag up to a maximum lag K . We then use Bartlett's formula to estimate the variance of the sample mean (Brockwell and Davis, 1987):

TABLE 4.2: The results of our ACF test. The maximum value of the ACF function normalized to its mean, for $\tau_{150} \max(|ACF|)_{\tau_{150}}$ and $\max(|ACF|)_{\alpha}$, are presented, for both ISM models. The bold values show the cases where the normalized values are at least equal to 1.

PSR	anisotropic		isotropic	
	$\max(ACF)_{\tau_{150}}$	$\max(ACF)_{\alpha}$	$\max(ACF)_{\tau_{150}}$	$\max(ACF)_{\alpha}$
J0139+5814	3.48	2.72	3.63	2.56
J0358+5413	1.09	1.61	0.33	0.46
J0525+1115	0.92	0.71	1	1.13
J0614+2229	2.48	1.38	1.80	2.66
J1543+0929	1.92	1.47	1.47	2.01
J1645-0317	1.64	0.88	0.56	2.03
J1913-0440	2.74	2.05	2.86	2.58
J1917+1353	2.07	0.83	2.47	1.97
J2113+4644	0.97	1.48	2.69	1.62
J2139+2242	1.16	0.87	0.89	0.49

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{k=1}^K w_k S_k, \quad (4.3)$$

where S_k is the sample autocorrelation function at lag k , w_k is a weight assigned to each lag (proportional to the inverse of the lag) and n is the sample size.

Tab. 4.2, shows the maximum absolute values of the ACF function for each pulsar (calculated over a 95% confidence interval) normalized by the mean of the absolute ACF values.

The Runs Test

The Runs test (Wald and Wolfowitz, 1940) is a statistical test used to determine whether a time series is random noise or not. It is a non-parametric test that is used to detect the presence of a trend or a pattern in a time series data.

The Runs test is based on the number of runs in a time series data. A run is defined as a sequence of consecutive observations that are either increasing or decreasing. The Runs test examines whether the observed number of runs is consistent with what would be expected if the data were random.

The test is performed by calculating the total number of runs in the time series data, and comparing this value to a critical value obtained from a Runs distribution table. The critical value is determined by the sample size and the significance level of the test.

$$R = \frac{(N - E(R))}{\sqrt{V(R)}}, \quad (4.4)$$

where R is the Runs test statistic, N is the number of runs in the time series data, $E(R)$ is the expected number of runs under the null hypothesis of randomness and $V(R)$ is the variance of the number of runs under the null hypothesis of randomness.

Under the null hypothesis of randomness, the expected number of runs is given by:

$$E(R) = \frac{(2N_1N_2)}{N} + 1, \quad (4.5)$$

where N_1 is the total number of observations larger than $E(R)$ and N_2 the total number of observations smaller than $E(R)$ and $N = N_1 + N_2$. The variance of the number of runs is given by:

$$V(R) = \frac{(2N_1N_2)(2N_1N_2 - N)}{N^2(N - 1)} \quad (4.6)$$

The Runs test is a one-tailed test. The null hypothesis is that the data is random, and the alternative hypothesis is that the data is not random.

If the calculated Runs test statistic (z) is greater than the critical value ($z_{critical}$), we reject the null hypothesis and conclude that the time series is not random. Conversely, if the calculated Runs test statistic is less than the critical value, we fail to reject the null hypothesis and conclude that the time series is random.

The Runs test was applied to our α and τ_{150} estimations from both the isotropic and anisotropic models. For our datasets, z was always $z < z_{critical} = 1.96$ for a 95% confidence interval.

After exploring the idea of using the Runs test to detect white noise in our data, it turns out it is not an appropriate test for it, because our data is a mixture of true deterministic variations and the white noise from measurement uncertainty.

The Ljung-Box Test

The Ljung-Box test can be defined by the following null and alternative hypotheses:

- Null hypothesis (H_0): The data are independently distributed, which means that any observed correlations in the data are due to random sampling and not from correlations in the underlying population.
- Alternative hypothesis (H_a): The data are not independently distributed, and exhibit serial correlation.

$$Q = n(n+2) \sum_{k=1}^h \frac{\rho_k^2}{n-k}, \quad (4.7)$$

where the sample size is denoted by n , ρ_k represents the sample autocorrelation at lag k , and h indicates the number of lags that are being used for testing.

We performed this test for each α and τ_{150} dataset for different time lags. The maximum number of time lags was set equal to the number of data points. In Tab. 4.3 we present the largest p -value found for each pulsar, after calculating it for all the time lags.

4.2 Variability of the scattering parameters

The ACF test (Sec. 4.1.1) indicates that almost all α and τ_{150} datasets for *PSR J0139+5814*, *PSR J0614+2229*, *J1543+0929* and *PSR J1913+1353* are not white noise. The same is true for

TABLE 4.3: Results of the Ljung-Box test. The largest p -value found for each pulsar, after calculating it for all the time lags. In bold, are the cases that the test can reject as not being white noise, when using the threshold of $p - value = 0.05$. The asterisk (*) denotes values that have been approximated as 0.00 due to their being significantly smaller in magnitude compared to the other values listed in the table.

PSR	anisotropic		isotropic	
	τ_{150} p- value	α p- value	τ_{150} p- value	α p- value
J0139+5814	0.00*	0.02	0.00*	0.00*
J0358+5413	0.53	0.56	0.84	0.73
J0525+1115	0.11	0.33	0.98	0.90
J0614+2229	0.32	0.97	0.96	0.5
J1543+0929	0.09	0.65	0.8	0.72
J1645-0317	0.26	0.92	0.76	0.19
J1913-0440	0.37	0.04	0.07	0.06
J1917+1353	0.04	0.98	0.003	0.14
J2113+4644	0.99	0.9	0.002	0.97
J2139+2242	0.63	0.53	0.47	0.81

the anisotropic τ_{150} estimations of *PSR J1645-0317* and *PSR J2139+2242* and all the anisotropic estimations of *PSR J1917+1353*. *PSR J0525+1115* is not identified as noise, but it is close to the decision limit. The Ljung-Box text indicates that the datasets of τ_{150} and α for *PSR J0139+5814* and the τ_{150} datasets for *PSR J1917+1353* for both scattering models are not random white noise. The same applies for the anisotropic α dataset of *PSR J1913-0440* and the isotropic τ_{150} dataset of *PSR J2113+4644*. As described in the previous section, the Runs test has failed to provide clear results. A quick inspection of Figs. 4.4 and 4.2 show that the isotropic and anisotropic α variability of *PSR J0139+5814*, *PSR J1917+1353* is strong.

The following plots presented depict our isotropic and anisotropic α and τ_{150} plotted against time. In the isotropic case *PSR J0139+5814* shows a noticeable variability for both α and τ_{150} estimations. *PSR J0525+1115*, *PSR J0614+2229*, *PSR J1917+1353* and *PSR J1913-0440* also show a noticeable α variability whereas τ_{150} varies but its values are not much further from the weighted mean value.

PSR J0358+5413, *PSR J1645-0317*, *PSR J2139+2242*, *PSR J2113+4644* do not have many good quality points so no clear variability can be confirmed. *PSR J1543+0929* only shows α variability.

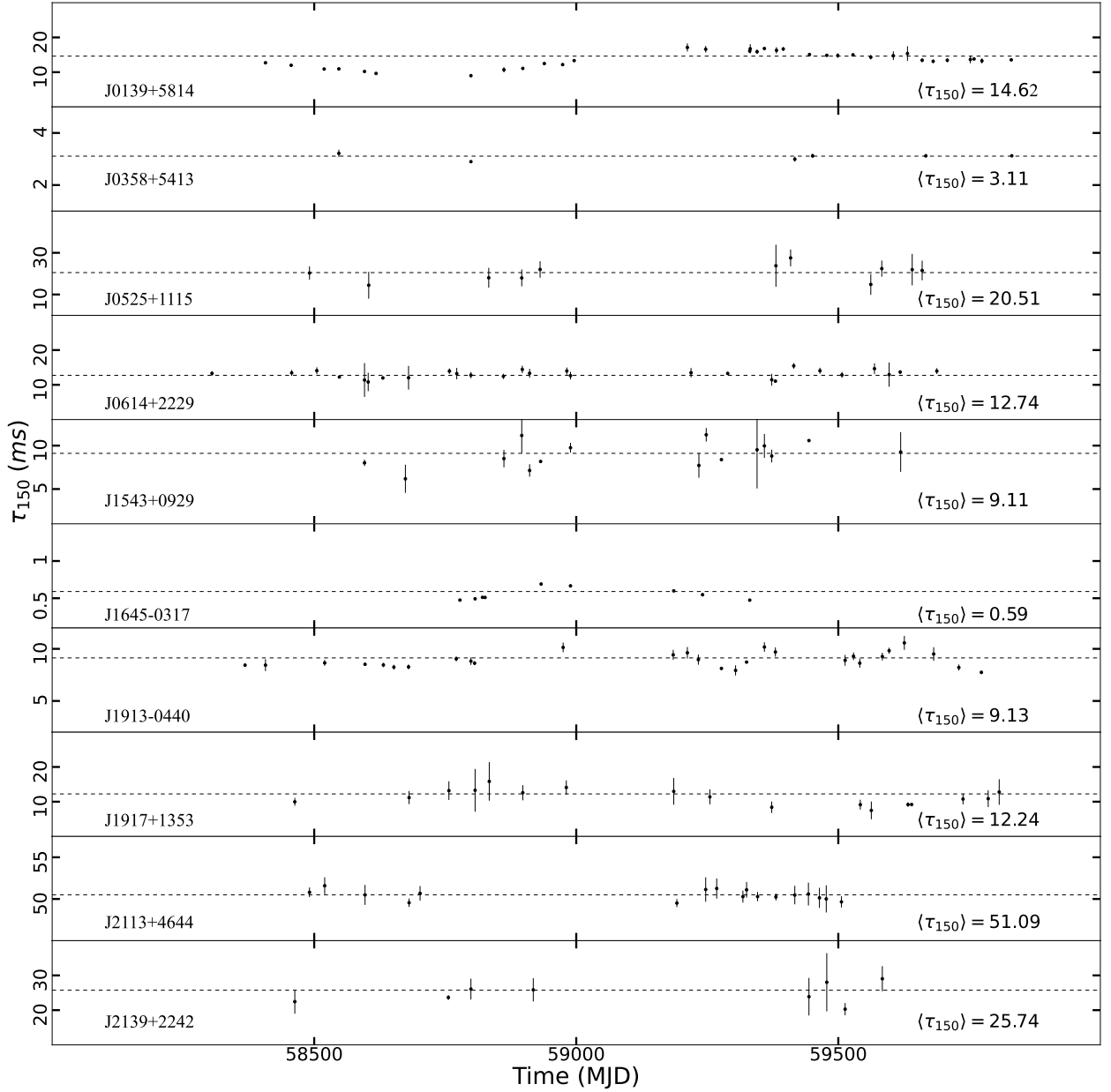


FIGURE 4.1: The isotropic τ_{150} variation. The international LOFAR station PL611 was used for the gathering of these datasets, which span a range of ≈ 4 years.

Our anisotropic estimations show a strong α and τ_{150} , just as in the isotropic case. *PSR J1913-0440* also shows α and τ_{150} variability. No clear τ_{150} variability is seen for *PSR J0358+5413*, *PSR J0614+2229*, *PSR J1543+0929*, *PSR J1645-0317*. However, their α estimations show some degree of variability. *PSR J2113+4644*, although does not have a lot of data points, shows a small degree of τ_{150} variability. It is interesting that our α estimations show more clear variations very close to the theoretical range. It is more difficult to judge if any variability is seen for *PSR J0525+1115*, *PSR J1917+1353* and *PSR J2139+2242*. There are not enough good quality data points and although α shows some variability trend the error bars prohibit us from a final verdict. In general, these plots also show that there is a noticeable discrepancy between the obtained values and the anticipated theoretical values of 4.4 or 4.0.

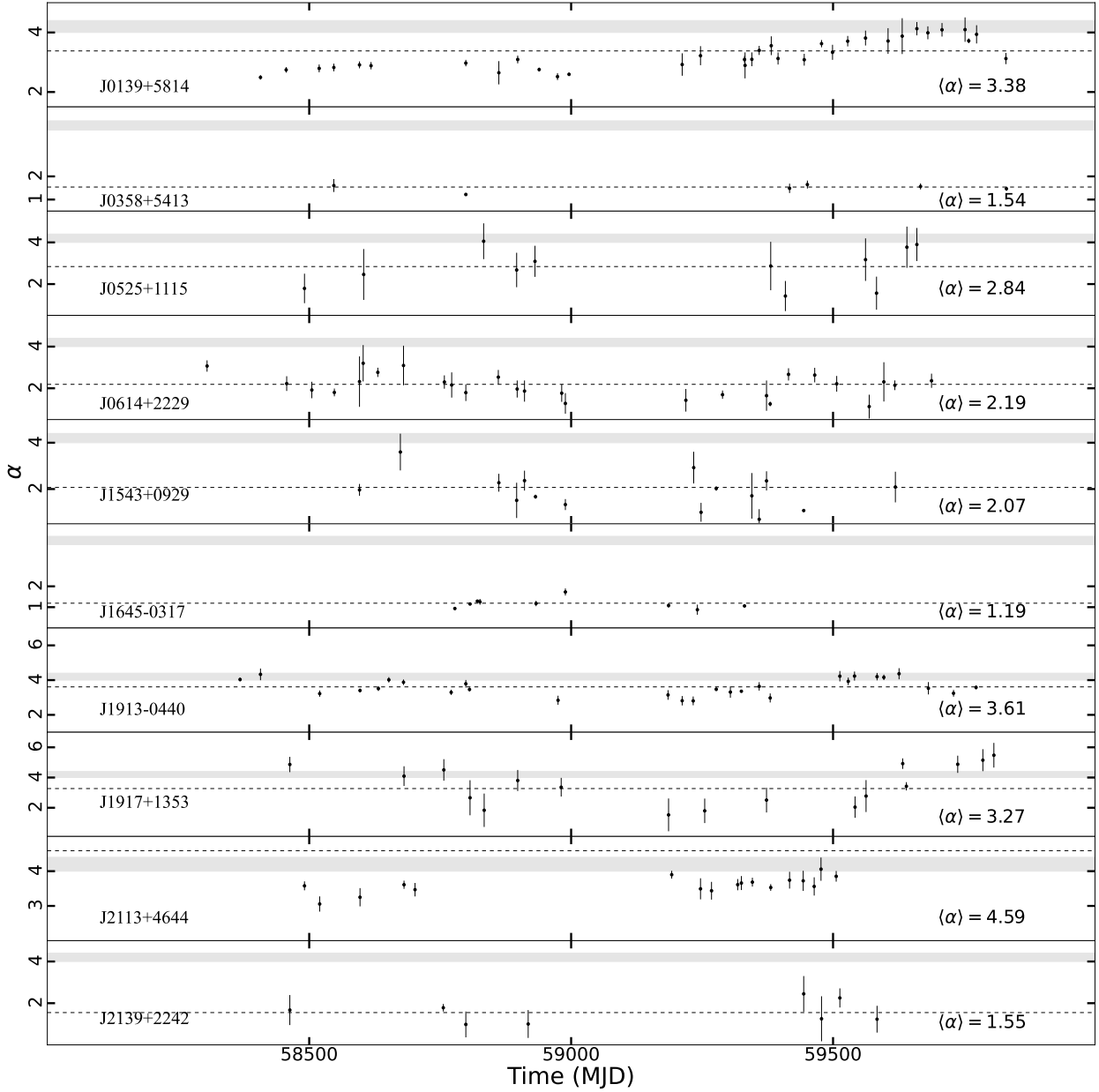


FIGURE 4.2: The isotropic α variation. The international LOFAR station PL611 was used for the gathering of these datasets, which span a range of 4 years. It is immediately noticed that the α values can deviate a lot from the theoretical values of $\alpha = 4.0$ or 4.4

4.3 Modelling of variability with Deep Learning Methods

In this section, we model the variability trends of the isotropic and anisotropic α and τ_{150} variations. The trends are studied by performing non linear regression using the Gaussian Processes (2.8.1), Random Forest (2.9) and Neural Networks (2.10) methods. Besides GP the rest of the methods have not been extensively used in pulsar studies. We show that they can be used to describe the variability of various parameters estimations as well as detect outliers. A more detailed description has been given in Chapter 2. Our main aim is not to capture the inherent

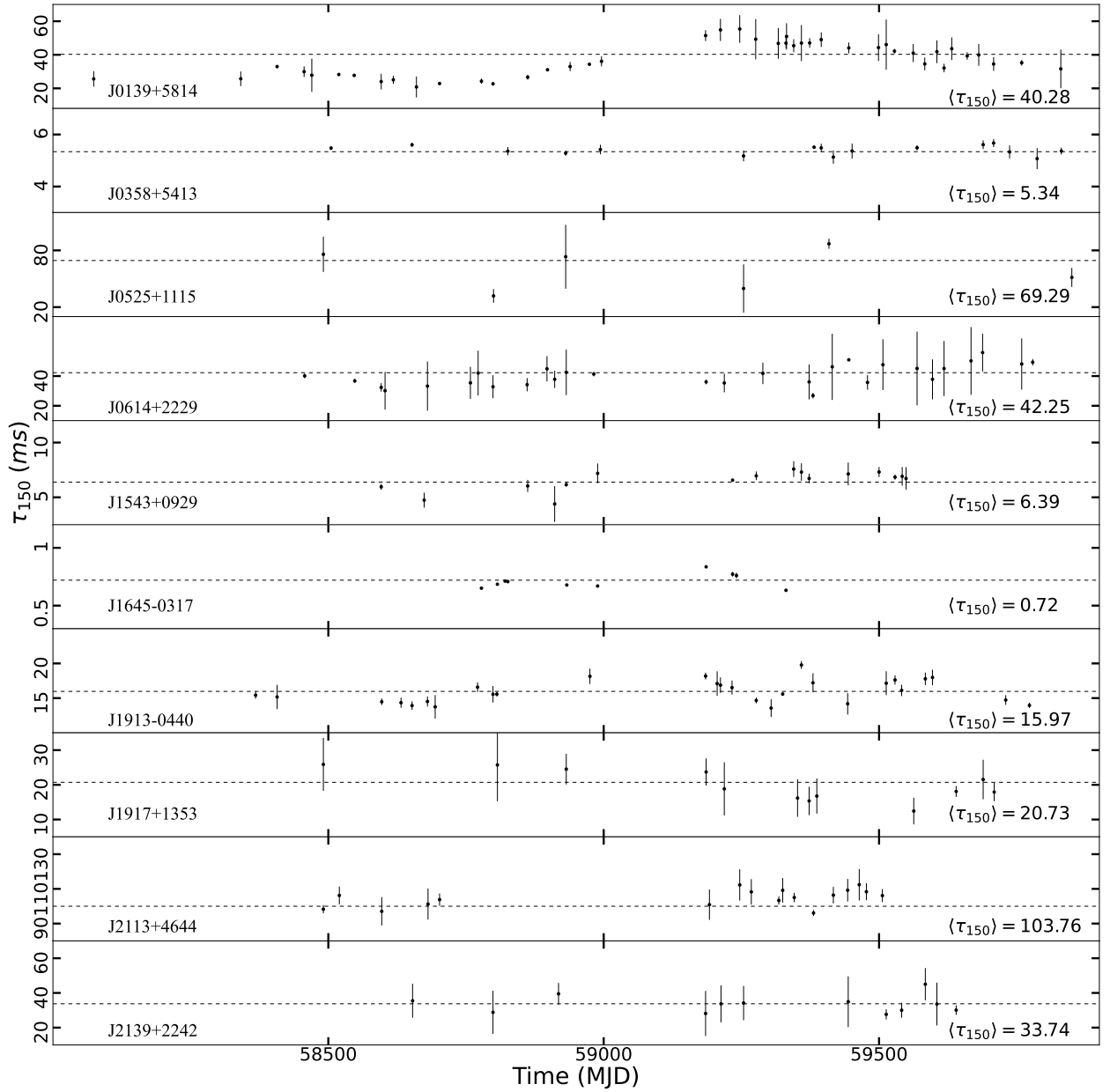


FIGURE 4.3: The anisotropic τ_{150} variation. The international LOFAR station PL611 was used for the gathering of this datasets, which span a range of 4 years.

variability within each dataset. We're more interested in using models, their confidence levels, and the uncertainties they provide in order to figure out if the variation we see is real or just random noise. We want to separate the actual patterns from the random ups and downs in our data. To do that, we compare how much something varies with how sure our model is about that variation. When the variation of a parameter significantly exceeds the uncertainty estimated by the fitted model, it serves as a strong indicator that the observed variability is likely a true, meaningful phenomenon, rather than random noise. This approach allows us to sift through data, identifying and validating patterns that carry genuine information and insight, while disregarding fluctuations that do not contribute meaningfully to our understanding of the underlying processes.

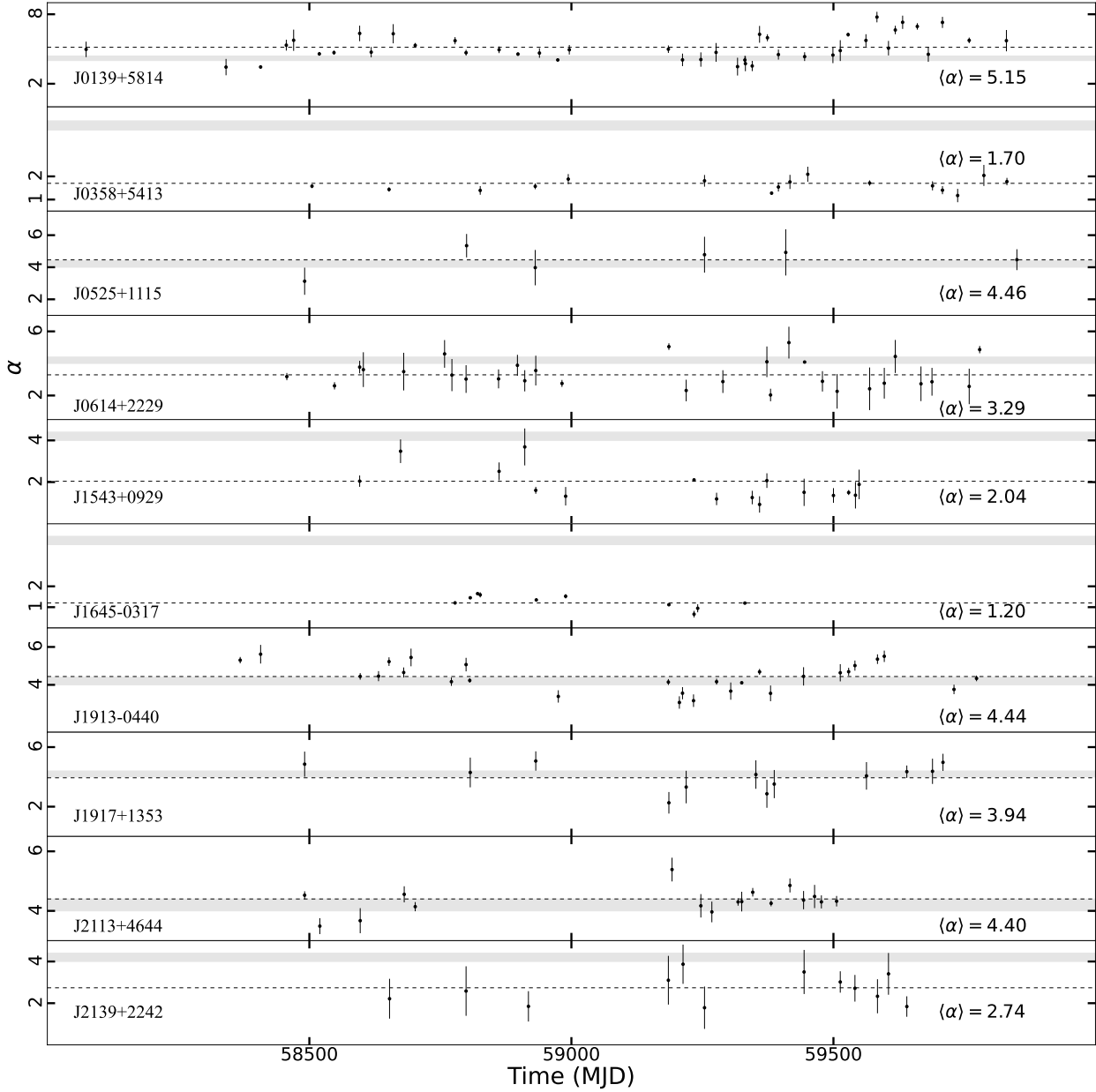


FIGURE 4.4: The anisotropic α variation. The international LOFAR station PL611 was used for the gathering of this datasets, which span a range of 4 years. α values can vary a lot compared to the theoretical values of $\alpha = 4.0$ or 4.4

4.4 Selection of the most suitable regression method

For the presenting simplicity of the methods' comparison, we show the results of fitting the isotropic α estimations of PSR J0139+5814 (Fig. 4.7). For each method, the variance of the predicted values was compared to the variance of our estimations. We introduce this ratio as a criterion about the fitting quality, because it describes the difference between the expected value ranges of the dataset and the model. We are interested in the ability of the model to interpolate, extrapolate and study variation trends and not on its fitting error. That is why we are not using the χ^2 as the model's quality criterion.

In Fig. 4.7 (a) we see an example of a GP (Sec. 2.8.1) regression result. The gray area represents the 95% confidence interval. For most of the part, the gray area is 100% equal to the dataset's variance. This means that the model is trained very well on the given data, but it will fail in interpolating/extrapolating values. The broader gray areas, are the regions where the model is fitting very poorly the dataset. In these regions, the variance of the model's predicted values is 120% of our dataset's variance. GPs are not a good choice for modelling our data because the time series data of α , τ_{150} and DM are not stationary, which means that their mean values are not constant over time. This might be attributed to the existence of trends in the time series (either increasing or decreasing). A more detailed reference to the assumptions required for GPs are described in Sec. 2.8.1.

In Fig.4.5(b) we see an example of regression with the RFR method (Sec. 2.9). In this case, the model's variance is 32% of the dataset's variance. We can argue that this means that the RFR method is overfitting the data. The degree of overfitting is ~ 3 times less compared to the GP regression, but still overfitting is substantial. The data was split randomly into 100 subsets and each one of them was fed to a decision tree. Each decision tree, performed curve fitting using the loss function Eq. 2.14. The final result was a combination of all the decision trees.

In Fig. 4.5(c), we used the NN regression method (Sec. 2.10.2). After 1000 training epochs (one complete forward and backward propagation is one training epoch), the NN converged to an approximation of our dataset. In this case, the model's variance is 65% of the dataset's variance. This is an intermediate value compared to the other two methods and is the one that seems to be the most useful, as it is neither over/under-fitting to a great degree.

As mentioned in the previous section, Sec. 2.10.2, the output layer of a NN consists of one neuron, when dealing with regression tasks. In our case, a second node is introduced in the output layer ¹. This extra node estimates the fitting error as calculated by a modified version of MSELF (Eq.2.14), known as variance attenuation (Valdenegro-Toro and Saromo, 2022). NNs, in theory, need a large amount of data, in order to "learn" from them and provide a function that fits well the input(training) data. In our example case, the dataset consists of only 35 measurements. Despite this small population of data points, we show that we can achieve a reasonable fit. Our NN consists of the input layer, three hidden layers (with 32 neurons each) and an output layer with two neurons. The architecture of our NN is represented in Fig. 4.6 ².

As mentioned in Sec: 2.10.2, the NN regression method does not need any assumptions, about the data or the fitted function. The UAT (Sec. 2.10.1) states that the data can be approximated by a function. Although deciding on the values of the model variance to the data variation, ratio is subjective, we argue that regression with neural networks has provided better results, since it is avoiding the extreme cases of over/under-fitting.

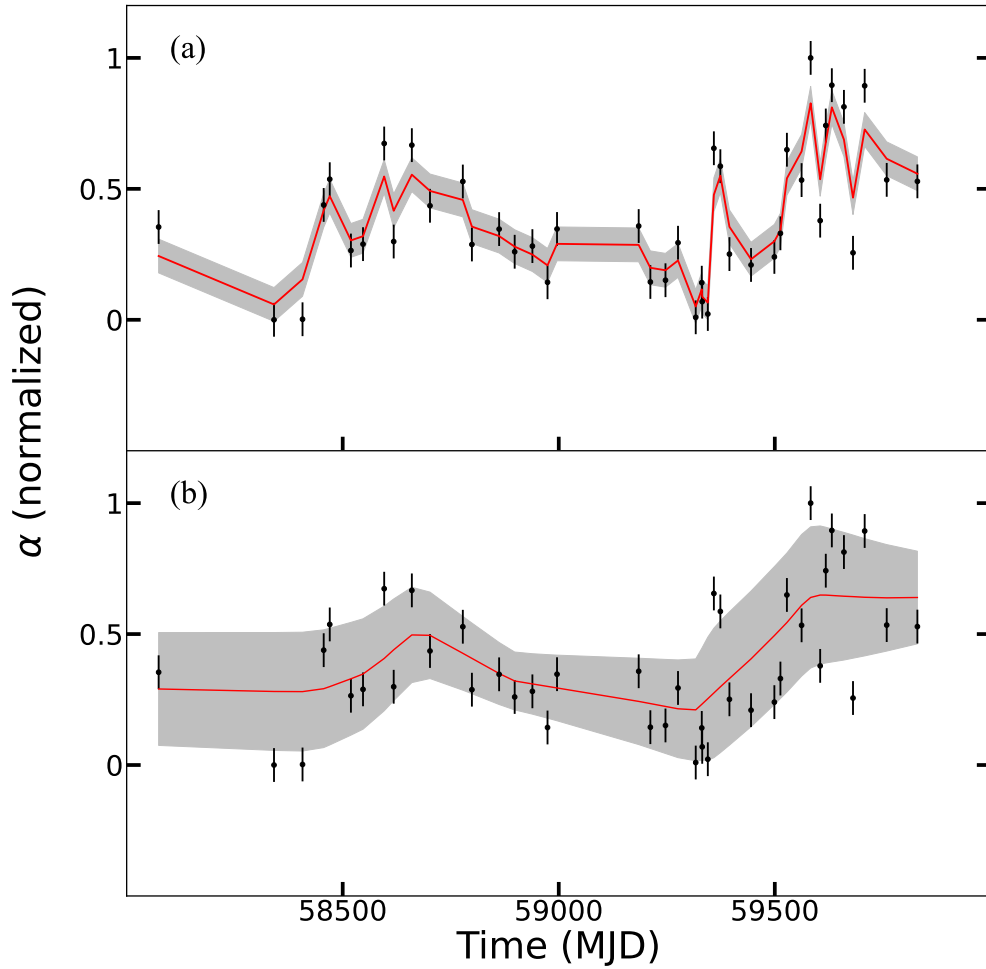


FIGURE 4.5: A comparison of non linear Deep Learning regression techniques. The variation of the isotropic estimated α for PSR J0139+5814 is modelled with two different non linear Deep Learning regression techniques mentioned in the text. Subplot (a) corresponds to the NN method (Sec. 2.10.2. Subplot (b) corresponds to the RFR. The gray shaded areas represent the 95% confidence interval. These plots are useful in identifying trends and outliers, which bias the average values of the studied parameters. The α values and their errors have been normalized by their respective maximum values, for more clear illustration purposes.

4.5 Conclusions on using Deep Learning Methods for α and τ_{150} variability modelling

As discussed in the previous section, in our case the NN method is the best choice for describing the variability of our α and τ_{150} data because there was no over/under-fitting.

This method can help us assess whether the observed variability in a dataset is genuine or if it might be due to random chance or measurement error. The grey-colored area in Fig. 4.5 represents its confidence interval that indicates the range within which the true values are likely to fall. To determine if the observed variability is likely to be true or if it's just random noise,

¹<https://github.com/Garve?tab=repositories>

²<https://github.com/RedaOps/ann-visualizer>

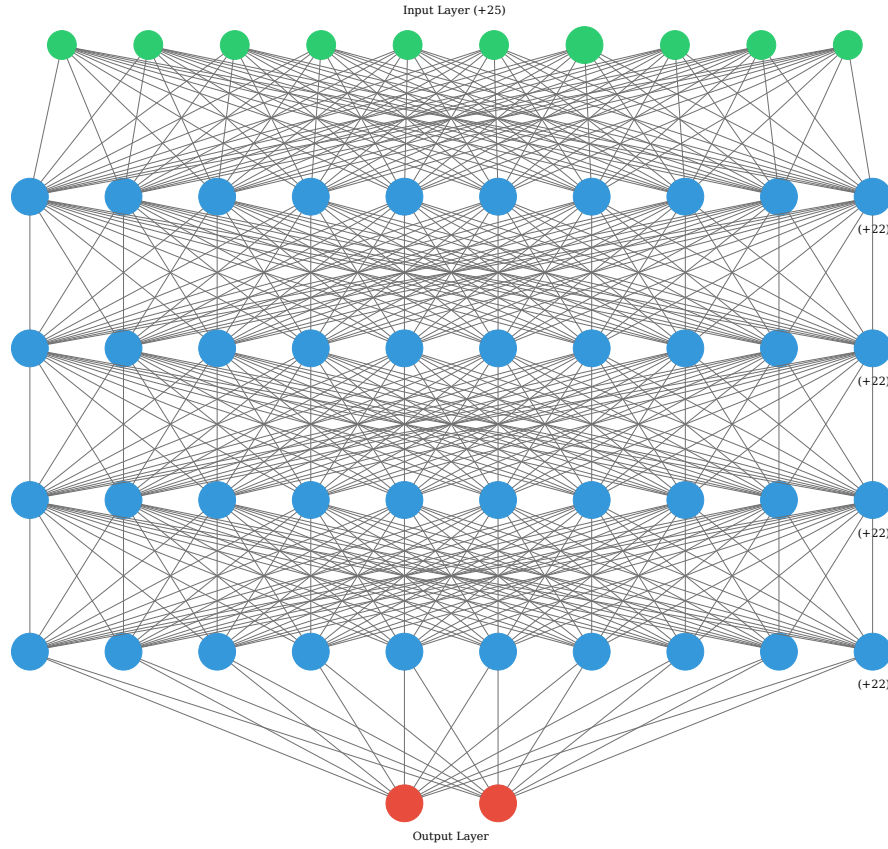


FIGURE 4.6: The architecture of the neural network used for our regression study. The green filled circles represent the input layer neurons, the blue filled circles the hidden layers neurons and the red, the two output neurons. One of them outputs an estimated value and the other one the error of this estimation.

we can compare the width or range of the grey-colored model confidence area to the observed variability. If the grey-colored area is significantly narrower than the observed variability, it suggests that the model is making precise predictions, and the observed variations are well within the expected range. This implies that the variability is consistent with the model's predictions, and there's a higher likelihood that it's real and not due to random fluctuations or measurement error. Conversely, if the grey-colored area is much wider than the observed variability, it indicates that the model is not making precise predictions, and the observed variations are larger than what the model expected. This might suggest that the variations are beyond what the model can explain, which could be an indication that they are not real and might be the result of random noise or other unaccounted-for factors.

In the case of the isotropic τ_{150} variability, *PSR J0139+5814*, *PSR J0614+2229*, *PSR J1543+0929*, *PSR J1913-0440*, *PSR J1917+1353* are modelled very well, by the NN method. Due to a low number of good quality data points, the method compensates this problem by overfitting the datasets for *PSR J1645-0317* and *PSR J2113+4644*. For *PSR J0358+5413*, *PSR J0525+1115*, and *PSR J2139+2242*, initially underfitting and later overfitting is performed.

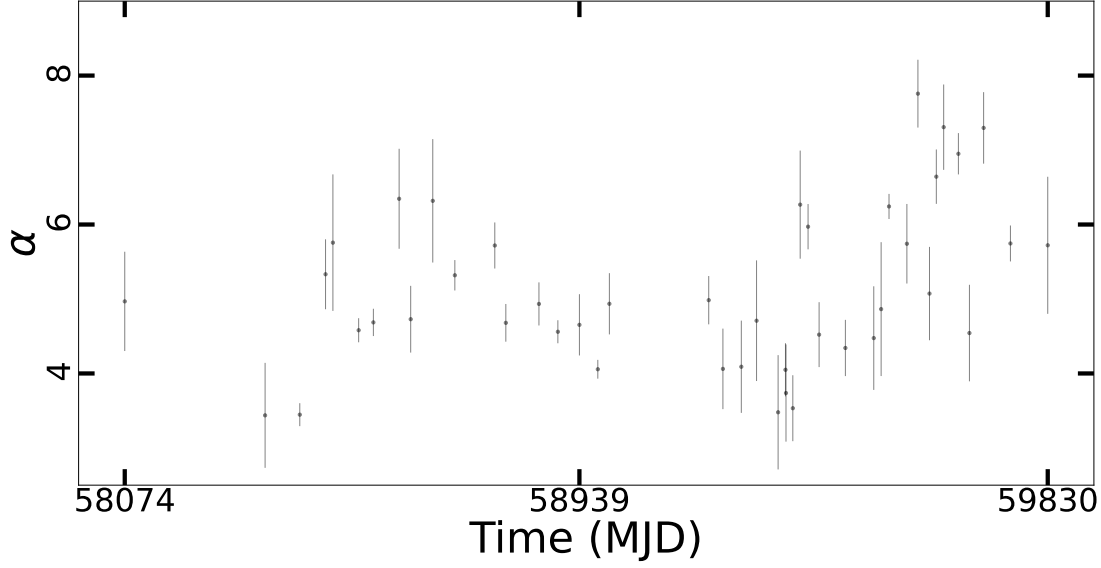


FIGURE 4.7: The dataset of isotropic α estimations for *PSR J0139+5814*, used as a representative example for the comparison of the regression methods mentioned in the text.

In the case of the anisotropic τ_{150} variability, the NN methods performs well for *PSR J0139+5814*, *PSR J0614+2229*, *PSR J1543+0929*, *PSR J1913+1353*, *PSR J1917+1353* and *PSR J2113+4644*. Overfitting is seen for *PSR J0525+1115*. Judging the fitting quality for *PSR J0358+5413*, *PSR J2139+2242* is a bit more difficult, because of the few data points, but we could argue that the fit is reasonably good.

The NN method models well the isotropic α variability of *PSR J0139+5814*, *PSR J0614+2229*, *PSR J1913+1353* and *PSR J2113+4644*. *PSR J1543+0929*, *PSR J1645+0317* and *PSR J2139+2242* are modelled reasonably well. However, we notice overfitting for the cases of *PSR J0358+5413*, *PSR J0525+1115* and *PSR J1917+1353*.

The variability of the anisotropic α estimations is modelled well for the cases of *PSR J0139+5814*, *PSR J0525+1115*, *PSR J0614+2229*, *PSR J1543+0929* and *PSR J1913-0440*. Overfitting is seen for *PSR J2113+4644* and a combination of underfitting for the earlier and overfitting for the later points, for *PSR J0358+5413*, *PSR J1645-0317*, *PSR J1917+1353* and *PSR J2139+2242*.

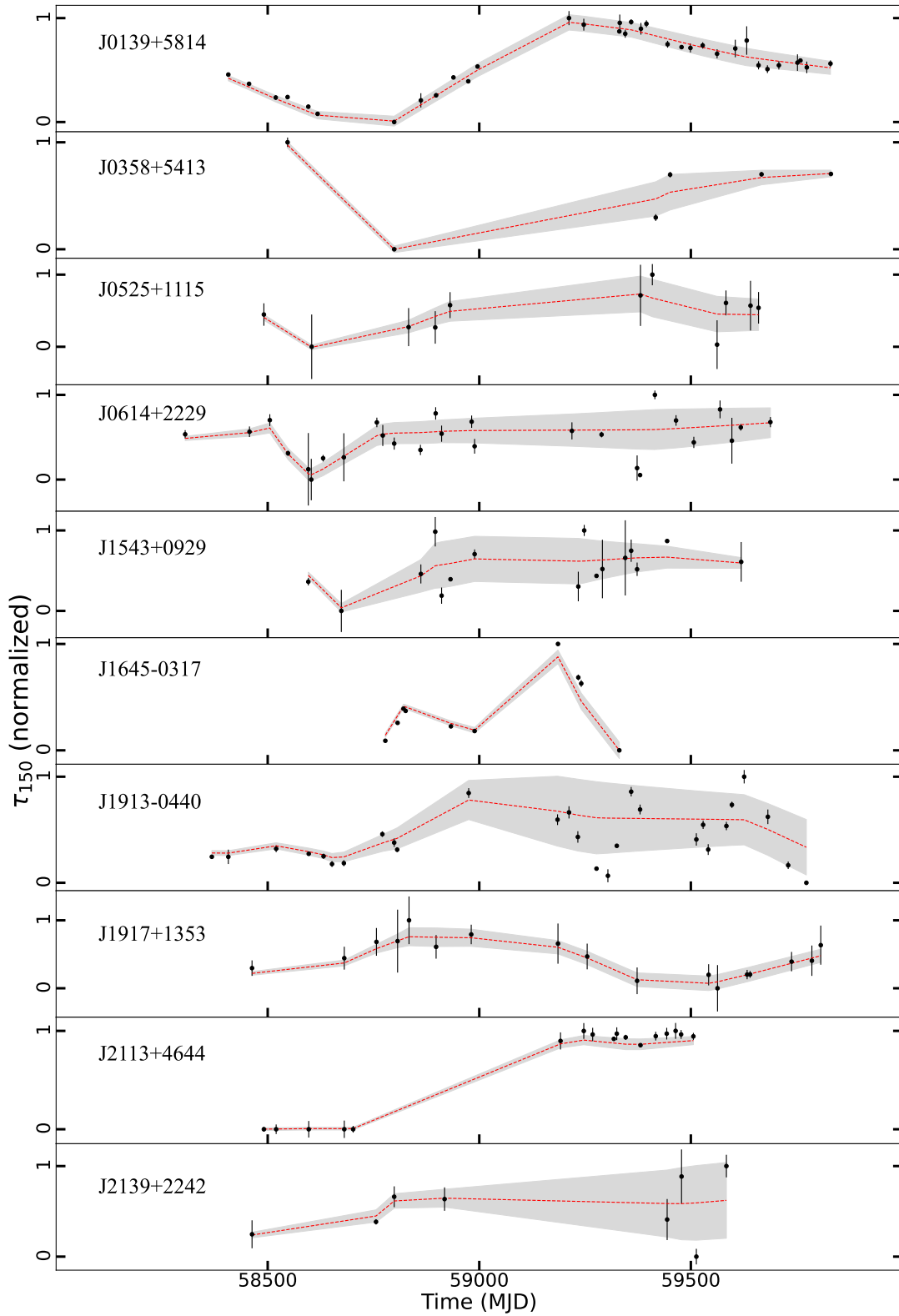


FIGURE 4.8: Time variability of τ_{150} obtained from the isotropic model, analysed using the NN methods. The grey area represents the confidence interval of the loss function used for fitting (See. Sec. 2.10.2 and 4.3). The τ_{150} values and their errors have been normalized by their respective maximum values, for more clear illustration purposes. A significant variability is visible in pulsars PSR J0139+5814, PSR J0614+2229 and PSR J1913-0440.

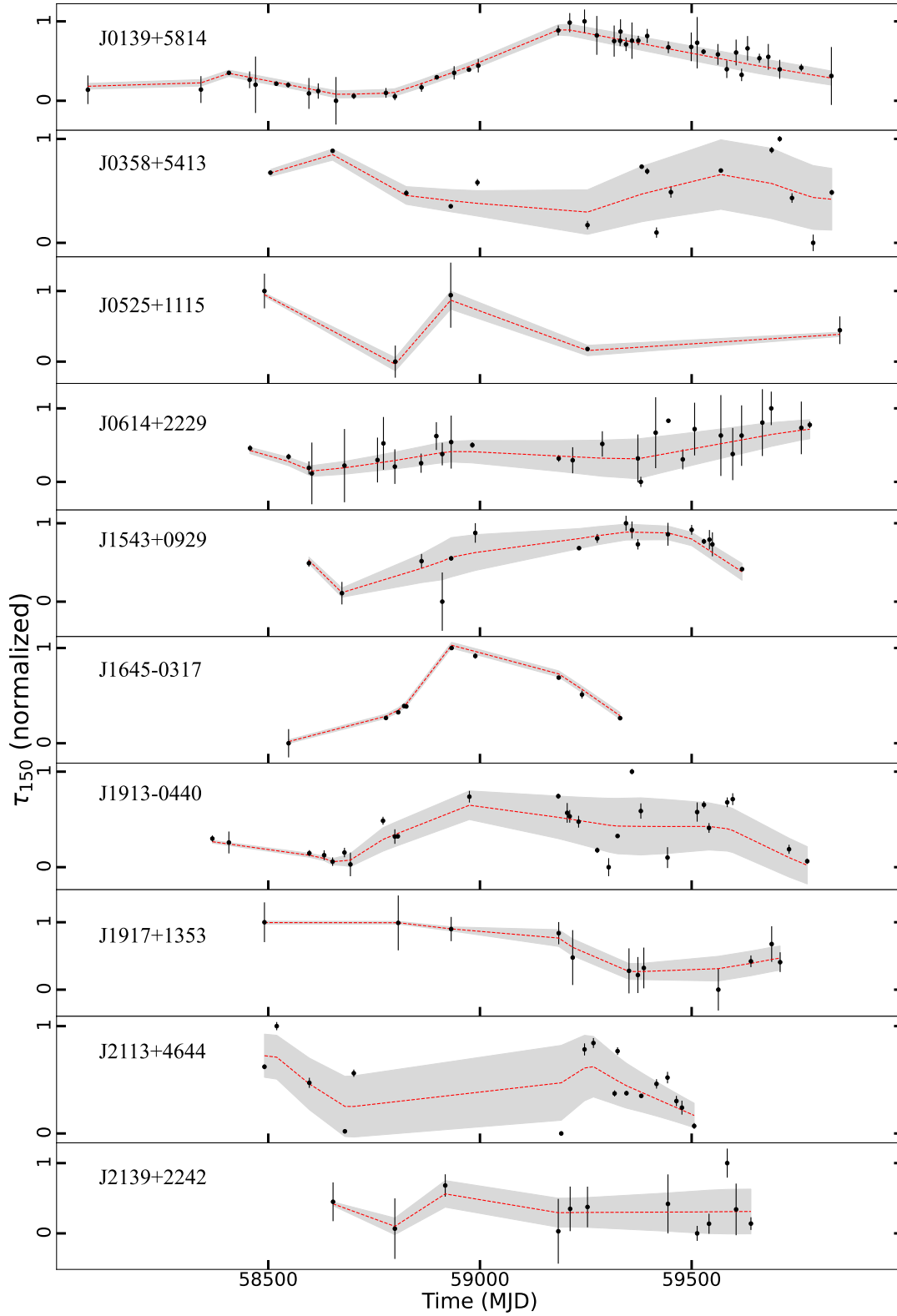


FIGURE 4.9: Time variability of τ_{150} obtained from the anisotropic model, analysed using the NN methods. The grey area represents the confidence interval of the loss function used for fitting (See. Sec. 2.10.2 and 4.3). The τ_{150} values and their errors have been normalized by their respective maximum values, for more clear illustration purposes. A significant variability is visible in pulsars PSR J0139+5814, PSR J0614+2229, PSR J1913-0440, PSR J1917+1353 and PSR J2113+4644.

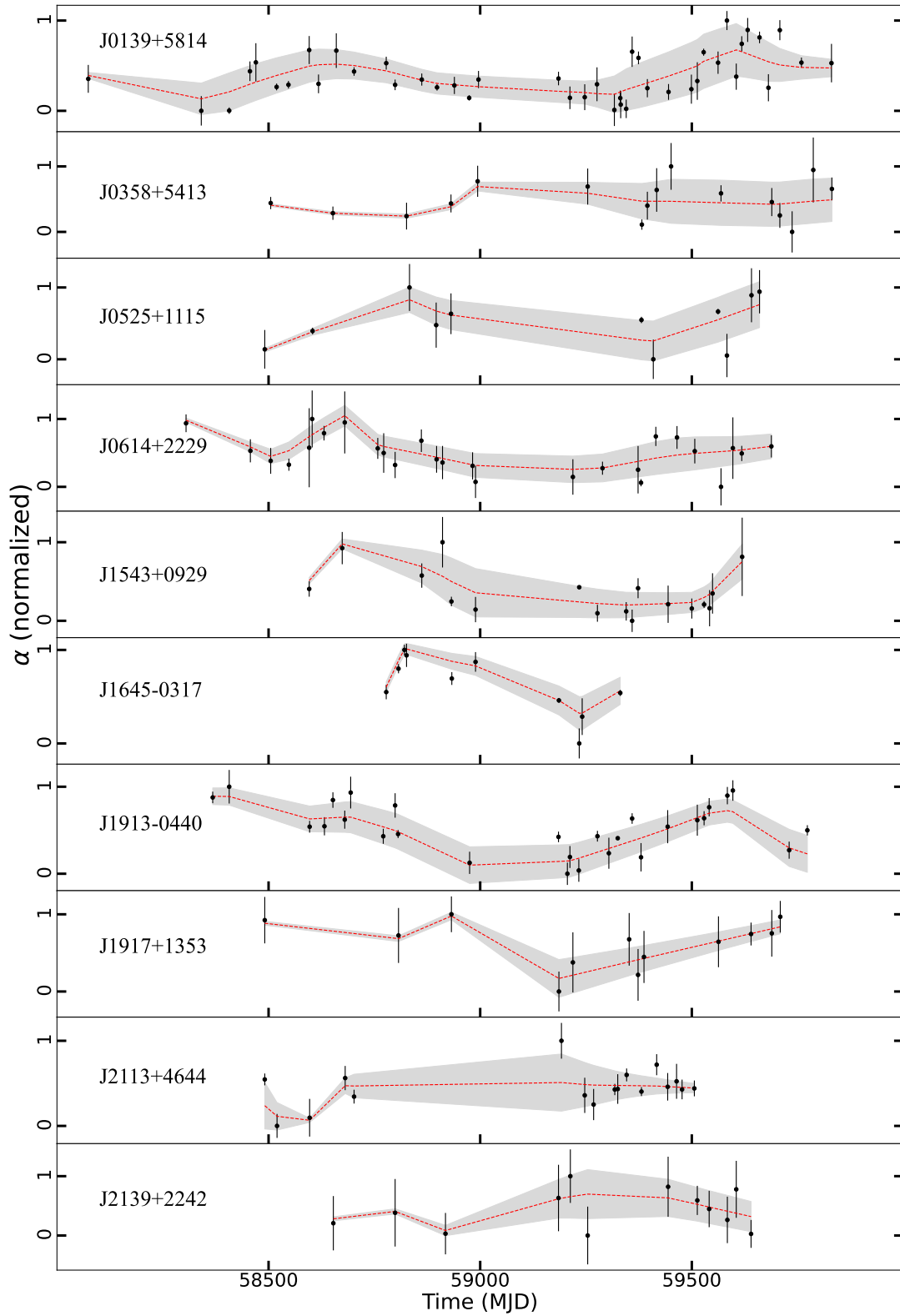


FIGURE 4.10: Time variability of α obtained from the isotropic model, analysed using the NN methods. The grey area represents the confidence interval of the loss function used for fitting (See. Sec. 2.10.2 and 4.3). A significant variability is visible in pulsars *PSR J0139+5814*, *PSR J0358+5413*, *PSR J1913-0440*, *PSR J1917+1353* and *PSR J2113+4644*.

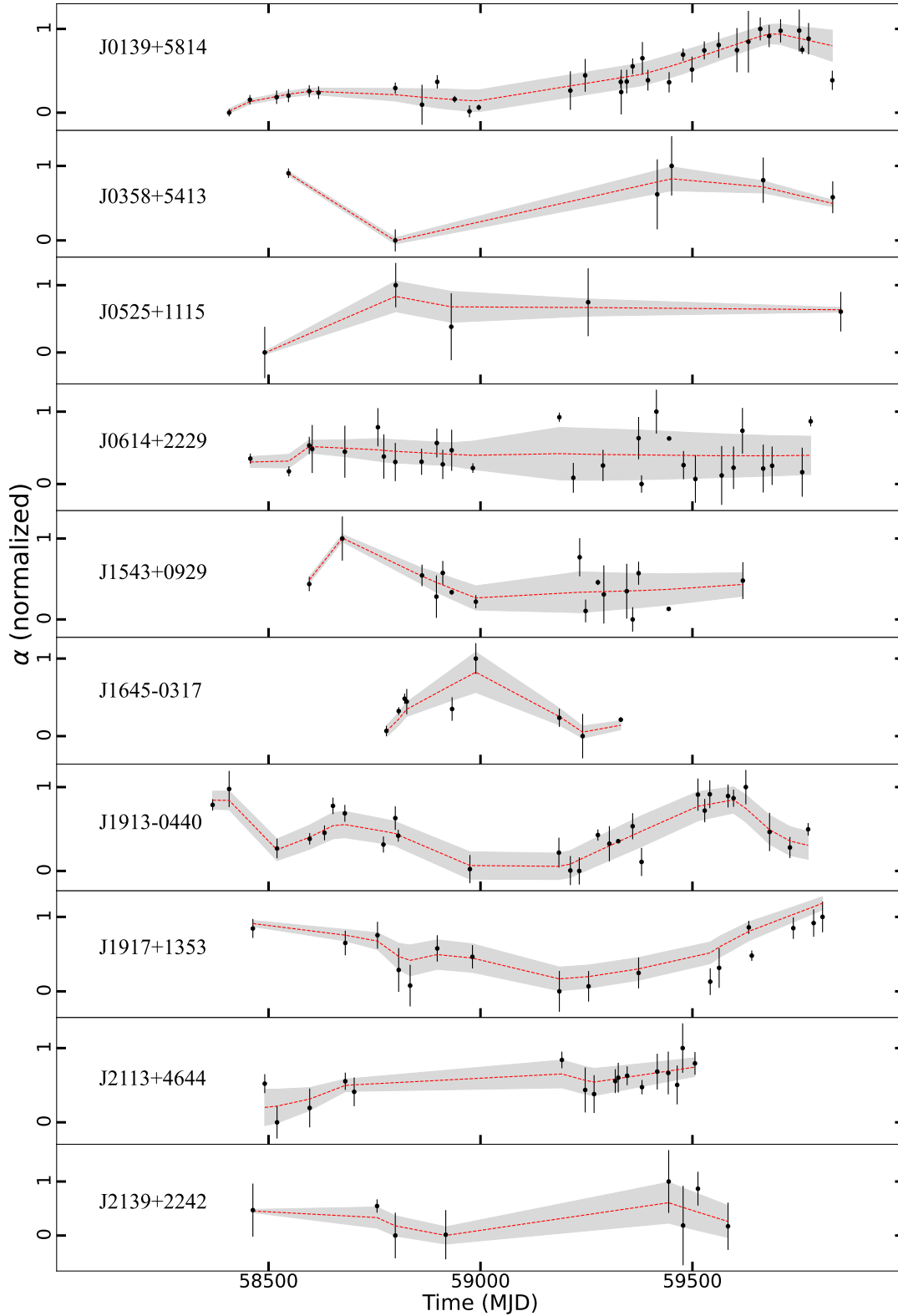


FIGURE 4.11: Time variability of α obtained from the anisotropic model, analysed using the NN methods. The grey area represents the confidence interval of the loss function used for fitting (See. Sec. 2.10.2 and 4.3). The α values and their errors have been normalized by their respective maximum values, for more clear illustration purposes. A significant variability is visible in pulsars PSR J0139+5814, PSR J0525+1115, PSR J0614+2229, PSR J1913-0440, PSR J1917+1353 and PSR J2139+2242.

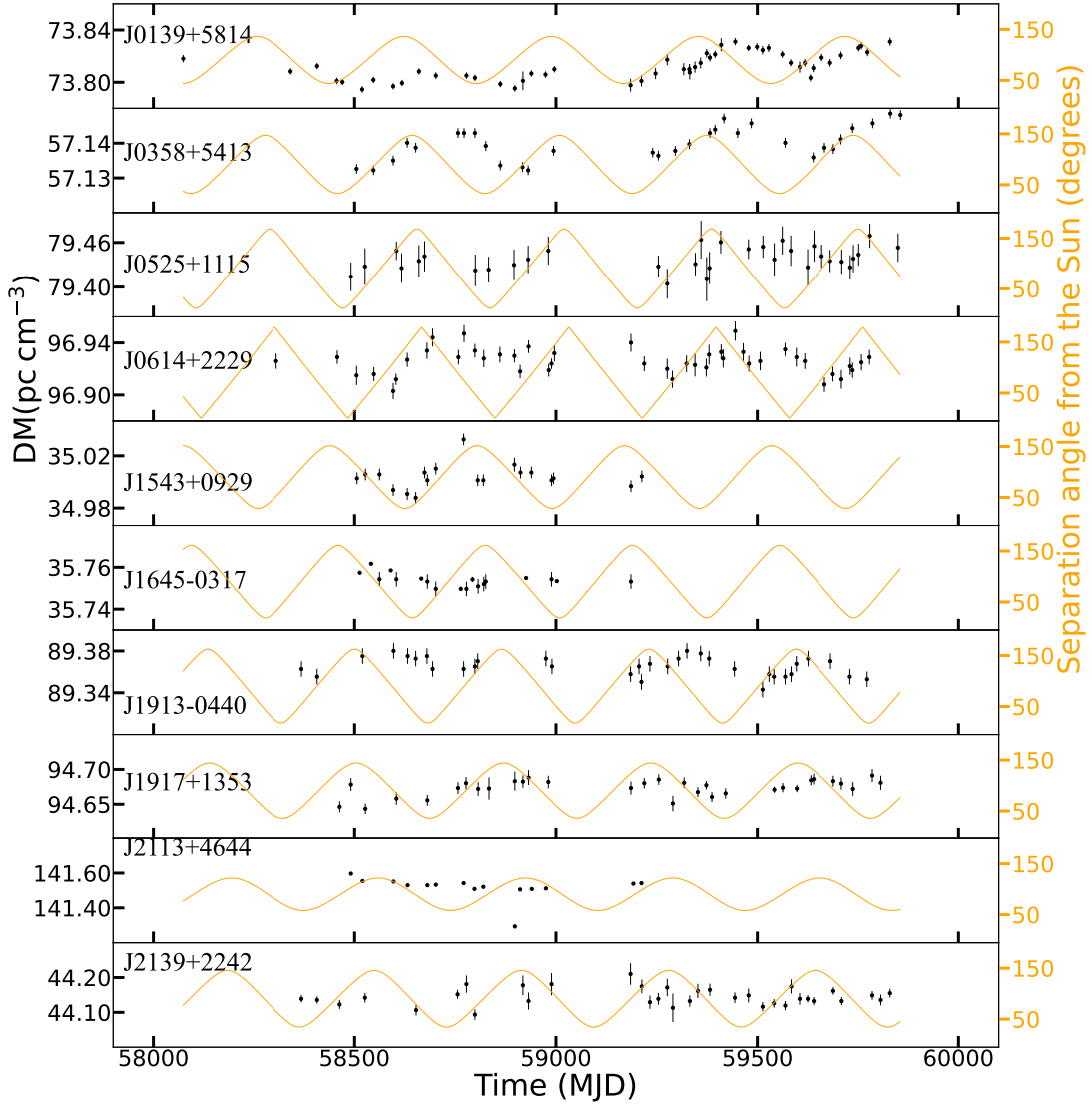


FIGURE 4.12: The DM variation for the targets that show scattering in our observations. The orange curve represents the Sun separation angle of the corresponding pulsar as frequency of time. No correlation is seen between these two parameters. This indicates that the solar wind has no noticeable effect on the DM measurements.

4.6 Dispersion Measure variability

Due to the low observing frequency range and the wide bandwidth of LOFAR (Sec. 2.1), we can conduct accurate DM measurements. After an observation is completed the DM is approximated incoherently using the **PSRCHIVE** tool **pdmp**³, which maximizes the SNR of the profile. This step gives us the final corrections to the estimated DM values.

In comparison, the coherent dedispersion technique deconvolves the transfer function of dispersion from the recorded voltage signal (which is removed from our archives after some period of time), whereas the incoherent dedispersion method aligns small frequency channels

³<https://psrchive.sourceforge.net/manuals/pdmp/>

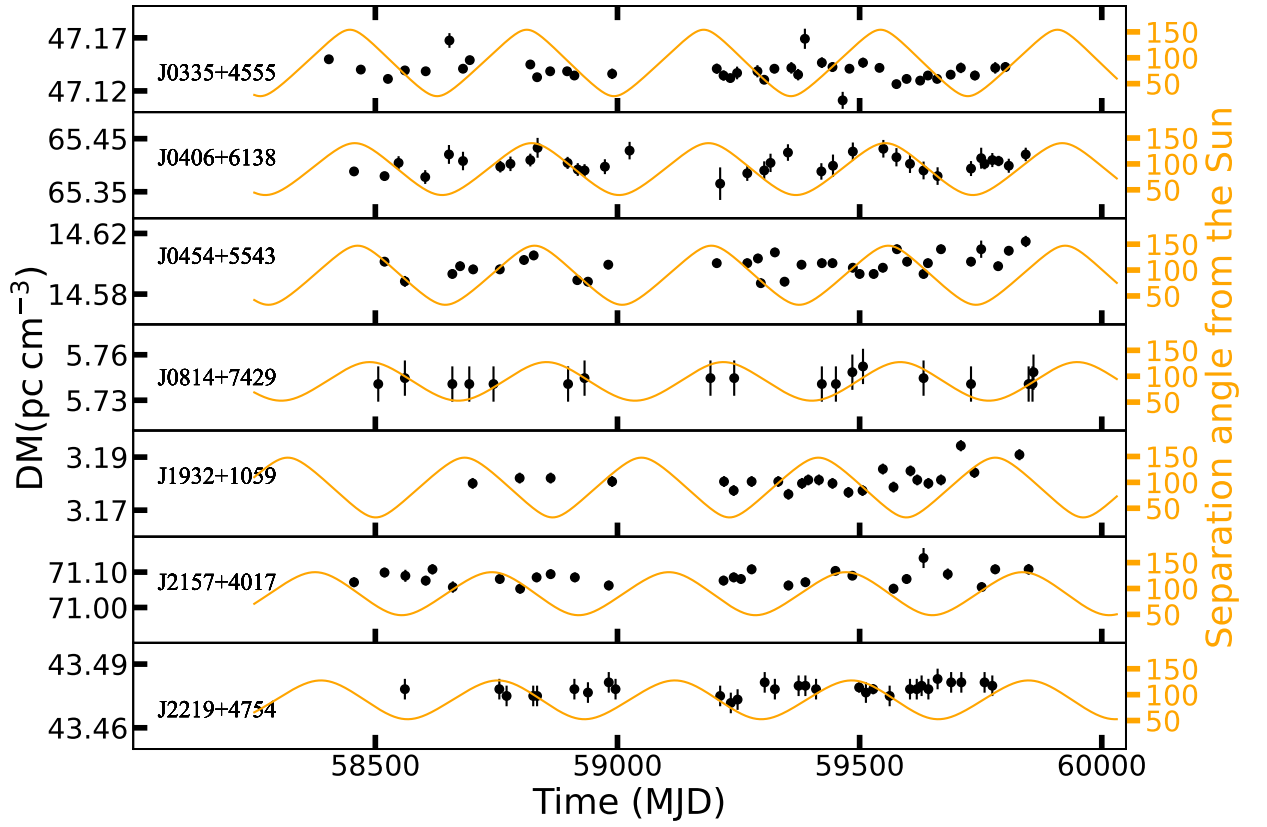


FIGURE 4.13: The DM variation for the targets that don not show significant scattering in our observations. As in Fig. 4.12, the orange curve represents the Sun separation angle of the corresponding pulsar as frequency of time. Again, it does not seem that the solar wind is affecting our DM estimations.

in time (Liu et al., 2022). In the case of coherent dedispersion, the delay is applied to the signal as a whole, meaning that all the frequencies are delayed by the same amount. This is done by applying the dedispersion filter to the data before Fourier transforming it into frequency space. This results in a single spectral channel where the signal is corrected for dispersion.

However, incoherent dedispersion involves dedispersing each individual spectral channel separately. This means that each channel is corrected for dispersion, but the resulting data is not combined into a single channel. This creates a filterbank-like dataset, where each channel contains a different frequency range that has been corrected for dispersion.

To find the proper dispersion measure (DM) of the pulsar, which is a measure of the amount of dispersion caused by the ISM, the filterbank-like dataset can be used. Various methods such as a brute-force search or a matched filtering approach can be applied to the data to find the best-fit DM that maximizes the signal-to-noise ratio of the pulsar's signal. Once the proper DM is determined, the data can be coherently dedispersed at that DM value, resulting in a single spectral channel with a corrected signal that is optimized for the particular pulsar being observed.

TABLE 4.4: The correlations between the separation angle and Dispersion Measure for all pulsars in our dataset.

pulsar name	r(sep.angle, DM)	p-value
J0139+5814	-0.11	0.44
J0335+4555	0.10	0.54
J0358+5413	0.06	0.76
J0454+6138	-0.04	0.81
J0525+1115	0.00	1.00
J0614+2229	0.38	0.01
J0814+7429	0.23	0.19
J1543+0929	-0.36	0.14
J1645-0317	-0.09	0.66
J1913-0440	-0.56	0.01
J1917+1353	-0.06	0.73
J1932+1059	0.44	0.07
J2113+4644	0.28	0.30
J2139+2242	-0.06	0.72
J2157+4017	-0.10	0.65
J2219+4754	0.14	0.43

4.7 Factors impacting the estimation of Dispersion Measurement

The solar activity, is another factor that can influence our DM estimations. When a pulsar appears to be close to the Sun (when the separation angles is small), the solar wind increases the free electrons density, thus affecting the DM value (Tiburzi et al., 2019). However, this effect seems to be unnoticeable in our pulsar sample. In our data, the separation angle (the phenomenal distance between the pulsar and the Sun) of each pulsar is never less than 40° .

For each observation, the solar angle was calculated using the *astropy* (Astropy Collaboration et al., 2013; Astropy Collaboration et al., 2018; Astropy Collaboration et al., 2022) python module then plotted it with the corresponding DM values. Inspection of Figs. 4.12, 4.13 reveals that there was no significant influence of the solar wind.

In Figs. 4.12 and 4.13 no clear connection is visible between DM and the separation angle. Also, table 4.4 does not show any correlation between our targets' separation angles from the Sun and DM.

Also, Earth's ionosphere is not affecting our DM estimations. DM fluctuations in Earth's ionosphere are at most of the order of $5 \times 10^{-4} pc\ cm^{-3}$ (Tiburzi et al., 2019). These fluctuations are too small to be detected by modern radio telescopes, however they will be very important for observations with the next generation radio telescopes, such as the Square Kilometre Array (SKA) (Weltman et al., 2020).

4.7.1 Turbulence

Previous studies have shown that DM variations are slow and they appear as linear components in time series (Keith et al., 2013; Coles et al., 2015). These variations can be useful in studying

turbulence in the ISM (Armstrong, Rickett, and Spangler, 1995; Bilous et al., 2016). Since turbulence inside the ISM affects the received signals, we are interested in studying the n_e fluctuations, which can be, indirectly, done by observing the DM variations. Because of Eq. 1.14, the mean electron density can be calculated as:

$$\langle n_e \rangle = \frac{DM}{D}, \quad (4.8)$$

where D is the distance to the pulsar and considered constant. The rate of change of the average electron density n_e can be inferred from the rate of change of the dispersion measure value, which is often denoted as ΔDM , and usually expressed as a rate of change of the measured DM value per year.

4.7.2 The estimation of the ΔDM values

ΔDM has been determined for each pulsar in our sample. Previous works, whose data are cover a period of maximum 2 years, such as Hobbs et al. (2004) and Donner et al. (2020) assume that the value of ΔDM is equal to the slope of a linear fit of DM versus time.

However, since in our project we were monitoring the dispersion measure for extended periods of time, in the case of our observations - as it can be seen in Figures 4.12, 4.13 - the slope of change in DM usually switches from positive to negative at least a few times for each pulsar. This is the reason why in our analysis the time series of each pulsar was split into smaller time periods and then the ΔDM was calculated for each one of these constant slope sections of the data. Since we are only interested in selecting them in such a way, that during the individual time-span the direction of the DM variation is the same, we were only interested in its absolute value ($|\Delta DM|$). Then, the combined average of $|\Delta DM|$ for all the periods was calculated. The combined average is similar to the weighted average, but the weights are the number of measurements during each period. The resulting values are shown in Table 4.5.

Having obtained the average $|\Delta DM|$ values for each pulsar we investigated the relation of a correlation between the absolute values of ΔDM and DM. The wedge model tries to account for variations in the DM that arise from inhomogeneities in the ISM along the line of sight to the pulsar (Backer et al., 1993). It assumes that the ISM can be divided into a series of thin, wedge-shaped slices, each with a uniform electron density. The variations in DM arise from changes in the electron density in each slice, and the plasma wedge model allows for the estimation of these variations based on measurements of the pulsar's pulse arrival times at different frequencies. It predicts that: $|dDM/dt|$ is proportional to the square root of DM, as in Eq. 4.9:

$$|\Delta DM| \approx a DM^b, \quad (4.9)$$

with $b=0.5$. A few years later, Hobbs et al. (2004) used long term observations, in various frequencies between 408-1630 MHz, of 374 pulsars and estimated the value of the parameter b in Eq. 4.9 to be equal to 0.57 ± 0.02 . If $b = 0.5$, as estimated in Backer et al. (1993), then $a = 0.0002$.

Donner et al., 2020 used 150 MHz LOFAR observations of 36 millisecond pulsars with $P < 30$ ms and estimated that $b = 0.7 \pm 0.6$ and if $b = 0.5$ then $a = 24 \pm 5 \times 10^{-6}$.

TABLE 4.5: The dispersion measure derivative values of our complete pulsar sample. DM and distance from Manchester et al. (2005).⁴

Pulsar	DM ($pc\ cm^{-3}$)	Distance (kpc)	$ \Delta DM $ ($pc\ cm^{-3} yr^{-1}$) $\times 10^{-2}$	$ \Delta_{err} $ ($pc\ cm^{-3} yr^{-1}$) $\times 10^{-4}$	Time span (MJD)
J0139+5814	73.81	2.600	6.28	8.42	58074 - 59830
J0335+4555	47.15	2.439	3.27	6.73	58404 - 59801
J0358+5413	57.14	1.000	3.62	35.1	58505 - 59856
J0406+6138	65.41	4.545	1.96	37.4	58456 - 59843
J0454+5543	14.59	1.180	2.91	11.0	58519 - 59843
J0525+1115	79.42	1.841	7.71	20.7	58491 - 59850
J0614+2229	96.91	3.571	4.53	150.00	58305 - 59779
J0814+7429	5.75	0.432	2.46	19.9	58506 - 59859
J1543+0929	34.98	5.900	1.70	0.866	58596 - 59619
J1645-0317	35.76	3.846	1.53	8.49	58340 - 59786
J1913-0440	89.39	4.041	5.69	61.2	58368 - 59773
J1917+1353	94.54	5.882	5.14	9.73	58463 - 59807
J1932+1059	3.18	0.310	1.55	94.7	58701 - 59830
J2113+4644	141.26	2.174	2.23	0.183	58491 - 59506
J2139+2242	44.16	4.933	1.96	34.6	58368 - 59830
J2157+4017	71.12	2.900	1.57	9.80	58456 - 59849
J2219+4754	43.50	2.387	1.64	6.92	58561 - 59774

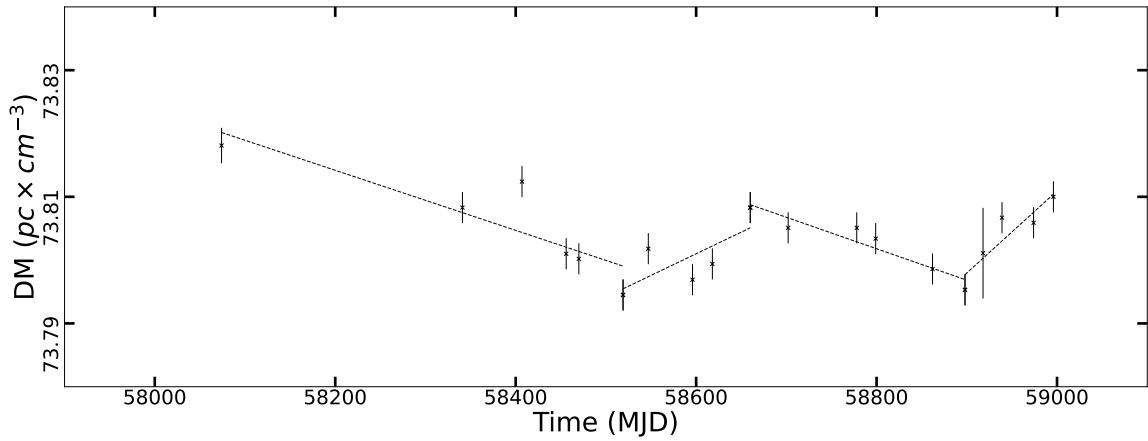


FIGURE 4.14: ΔDM estimation. We illustrate the practice of dividing our DM estimations into smaller time intervals to calculate ΔDM individually for each interval. The dashed lines depict the linear models fitted to each specific period, with the slope of each line representing ΔDM .

We fitted Eq. 4.9 to our averaged data (Table 4.5) and estimate that $a = 1.43 \pm 0.17 \times 10^{-06}$ and $b = 0.4 \pm 0.28$. Differences among different observations are expected, since we are using different data and different frequencies.

This method does not seem appropriate for our sample. One of the basic assumptions is that the DM variability follows a linear trend (Backer et al., 1993). However, this is not the case, as seen in Figures 4.12 and 4.13. Also, the ΔDM values are dependent on the time range over which they are calculated.

⁴atnf.csiro.au/research/pulsar/psrcat

TABLE 4.6: Correlations between the temporal variations of DM, τ_{150} and α , estimated with the isotropic model. The asterisk (*) denotes values that have been approximated as 0.00 due to their being significantly smaller in magnitude compared to the other values listed in the table.

pulsar	$r(\alpha, \tau)$	$r(\alpha, \text{DM})$	$r(\tau, \text{DM})$
J0139+5814	-0.22/0.17	0.14/0.37	0.28/0.072
J0358+5413	-0.49/0.05	0.13/0.63	-0.25/0.38
J0525+1115	-0.54/0.27	0.36/0.48	0.37/0.47
J0614+2229	0.11/0.56	0.23/0.22	0.087/0.65
J1543+0929	-0.92/0.00*	0.21/0.39	-0.20/0.41
J1645-0317	-0.03/0.91	0.32/0.29	0.45/0.13
J1913-0440	-0.13/0.51	-0.144/0.47	-0.10/0.60
J1917+1353	0.23/0.48	0.50/0.10	0.31/0.33
J2113+4644	0.82/0.00*	0.005/0.84	0.13/0.61
J2139+2242	-0.28/0.36	0.16/0.60	0.34/0.26

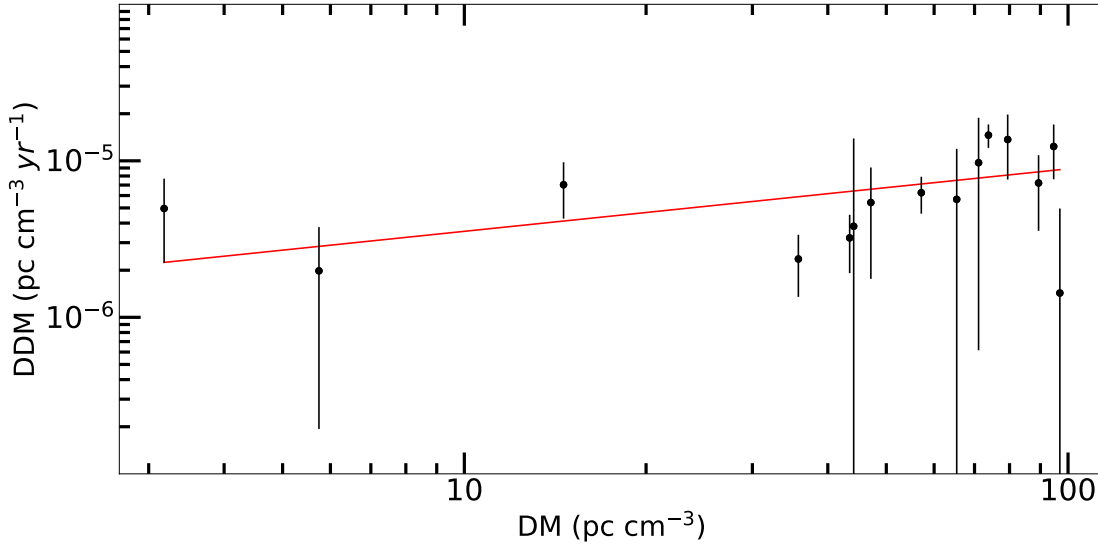


FIGURE 4.15: Relation between DM and ΔDM , according to Eq. 4.9, presented in logarithmic scale.

4.8 ISM parameters correlations

In Tables 4.6 and 4.7 we present the Pearson correlation factors and their corresponding p-values, between the temporal variations of α , τ and DM, for both the isotropic and anisotropic models. The null hypothesis of the Pearson correlation test, is that the distributions underlying the samples are uncorrelated and normally distributed. It must be noted that the p-values for most of the correlations are too large, thus rendering them statistically insignificant.

4.9 A few thoughts about performing regression, with neural networks on our datasets

The robustness of neural networks and random forests, despite the noise in the data is a key advantage in regression tasks, thus making them a very good choice for many machine learning

TABLE 4.7: Correlations between the temporal variations of DM, τ_{150} and α , estimated with the anisotropic model. The asterisk (*) denotes values that have been approximated as 0.00 due to their being significantly smaller in magnitude compared to the other values listed in the table.

pulsar	$r(\alpha, \tau)$	$r(\alpha, \text{DM})$	$r(\tau, \text{DM})$
J0139+5814	0.32/0.069	0.57/0.00*	0.46/0.01
J0358+5413	0.85/0.03	-0.37/0.48	-0.53/0.28
J0525+1115	-0.30/0.34	0.02/0.95	0.05/0.87
J0614+2229	-0.11/0.56	0.003/0.99	0.26/0.20
J1543+0929	-0.93/0.00*	0.02/0.94	-0.04/0.90
J1645-0317	-0.11/0.54	0.11/0.72	0.53/0.05
J1913-0440	-0.047/0.81	-0.23/0.24	0.12/0.55
J1917+1353	-0.11/0.67	-0.03/0.90	0.04/0.89
J2113+4644	0.17/0.51	0.22/0.40	-0.10/0.70
J2139+2242	-0.72/0.043	-0.20/0.64	0.49/0.22

applications where the data is noisy or incomplete.

We show that the Deep Learning methods described in Chapter 2 can be applied to the trend analysis of various parameters, in our case α , τ_{150} and DM. They can even be used to interpolate missing values and extrapolate future ones, together with an error estimation for each prediction. They also help identify outliers, mostly due to low quality observations. Regression with either the help of neural networks, or random forests, seems to be the most efficient in terms of computational cost, good quality fitting results and reasonable error estimations. In our work, we found out that NN regression is providing better error estimations than the RF algorithm.

The use of neural networks for regression offers several advantages, including flexibility, large-scale learning, robustness to noise, automated feature extraction, and real-time predictions, which make them a popular choice for many machine learning applications. None of these methods showed a trend that could be traced back to the solar wind.

4.10 Discussion of our results for each individual pulsar

In this section, I present the results for individual sources concerning the scatter time and frequency scaling index. Furthermore, I conduct a comparative analysis by juxtaposing these findings with the similar values previously documented in the scientific literature.

PSR J0139+5814

PSR J0139+5814 is a relatively bright pulsar at LOFAR HBA frequencies with the average flux of 197 mJy at 150 MHz, according to the recent LOTASS survey (Sanidas et al., 2019a). In the past, there were limited observations carried out to assess the scattering of this pulsar (Lewandowski, Kowalińska, and Kijak, 2015), and no dependable estimation of the frequency scaling index was derived. The average value of the isotropic τ_{150} we obtained is 14.6 ± 0.5 , which is in a relatively good agreement with what can be found in Lewandowski, Kowalińska, and Kijak (2015, see Appendix), where they report a measurement of $\tau = 25 \pm 6$ ms at 102 MHz, based upon a

pulse profile from the European Pulsar Network (EPN) Database⁵. The estimated anisotropic τ_{150} value is 40.28 ± 0.69 , which is considerably higher than its isotropic counterpart. Such a discrepancy has also been observed in Geyer and Karastergiou (2016), where the general rule is that the anisotropic τ estimations are larger than the isotropic ones. This can also be seen in Tab. 3.1.

I estimated the isotropic α value at 3.38 ± 0.20 and the anisotropic at 5.15 ± 0.25 . Both cases demonstrate a notable deviation from the theoretical forecasts of the thin screen model. It is interesting, that this pulsar's τ_{150} and α estimations are highly variable (Sec. 4.2). During the nearly 5-year observation period, the scatter time measurement undergoes noticeable shifts. Interestingly, no discernible correlation between the variations in τ and α was identified.

PSR J0358+5413

PSR J0358+5413 has a 88 mJy flux at 133 MHz. Sanidas et al. (2019b) estimate that $\alpha = 4.6 \pm 0.8$ and $\tau = 0.1 \pm 0.1$, based on scintillation analysis observed at 135.25 ± 31.64 MHz. After combining their observations, at 318 MHz, with older data at frequencies higher than 400 MHz (Cordes, Weisberg, and Boriakoff, 1985), they estimate α to be equal to 3.5 ± 0.1 . Its line of sight is associated to the emission nebula Sh2-205, so it very likely that α and τ estimations are affected by turbulence inside it. This is a pulsar that has not been studied extensively, in terms of pulsar scattering. Our α estimations, are much lower, 1.7 ± 0.19 and 1.54 ± 0.15 for the anisotropic and isotropic model respectively. Besides, the different observation periods, a reason for the disagreement between our estimations is the different analysis methods used. In Sanidas et al. (2019b) they used the CLEAN deconvolution method to estimate the scattering time delay (Bhat, Cordes, and Chatterjee, 2003). Cordes, Weisberg, and Boriakoff (1985) used scintillation studies for their scattering analysis. It should be noted, that this pulsar's intrinsic profile is asymmetric, with a much wider right side. This can influence the analysis of the CLEAN-deconvolution method, because one of its fitting quality criteria is the symmetry of the final fitted profile (Bhat, Cordes, and Chatterjee, 2003; Bhat et al., 2004). Furthermore, this asymmetry can influence the scattering time estimations if significant Radius-to-Frequency mapping is present. It might be beneficial to conduct scattering observations of this target in lower frequencies where more profile components might become more obvious.

PSR J0525+1115

PSR J0525+1115 is the faintest of our targets. Its line of sight is associated to the emission nebula Sh2-264. According to Sanidas et al. (2019b) it has a 35 mJy flux at 150 MHz, $\alpha = 3.4 \pm 0.4$, $\tau = 4.4 \pm 0.8$. Sanidas et al. (2019b) did not find any indication that scattering measurements are affected by profile evolution. Despite its low flux we managed to get enough good quality isotropic α estimations to show that it follows variation patterns in time scale of 1-1.5 years (See Fig. 4.2). The weighted mean anisotropic value for $\alpha = 4.45 \pm 0.31$, which is in a very good agreement with theory. Our isotropic weighted mean estimation is $\alpha = 2.84 \pm 0.28$ which agrees, within error bars, with the Sanidas et al. (2019b) estimation. We suggest that the anisotropic

⁵<https://www.jb.man.ac.uk/pulsar/Resources/epn/>

model explains better this pulsar's scattering behaviour, probably due to its association to the emission nebula Sh2-264, as the anisotropy might be caused by it.

PSR J0614+2229

PSR J0614+2229 has a mean flux of 66 mJy at 150 MHz, estimated by the LOTASS survey. This pulsar's scattering properties have been observed before at similar low frequencies. Kuzmin and Losovsky (2007) reported $\tau = 40 \pm 10$ at the frequency of 111 MHz which agrees with our anisotropic τ_{150} estimation of 42.25 ± 0.39 . Geyer et al. (2017b) analysed newer data from LOFAR Core observations of this pulsar and estimated $\tau = 15 \pm 1$ ms and $\alpha = 1.9 \pm 0.1$ at 150 MHz. These estimations are in agreement with our isotropic estimations of $\tau_{150} = 12.74 \pm 0.2$ and $\alpha = 2.19 \pm 0.17$. Krishnakumar, Joshi, and Manoharan (2017) observed this target using the Giant Metrewave Radio Telescope (GMRT), the Ooty Radiotelescope, and the Long Wavelength Array (LWA) observations with a total frequency range between 148 and 410 MHz, to observe this pulsar. They estimate $\alpha = 2.9 \pm 0.1$ and $\tau = 23 \pm 3$ at 148 MHz, which lay between our isotropic and anisotropic estimations.

Our measurement of the average $\tau_{150} = 12.7 \pm 0.2$ ms is in a relatively good agreement with Geyer et al. (2017b), as is the value of $\alpha = 2.2 \pm 0.2$. The values presented by Krishnakumar, Joshi, and Manoharan (2017) are closer to our anisotropic estimations, especially for α . Krishnakumar, Joshi, and Manoharan (2017) used data from different telescopes at separate observing frequencies, and with much wider frequency range. Their strategy was similar to earlier scattering measurement projects, like Lewandowski et al. 2013; Lewandowski, Kowal-ńska, and Kijak 2015; Lewandowski et al. 2015, and the possible reasons for erroneous scatter time measurements (which leads to large uncertainties of α) were widely discussed in these papers. One of such reasons may be the radius-to-frequency mapping (RFM Kijak and Gil, 2003), i.e. the changes to the intrinsic profile width which may simulate scattering-like behaviour, and affect the measured length of the scattering tail, especially at higher frequencies where the scattering is barely noticeable. This effect may cause an underestimate or overestimate the value of the scatter time measurement when the effect is small, compared to the intrinsic profile width, that could lead to wrong values of α . Although our anisotropic α estimation $\alpha = 3.29 \pm 0.18$, it is still not close enough to the theoretical values of 4.0 or 4.4. Thus it is not obvious which model describes better this pulsar's scattering behaviour.

PSR J1543+0929

PSR J1543+0929 is another bright pulsar with a relatively narrow profile, and its mean flux is around 570 mJy at 150 MHz (Sanidas et al., 2019b). For this pulsar we obtained an unusually low value of isotropic $\alpha = 2.07 \pm 0.38$ and the isotropic $\tau_{150} = 9.1 \pm 0.3$ ms. Our anisotropic estimations are similar, $\alpha = 2.04 \pm 0.38$ and $\tau_{150} = 6.39 \pm 0.36$. Such α estimations suggest, that what we measured is not the actual scattering of the pulsar, but rather its profile evolution due to RFM. That assumption could be supported by the results from Krishnakumar et al. (2019) (See Sec. 1.3), who reported $\alpha = 3.8 \pm 0.6$ and $\tau_{200} = 0.7 \pm 0.4$, but their estimates are based on very low frequency LWA observations (between 52 and 86 MHz). By extrapolating their values

to the LOFAR HBA frequency range one can easily note that the expected scattering at 150 MHz is around 3 ms, much lower than the value we obtained from both models. In that regard it is not surprising than the τ and α variability plots (Sec. 4.2) show only white noise patterns.

PSR J1645-0317

PSR J1645-0317 is the brightest pulsar of our sample (830 mJy flux at 133 MHz Sanidas et al. 2019b). Cordes, Weisberg, and Boriakoff (1985) estimated that $\alpha = 4.5 \pm 0.4$ and Krishnakumar et al. (2019) reported (with more recent observations) values of $\alpha = 4.1 \pm 0.6$ and $\tau = 0.011 \pm 0.004$ ms, using LOFAR observations conducted between 42.92-66.95 MHz. Our estimations are very far from these values, for both models. Our dataset consists of very few points (less than 10) and they are all concentrated during the middle of our 4 year observational run. Our τ estimations are low, but 10 times larger than the ones reported by Krishnakumar et al. (2019). On the contrary, their α estimations are 2 times larger than ours. Given the fact that scattering is stronger in lower frequencies, it seems that is another case, where what we have measured is not pulsar scattering but profile evolution due to RFM.

PSR J1913-0440

PSR J1913-0440, a pulsar with a mean flux of 222 mJy at 150 MHz according to the recent LOTASS survey (Sanidas et al., 2019b) was observed frequently in our projects and we managed to gather 29 estimates of the scatter time and frequency scaling index over the period of almost 4 years (see Table 3.1). Our average isotropic values of τ_{150} is 9.1 ± 0.3 ms and of α is 3.6 ± 0.2 , whereas the anisotropic $\alpha = 4.44 \pm 0.3$ estimation is very close to the thin screen approximation and $\tau_{150} = 15.97 \pm 0.32$. Our isotropic estimations are in close agreement to the isotropic estimations of Geyer (2017), using LOFAR Core observations, where they estimated $\alpha = 2.7 \pm 0.2$ and $\tau_{150} = 9 \pm 0$ s using the isotropic model and $\alpha = 3.5 \pm 0.3$ and $\tau_{150} = 16 \pm 10$ s using the anisotropic.

A statistically significant variation of the scatter time can be seen, for both models, in Fig.4.3 and Fig. 4.1, especially in the second half of our project, with a quasi-periodic behaviour, which timescale is below 500 days - however, this does not seem to be correlated (nor anti-correlated) with the pulsar distance from the Sun (which would indicate the possible influence of the interplanetary medium). The variation of α also seems to be statistically significant, however the changes are much slower than for τ (with a timescale exceeding 500 days) and they are not correlated with the variations of the scatter time. The scattering properties of this pulsar were studied before. Kuzmin and Losovsky (2007) reported $\tau_{111} = 35 \pm 15$ ms, and in earlier observations Slee, Dulk, and Otrupcek (1980) reported $\tau_{160} = 32 \pm 5$ ms and Alurkar, Bobra, and Slee (1986) reported $\tau_{160} = 16.7 \pm 1.8$ ms. The frequency dependence of scattering for this pulsar was also studied in the past: Lewandowski et al. (2015) combined their own measurements from the GMRT with the data published earlier and estimated $\alpha = 4.18 \pm 0.44$ and $\tau_{1\text{GHz}} = 0.0067$ ms (which given the value of α translates to $\tau_{150} = 18.6$ ms) and a measured value at 150 MHz from GMRT of 19.9 ± 0.8 ms.

This pulsar was also studied by Geyer et al. (2017b) who show that $\tau_{150} = 7 \pm 0$ and $\alpha = 3.3 \pm 0.1$ based on their LOFAR Core data, and by Krishnakumar et al. (2019) where they presented the values of $\tau_{200} = 4 \pm 3$ ms and $\alpha = 4.5^{+0.3}_{-0.5}$ based on LOFAR, LWA and ORT data. Our data seems to agree with the estimates by Geyer et al. (2017b) as the values they reported lay within the range of variations we observed for both α and τ . The values of alpha reported by Lewandowski et al. (2015) and Krishnakumar et al. (2019) are significantly larger than what was observed using LOFAR-only data (ours and Geyer et al. 2017b), but given the observed variation of α in our project (see Fig.4.2 and Fig. 4.4) they are still within the range of possible agreement. According to Krishnakumar et al. (2019) their α estimation might be steeper than the models that I used, because of the lack of observations below 148 Mhz. What is more puzzling is the fact, that all the early estimates of the scatter time are much larger than what we observed, with τ_{150} exceeding 30 ms in the early observations (see above the values from Kuzmin and Losovsky 2007 and Slee, Dulk, and Otrupcek 1980), and in some other exceeding 15 ms (Lewandowski et al., 2015; Alurkar, Bobra, and Slee, 1986). This discrepancy could be attributed to either the analysis different methods used or to intrinsic variations of this pulsar's scattering properties, over our project's 4 year span.

PSR J1917+1353

PSR J1917+1353, a pulsar with 198 mJy flux at 150 MHz (Sanidas et al., 2019b) was studied in the past by various researchers. Slee, Dulk, and Otrupcek (1980) published the scatter time value $\tau_{160} = 12 \pm 3$ ms, later Alurkar, Bobra, and Slee (1986) reported $\tau_{160} = 11.7 \pm 1.9$ ms and Kuzmin and Losovsky (2007) a value of $\tau_{111} = 40 \pm 20$ ms. Based on these estimates and their own analysis of EPN database of pulse profiles Lewandowski, Kowalińska, and Kijak (2015) estimated the frequency scaling of this pulsar to be $\alpha = 2.11 \pm 0.05$, however they caution that this value is unreliable. Geyer (2017) and Geyer et al. (2017b) studied this pulsar, using LOFAR Core observations data. Our isotropic τ estimation agrees with all the τ estimations from the previous studies (see Table 3.1). Compared to Geyer et al. (2017b) and Geyer (2017) there is a very good agreement for both α and τ estimations. Our isotropic estimations are $\alpha = 3.27 \pm 0.38$, $\tau = 12.24 \pm 0.35$ agree to the isotropic estimations of Geyer et al. (2017b), where $\alpha = 2.8 \pm 0.4$ and $\tau = 11 \pm 1$. Also, our anisotropic estimations of $\alpha = 3.94 \pm 0.27$, $\tau = 20.73 \pm 0.56$ agree with the the anisotropic estimations of Geyer (2017), where $\alpha = 3.6 \pm 0.6$, $\tau = 21 \pm 2$. For both cases, our α estimations are slightly larger and closer to the theoretically predicted range. The variability of the scatter time (see Sec. 4.2) seems to indicate changes happening on a timescale exceeding 500 days, however due to relatively large τ_{150} uncertainty estimates for some of the observational epochs and a relatively low number of good quality observations all the statistical methods I applied (Sec. 4.1.1) cannot reject the possibility that these variations are white noise.

PSR J2113+4644

PSR J2113+4644 is the highest DM pulsar in our sample (141 pc cm^{-3}) and both the scattering itself and its change with the observing frequency are clearly evident (conf. Fig. ??). This pulsar is relatively bright (244 mJy at 150 MHz, Sanidas et al. 2019b). No significant variability was

detected in the scattering parameters, with the average values of $\tau_{150} = 51 \pm 1$ ms and $\alpha = 3.7 \pm 0.2$ for the isotropic model and $\tau_{150} = 100 \pm 5$ ms and $\alpha = 4.4 \pm 0.2$ for the anisotropic model. Both of our α estimations are very close to the expected theoretical values. This pulsar was only sparsely observed before at similar frequencies: Kuzmin and Losovsky (2007) reported a value of $\tau_{111} = 120 \pm 40$ which is closer to our anisotropic result. Krishnakumar, Joshi, and Manoharan (2017) reported $\tau_{148} = 50 \pm 2$ ms, and the frequency scaling index of 4.2 ± 0.1 based on observations with a total frequency range of 148 to 327 MHz. This is in better agreement with our isotropic estimations. However, this is another case where our analysis suggests that the anisotropic model describes better the scattering properties of this pulsar, since we have an exact agreement to the Kolmogorov theoretical value of 4.4.

PSR J2139+2242

PSR J2139+2242 has a 46 mJy flux at 150 MHz (Sanidas et al., 2019b). Bilous et al. (2020) observed it as part of a pulsar population study, using the LBAs of the LOFAR core stations operating between 30-89 MHz. They used the Galactic electron density model from **Ya2017** estimated at 60 MHz that the scatter time delay normalized over the pulsar's period is 6% which translates into $\tau = 66$ ms, whereas our results are $\tau = 33$ ms for the anisotropic model and $\tau = 26$ ms for the isotropic. Our results are much lower than their values. This could be because of the different method they followed for the scatter time delay. Their estimation is derived from scintillation studies at one frequency and also on a more limited time period compared to our observations. Bondonneau et al. (2020) have also observed this pulsar, using the international LOFAR station FR606 at an observational frequency range between 42.10-77.25 MHz. They estimated the normalized, over period (P), scatter time to be equal to $\tau/P = 6\%$.

4.11 Summary

The study of time variations in the interstellar medium (ISM) is very important when investigating its homogeneity and anisotropy. By investigating the temporal aspects of ISM, we gain a deeper understanding of the ionized ISM's structure along our lines of sight. Extending this study to a broader range of pulsars contributes to our comprehension of the overall picture of ionized ISM structure and its properties.

Variability analysis serves as a crucial tool, allowing us to estimate both the time scales of these variations and their magnitude. The assessment of time scales is particularly significant, as it enables us to convert the variation timescale into the spatial scale of ISM irregularities. This conversion becomes even more relevant when considering the pulsar's proper motion and the motion of the solar system, providing valuable insights into the ISM's dynamic nature.

The magnitude of these variations is closely tied to the differences in scattering and dispersion properties, primarily shedding light on density variations. While the relationship between density and dispersion measures is straightforward as the DM just quantifies the total electron content along the line of sight, the relation is more intricate for scattering. Scattering is strongly influenced by density gradients, accentuating the differences in density between adjacent ISM

regions. This phenomenon arises because scattering is most pronounced at the boundaries between regions with markedly distinct refractive indices, and these indices are largely contingent on density.

I have used Deep Learning methods that fit the α and τ_{150} variability. This approach presents an innovative method for a better understanding of the turbulence within the ISM. Previous studies have predominantly relied on statistical approaches based on individual observations of various pulsars. In some cases, long-term scintillation studies were conducted, often focused on one pulsar at a time, with only a handful of pulsars receiving such scrutiny. The adoption of time variation analysis, as presented here, opens new possibilities for comprehending ISM turbulence.

While the development of theoretical models capable of explaining the observed variations goes beyond the scope of this thesis, the results presented herein, as well as those from future projects with a similar focus, are expected to contribute significantly to the refinement of such models. This research represents a crucial step toward unravelling the intricacies of the ISM and advancing our understanding of this dynamic and complex cosmic medium.

5 Summary and Conclusions

5.1 The applicability of interstellar scattering models

We introduced and engaged upon the preliminary outcomes of our interstellar scattering measurements for ten pulsars as part of our ongoing pulsar observing campaign carried out with the PL611 LOFAR station. In Table 3.1, we present the weighted average parameters for these pulsars. It's noteworthy that the frequency scaling index for most of these pulsars falls notably below the range 4 -4.4, at least for the isotropic scattering model which is predicted by the basic scattering theory. Only, the anisotropic case of *PSR J0139+5814* has an α value larger than this theoretical range. This discrepancy has been previously observed at higher frequencies in the case of pulsars with higher dispersion measures (DM) and those located at greater distances, as it has already been mentioned before, for example, at Lewandowski, Kowalińska, and Kijak (2015) and Krishnakumar et al. (2015). It has also been noticed at lower frequency observations (Geyer and Karastergiou, 2016; Bansal et al., 2019a; Krishnakumar et al., 2019).

In the study conducted by Lewandowski et al. (2013) various potential factors contributing to such deviations were discussed. These factors encompassed the non-Kolmogorov nature of the Interstellar Medium plasma turbulence, anisotropic scattering effects, and the possibility of multiple scattering events occurring along the line-of-sight. At that time, the proposition of multiple scattering events seemed plausible, particularly given the available data, which primarily exhibited deviations in high Dispersion Measure(DM) or distant pulsars.

However, in light of the findings presented in this study and the results of other projects focusing on low-frequency observations, the likelihood of this possibility appears extremely remote. The probability of encountering multiple scattering screens positioned along the line of sight is exceedingly low, even for more distant celestial objects. Moreover, for pulsars that are much closer and exhibit significant scattering primarily within the low observing frequency range, such as the LOFAR frequency band, the likelihood of multiple scattering events is essentially negligible. Another piece of evidence indicating that the multiple scattering model cannot account for the observed deviations is the consistent presence of these deviations across all the pulsars discussed in this study, as well as in the majority of objects examined in the works of Geyer and Karastergiou (2016), Bansal et al. (2019a), and Krishnakumar et al. (2019). The probability that radio waves from all these celestial objects would experience multiple scattering events during their journey is exceedingly improbable, if not practically impossible.

Therefore, we conclude that this hypothesis can no longer be considered a viable explanation for the observed discrepancies. This leaves us with two remaining possibilities: the non-Kolmogorov nature of the Interstellar Medium turbulence and the presence of anisotropy

and/or inhomogeneity in the scattering, with the latter appearing more plausible in our assessment. As demonstrated by Geyer and Karastergiou (2016), scattering anisotropy can lead to alterations in the shapes of observed scattering tails. These alterations can, in turn, result in inaccurate estimations of scatter time and consequently affect the estimates of the frequency scaling index. Furthermore, under certain circumstances, these effects of anisotropy or inhomogeneity may induce variations in the observed profile shapes. This variation could arise from the motion of the line-of-sight due to the pulsar's movement, the Solar System's motion, and/or fluctuations within the IISM region (including its inherent anisotropy). In such scenarios, the impact of anisotropy on the observed profile, and consequently the scatter time estimated using the standard model, as well as the resulting frequency-dependent scattering slope, could change over time. Such changes can only be discerned through long-term monitoring initiatives like our ongoing observation campaign. Within the scope of this work, we have amassed a substantial number of measurements related to pulse broadening effects and frequency scaling index estimations for the pulsars under examination.

5.2 Observations and Data Analysis

In Poland, there are three individual LOFAR stations (PL610, PL611, PL612) that are operational. Ever since these stations became functional, they have consistently focused on pulsars as their primary objective. For our research, we specifically utilized the HBA component of the PL611 International LOFAR station, which is located near Łazy, close to Krakow, Poland. Pulsar observations are typically carried out once a week, during the station's standalone mode. Our observations are centered at a frequency of 154.466 MHz, with a bandwidth of 71.4844 MHz. We exclude the signal from the extreme ends of the observing bandwidth, due to poor instrumental response and lack of valuable information. Consequently, we usually utilize only three frequency lanes positioned towards the center of the band. The collected data is transferred from the station to the computer center at the University of Zielona Góra for further processing. In this study, we specifically employed the PL611 station, which is located near Krakow, Poland. This station comprises 96 Low Band Antennae (LBA) and 48 High Band Antenna (HBA) tiles, a deviation from the typical count of 96 HBA tiles. In our pulsar scattering observations, we primarily employ the HBA part of the PL611 station. We monitor 17 pulsars using LBA and 15 pulsars using both HBA and LBA. Each pulsar is observed for a minimum of 3 hours every month.

We have used both isotropic and anisotropic scattering models (Geyer, 2017) of pulsar scattering, on these data and this has allowed us to compare them and study which one is more suitable to each observed pulsar separately. The results shown in this thesis and in our associated paper (submitted) clearly prove, that the individual LOFAR stations are a very useful tool for pulsar research, and especially so for long term monitoring projects, like the study of scattering presented here. Long term monitoring is also facilitated by the fact that it is generally easier and more straightforward to obtain dedicated observing time on a single LOFAR station, than on a large network of stations or arrays.

5.3 Deviation from the scattering theory

The wide bandwidth of LOFAR allows us to split an integrated profile into multiple good quality profiles each one with a narrower frequency range. Then we can proceed to fitting each one of them, with both the isotropic and anisotropic ISM models mentioned in Sec. 1.5.2. This is not only helpful but also necessary, since scattering is a frequency dependent phenomenon. Furthermore, as demonstrated in Sec. 4.1, certain pulsars exhibit a scattering time delay frequency dependence that deviates from the expected theoretical behaviour. This is a fact that has been studied before (Lewandowski et al., 2013; Lewandowski, Kowalińska, and Kijak, 2015; Geyer and Karastergiou, 2016; Oswald et al., 2021; Krishnakumar et al., 2015; Bansal et al., 2019a) and is an indication that the thin screen ISM model is not ideal as a general model of pulsar scattering, for all pulsars.

There are several plausible reasons for this disagreement between theory and observational results. To begin with, the thin-screen model assumes an infinitely thin screen, which appears to be valid for medium-range dispersion measures (DMs). However, for large DMs, the likelihood of encountering multiple screens increases, thus diminishing the validity of the thin-screen assumption. In the available literature, most studies on scattering (Lewandowski et al., 2013; Lewandowski, Kowalińska, and Kijak, 2015) have focused on high-DM pulsars ($100 \text{ pc cm}^{-3} < DM < 300 \text{ pc cm}^{-3}$), where α estimations less than 4 have been attributed to the presence of multiple finite scattering screens. On the other hand, in low-DM sources, the scattering screen can be finite, resulting in α values equal to or less than 4.0 (Cordes and Lazio, 2001) suggesting that α values less than 4 can arise due to finite scattering screens, while Stinebring et al. (2001) propose anisotropic scattering mechanisms as another contributing factor.

In addition to the dispersion measure, the position of a pulsar within the Galaxy is another significant factor that is expected to impact scattering. Previous research, such as the study conducted by Lewandowski, Kowalińska, and Kijak (2015), found no discernible connection between the spectral index of scattering and either the distance or the position of the source within our Galaxy. Likewise, according to Krishnakumar, Joshi, and Manoharan (2017) there is no indication of any pattern concerning the Galactic location, and they emphasized the necessity of DM-independent distance measurements to draw meaningful conclusions from such a plot. The expected scattered profile shape of the anisotropic model provides longer scattering time delay estimations, in comparison to the isotropic model, especially when observing in low frequencies.

5.4 Time variation of the scattering parameters

Studying temporal variations within the interstellar medium holds significant importance in exploring its inhomogeneity and anisotropy. This investigation of the temporal aspects of the ISM provides us with a deeper insight into the structure of the ionized ISM along our lines of observation. Expanding this research to encompass a broader spectrum of pulsars enhances our grasp of the overall landscape and properties of the ionized ISM. The analysis of variability plays a pivotal role as it enables us to estimate the time frames of these changes and through

that, taking into account the proper motion of the pulsar, we can estimate a spatial scale of the regions of altered density that cause these variations.

Fitting isotropic and anisotropic models to scattered pulsar profiles is a more recent way of examining turbulence properties within the ISM. Earlier conducting long-term scintillation studies, typically focused on one pulsar at a time, this only a limited number of pulsars were subjected to such detailed scrutiny using these methods. Examining time scales is a valuable tool in unraveling the spatial intricacies of the Interstellar Medium (ISM) through pulsar proper motion. Additionally, the extent of variation directly influences density variations in the case of DM fluctuations and has an indirect impact on scattering. This methodology provides fresh perspectives on investigating turbulence within the ISM. The creation of comprehensive models for anisotropic and inhomogeneous scattering is a task that falls outside the purview of this thesis. Nevertheless, the outcomes from observational programs, such as the one introduced in this work, as well as similar observations in the future, will play a pivotal role in the formulation and validation of such models. The implementation of Deep Learning methods for the variability modelling of the scattering parameters, allows us to look for patterns and also detect outliers, without any prior assumptions. Furthermore, these methods give us the capability to delve into higher time resolution estimations. This is needed in order to better understand the time span of variations into the spatial dimensions of ISM irregularities.

5.5 Dispersion Measurement Estimations

The extensive bandwidth of LOFAR, combined with its operation in the lower frequency range, allows us to perform DM measurements with high accuracy. This is important because DM is crucial in studying pulsars. It helps correct for the dispersion effect introduced by the ISM. Accurate DM measurements are essential to obtain clean and precise pulsar timing data, which is used for various astrophysical studies, including gravitational wave detection, testing general relativity, and exploring the pulsar population. Also, measuring DM allows us to probe the distribution of free electrons in the interstellar medium. Variations in DM across different lines of sight can provide valuable information about the electron density and distribution in the Milky Way. We used the wedge model (Sec. 4.7.2) to examine the relation between DM and its absolute rate values. The suitability of this method for our sample appears questionable. A fundamental assumption within this approach is that the DM variability adheres to a linear pattern. Nevertheless, our DM data, as evident in Figures 4.12 and 4.13 do not follow any linear trend. Furthermore, it's important to note that the ΔDM values are dependent on the specific time interval over which they are computed. Therefore we had to modify this approach, by estimating the ΔDM variation for multiple data subsets that were exhibiting a linear trend, and calculating the average value over all subsets. We believe that this approach allows us to estimate the overall rate of change of the DM for a given pulsar, and calculating it in the proposed way allows us to compare our results with the ones obtained for much shorter projects in the past. And indeed our data agrees with the previously published analyses, both for selected individual cases, as well as the general DM variation dependence on the dispersion measure itself.

5.5.1 The effect of the Solar Wind in our DM estimations

We do not detect an effect of the solar wind to our DM measurements. We attribute this to the angular distance (separation angle) of the pulsars in our sample and the sun, which was never less than 40° . As the separation angle decreases (e.g. less than 10°), the more the solar wind affects our observations (Bansal et al., 2019a).

5.6 Non Linear Regression in variability studies

Time series analysis is an essential field of study in various domains due to its numerous benefits and applications such as:

- **Trend Analysis and Forecasting:** Time series analysis enables the identification and extraction of trends from historical data, by understanding the underlying patterns and trends.
- **Seasonality and Cyclical Patterns:** Time series data often exhibit recurring seasonal or cyclical patterns. Analyzing and capturing these patterns helps in identifying regular fluctuations.
- **Anomaly Detection:** Time series analysis allows the detection of anomalies or outliers in data. These anomalies can represent critical events or noise.

Neural networks are particularly useful in regression tasks due to their ability to model complex, non-linear relationships between input variables and target outputs. They can consist of multiple interconnected layers of nodes (neurons) that can learn and represent non-linear relationships. The architecture of neural networks allows them to approximate complex functions by combining simpler, non-linear transformations at each layer. This enables them to capture intricate patterns, interactions, and dependencies in the data.

In the context of non-linear regression, neural networks can learn the mapping between the input variables and the target output by iteratively adjusting their weights and biases during the training process. By feeding the network with a set of input-output pairs, it can learn to generalize and make predictions on unseen data.

One of the challenges in time series analysis is dealing with heteroscedasticity, which refers to the phenomenon where the variability of the data changes over time. Traditional (non artificial intelligence) regression methods assume a constant error variance, which may not hold in many real-world scenarios. However, neural networks offer the advantage of instance-specific error calculation. Instance-specific error calculation allows the neural network to dynamically adjust its predictions based on the changing error characteristics within the time series. This adaptability helps the model account for outliers, anomalies, and other forms of heteroscedasticity that are commonly observed in time series data. As a result, the model can provide more reliable and accurate predictions, even in the presence of noise or irregularities. In Sec. 4.3 we show that this method works very well with our small sample α and τ_{150} datasets.

A Individual observation results

TABLE A.1: The list of our isotropic α and τ_{150} estimations for PSR J0139+5814.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58407	2.49	0.07	12.70	0.15
58456	2.74	0.09	11.98	0.18
58519	2.79	0.13	10.90	0.23
58547	2.82	0.13	10.94	0.23
58596	2.91	0.12	10.19	0.19
58618	2.88	0.12	9.63	0.19
58799	2.97	0.11	8.99	0.16
58862	2.65	0.39	10.69	0.74
58898	3.09	0.13	11.08	0.24
58939	2.75	0.06	12.48	0.13
58974	2.52	0.12	12.17	0.23
58996	2.59	0.05	13.33	0.11
59212	2.92	0.38	17.11	1.16
59247	3.22	0.32	16.60	0.95
59331	3.09	0.12	16.07	0.31
59332	2.89	0.44	16.74	1.34
59345	3.10	0.23	15.89	0.61
59359	3.40	0.15	16.81	0.43
59382	3.55	0.32	16.28	0.92
59395	3.13	0.20	16.67	0.56
59445	3.08	0.20	15.07	0.50
59478	3.62	0.12	14.85	0.30
59499	3.33	0.25	14.78	0.64
59528	3.71	0.18	14.98	0.45
59562	3.81	0.25	14.33	0.63
59605	3.71	0.43	14.74	1.24
59632	3.88	0.60	15.36	2.10
59660	4.13	0.22	13.43	0.53
59681	3.99	0.22	13.13	0.49
59708	4.09	0.23	13.43	0.54
59752	4.10	0.41	13.64	1.09
59759	3.72	0.08	13.80	0.18
59774	3.94	0.31	13.26	0.74

TABLE A.2: The list of our anisotropic α and τ_{150} estimations for PSR J0139+5814.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58074	4.97	0.67	25.63	4.62
58341	3.44	0.70	25.72	4.35
58407	3.45	0.15	32.98	0.86
58456	5.33	0.47	29.95	3.16
58470	5.76	0.92	27.81	9.88
58519	4.58	0.16	28.26	0.81
58547	4.69	0.18	27.73	0.92
58596	6.35	0.67	24.04	4.67
58618	4.73	0.45	25.08	2.43
58660	6.32	0.83	20.84	6.24
58702	5.32	0.20	22.90	0.87
58778	5.72	0.31	24.28	1.49
58799	4.68	0.25	22.69	1.07
58862	4.93	0.29	26.65	1.47
58898	4.56	0.15	31.07	0.84
58939	4.65	0.41	32.99	2.76
58974	4.06	0.13	34.37	0.74
58996	4.94	0.41	36.10	3.06
59185	4.98	0.32	51.43	3.24
59212	4.06	0.54	54.83	6.60
59247	4.09	0.62	55.39	8.25
59276	4.71	0.81	49.28	12.05
59317	3.48	0.77	46.80	9.17
59331	4.05	0.36	46.94	3.15
59332	3.74	0.65	50.92	7.89
59345	3.53	0.44	45.40	3.86
59359	6.27	0.73	46.98	10.76
59374	5.97	0.30	47.04	2.83
59395	4.52	0.44	49.03	4.36
59445	4.34	0.38	44.09	3.25
59499	4.47	0.70	44.28	7.99
59513	4.86	0.90	46.08	14.91
59528	6.24	0.17	42.18	1.37
59562	5.74	0.53	41.00	5.37
59583	7.76	0.46	34.55	3.82
59605	5.07	0.63	41.83	6.80
59618	6.64	0.37	32.10	2.53
59632	7.31	0.57	43.64	6.76
59660	6.95	0.28	39.33	2.24
59681	4.54	0.65	39.93	6.49
59708	7.30	0.48	34.52	4.12
59759	5.75	0.24	35.23	1.64

TABLE A.3: The list of our isotropic α and τ_{150} estimations for PSR J0358+5413.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58547	1.61	0.28	3.22	0.14
58799	1.22	0.06	2.90	0.03
59417	1.48	0.20	2.99	0.10
59451	1.65	0.17	3.12	0.08
59667	1.57	0.13	3.12	0.06

TABLE A.5: The list of our isotropic α and τ_{150} estimations for PSR J0525+1115.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58491	1.79	0.71	20.31	3.16
58604	2.46	1.22	14.44	6.46
58833	4.05	0.86	18.02	4.76
58896	2.68	0.83	17.96	4.05
58931	3.09	0.75	22.02	3.97
59381	2.87	1.16	23.79	10.09
59409	1.43	0.72	27.56	4.05
59562	3.17	1.02	14.83	5.01
59583	1.56	0.79	22.42	3.87
59641	3.77	0.99	21.92	7.55

TABLE A.4: The list of our anisotropic α and τ_{150} estimations for PSR J0358+5413.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58505	1.58	0.09	5.48	0.07
58652	1.44	0.09	5.60	0.08
58826	1.40	0.19	5.36	0.16
58931	1.57	0.13	5.28	0.10
58994	1.88	0.22	5.42	0.18
59254	1.81	0.25	5.18	0.21
59382	1.28	0.07	5.51	0.06
59395	1.54	0.19	5.49	0.16
59417	1.76	0.31	5.13	0.25
59451	2.09	0.33	5.37	0.29
59569	1.71	0.11	5.49	0.10
59689	1.59	0.20	5.61	0.17
59708	1.40	0.17	5.67	0.16
59737	1.17	0.29	5.33	0.25
59787	2.04	0.45	5.07	0.40

TABLE A.6: The list of our anisotropic α and τ_{150} estimations for PSR J0525+1115.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58491	3.13	0.85	75.86	18.66
58800	5.34	0.73	31.86	7.25
58931	3.98	1.10	73.26	33.77
59254	4.78	1.12	39.73	25.52
59409	4.92	1.44	87.01	5.35

TABLE A.7: The list of our isotropic α and τ_{150} estimations for PSR J0614+2229.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58305	3.07	0.27	13.29	0.65
58457	2.23	0.35	13.43	0.84
58505	1.92	0.40	14.07	0.99
58548	1.80	0.19	12.26	0.37
58596	2.32	1.21	11.38	4.87
58603	3.20	0.87	10.82	2.64
58631	2.77	0.22	11.99	0.46
58680	3.09	0.95	12.04	3.42
58758	2.30	0.32	13.94	0.79
58772	2.16	0.61	13.23	1.64
58799	1.79	0.40	12.79	0.91
58861	2.54	0.35	12.44	0.76
58897	1.96	0.41	14.43	1.06
58911	1.87	0.51	13.33	1.27
58982	1.77	0.42	13.98	1.02
58989	1.27	0.49	12.64	1.10
59219	1.43	0.54	13.48	1.36
59289	1.69	0.20	13.28	0.44
59373	1.65	0.73	11.45	1.71
59380	1.25	0.11	11.07	0.19
59415	2.67	0.29	15.45	0.79
59465	2.63	0.35	14.05	0.88
59507	2.22	0.38	12.85	0.85
59569	1.12	0.57	14.65	1.54
59597	2.31	0.94	12.94	3.50
59618	2.15	0.23	13.67	0.53

TABLE A.8: The list of our anisotropic α and τ_{150} estimations for PSR J0614+2229.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58457	3.18	0.23	40.10	1.60
58548	2.60	0.23	36.75	1.44
58596	3.77	0.39	32.36	2.85
58603	3.61	1.08	30.16	12.66
58680	3.49	1.17	33.23	16.56
58758	4.60	0.86	35.44	10.81
58772	3.27	1.00	42.00	15.06
58799	3.03	0.86	32.84	7.72
58861	3.04	0.59	34.19	4.46
58897	3.88	0.65	44.88	8.48
58911	2.92	0.66	37.79	5.81
58932	3.56	0.93	42.51	15.33
58982	2.75	0.22	41.34	1.54
59186	5.05	0.21	36.06	1.61
59219	2.31	0.67	35.35	6.26
59289	2.87	0.71	41.78	7.20
59373	4.10	0.96	36.06	11.76
59380	2.03	0.39	26.84	1.85
59415	5.30	0.99	46.20	22.28
59445	4.09	0.07	50.92	0.68
59479	2.88	0.64	35.69	4.86
59507	2.26	1.08	47.64	17.07
59569	2.42	1.33	45.09	24.80
59597	2.77	0.96	37.80	13.43
59618	4.44	1.03	45.06	18.54
59667	2.73	1.08	50.19	22.81
59688	2.86	0.87	55.80	12.71
59759	2.57	1.11	48.10	17.22

TABLE A.9: The list of our isotropic α and τ_{150} estimations for PSR J1543+0929.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58596	1.96	0.26	8.02	0.37
58674	3.60	0.79	6.18	1.62
58862	2.27	0.39	8.51	1.01
58896	1.51	0.76	11.17	2.04
58911	2.36	0.43	7.14	0.72
58932	1.67	0.09	8.17	0.12
58989	1.33	0.24	9.76	0.57
59234	2.93	0.68	7.72	1.42
59248	1.00	0.41	11.25	0.78
59277	2.02	0.10	8.39	0.20
59345	1.71	0.98	9.52	4.45
59359	0.69	0.44	9.97	1.38
59373	2.35	0.41	8.80	0.73
59444	1.07	0.03	10.58	0.09

TABLE A.10: The list of our anisotropic α and τ_{150} estimations for PSR J1543+0929.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58596	2.05	0.27	5.96	0.27
58674	3.48	0.57	4.74	0.68
58862	2.52	0.43	6.04	0.55
58911	3.69	0.89	4.39	1.63
58932	1.61	0.17	6.15	0.18
58989	1.32	0.44	7.19	0.90
59234	2.11	0.04	6.57	0.07
59277	1.19	0.30	6.97	0.39
59345	1.26	0.32	7.58	0.72
59359	0.93	0.39	7.31	0.80
59373	2.07	0.35	6.73	0.45
59444	1.51	0.65	7.13	1.04
59500	1.36	0.35	7.31	0.46
59529	1.50	0.12	6.84	0.23
59542	1.37	0.64	6.91	0.85

TABLE A.11: The list of our isotropic α and τ_{150} estimations for PSR J1645-0317.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58778	0.94	0.06	0.47	0.00
58807	1.15	0.04	0.49	0.00
58821	1.29	0.06	0.51	0.00
58826	1.26	0.14	0.51	0.01
58933	1.18	0.13	0.69	0.01
58989	1.73	0.17	0.67	0.02
59186	1.08	0.10	0.60	0.01
59241	0.88	0.24	0.55	0.02

TABLE A.12: The list of our anisotropic α and τ_{150} estimations for PSR J1645-0317.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58778	1.21	0.08	0.65	0.01
58807	1.45	0.05	0.69	0.01
58821	1.65	0.05	0.71	0.01
58826	1.59	0.12	0.71	0.01
58933	1.35	0.07	0.68	0.01
58989	1.52	0.10	0.67	0.01
59186	1.12	0.03	0.84	0.00
59234	0.66	0.16	0.77	0.02
59241	0.95	0.19	0.76	0.02

TABLE A.13: The list of our isotropic α and τ_{150} estimations for PSR J1913-0440.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58368	4.04	0.11	8.44	0.16
58407	4.33	0.34	8.44	0.57
58520	3.22	0.18	8.65	0.28
58597	3.40	0.11	8.52	0.16
58632	3.51	0.14	8.45	0.21
58652	4.02	0.15	8.24	0.23
58680	3.88	0.16	8.27	0.23
58771	3.30	0.15	9.04	0.24
58799	3.79	0.22	8.81	0.36
58806	3.46	0.11	8.63	0.20
58975	2.84	0.26	10.14	0.47
59185	3.15	0.28	9.43	0.48
59212	2.81	0.27	9.62	0.55
59233	2.80	0.25	8.97	0.49
59277	3.47	0.10	8.12	0.15
59304	3.31	0.33	7.93	0.48
59325	3.36	0.04	8.73	0.06
59359	3.64	0.24	10.17	0.45
59380	2.97	0.26	9.70	0.45
59513	4.23	0.30	8.90	0.52
59529	3.93	0.22	9.29	0.37
59541	4.23	0.26	8.63	0.44
59584	4.20	0.21	9.26	0.37
59597	4.16	0.16	9.82	0.29
59626	4.37	0.32	10.57	0.67
59682	3.53	0.36	9.50	0.65
59730	3.24	0.20	8.21	0.29

TABLE A.14: The list of our anisotropic α and τ_{150} estimations for PSR J1913-0440.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58368	5.30	0.17	15.42	0.51
58407	5.62	0.49	15.17	1.74
58597	4.45	0.17	14.47	0.47
58632	4.46	0.27	14.34	0.74
58652	5.23	0.23	13.92	0.62
58680	4.65	0.27	14.51	0.72
58694	5.45	0.47	13.74	1.70
58771	4.17	0.22	16.58	0.68
58799	5.07	0.36	15.55	1.18
58806	4.23	0.12	15.57	0.33
58975	3.39	0.33	18.14	1.11
59185	4.15	0.15	18.18	0.49
59206	3.07	0.33	17.09	1.76
59212	3.56	0.32	16.87	1.11
59233	3.16	0.33	16.52	1.02
59277	4.17	0.16	14.66	0.42
59304	3.67	0.45	13.56	1.28
59325	4.11	0.04	15.59	0.11
59359	4.69	0.15	19.77	0.57
59380	3.55	0.42	17.21	1.38
59443	4.45	0.48	14.18	1.55
59513	4.64	0.45	17.15	1.72
59529	4.69	0.21	17.62	0.66
59541	5.02	0.27	16.11	0.82
59584	5.36	0.25	17.78	0.91
59597	5.51	0.30	17.98	1.12
59730	3.76	0.25	14.73	0.69

TABLE A.15: The list of our isotropic α and τ_{150} estimations for PSR J1917+1353.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58463	4.86	0.51	9.96	1.14
58681	4.09	0.65	11.19	1.88
58757	4.51	0.71	13.19	2.69
58807	2.66	1.16	13.28	6.15
58834	1.83	1.11	15.84	5.55
58898	3.80	0.70	12.58	2.18
58981	3.36	0.62	14.10	2.03
59186	1.53	1.09	12.98	3.84
59255	1.80	0.81	11.39	2.15
59373	2.50	0.82	8.41	1.65
59542	2.04	0.71	9.14	1.43
59563	2.77	1.06	7.49	2.55
59633	4.92	0.35	9.17	0.63
59640	3.42	0.27	9.19	0.46
59738	4.88	0.57	10.76	1.53
59786	5.15	0.73	10.88	2.43

TABLE A.17: The list of our isotropic α and τ_{150} estimations for PSR J2113+4644.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58491	3.58	0.13	51.60	1.15
58520	3.06	0.22	53.17	2.03
58597	3.25	0.26	50.98	2.39
58681	3.61	0.12	49.10	1.00
58702	3.47	0.19	51.35	1.75
59192	3.90	0.11	49.02	0.97
59247	3.49	0.30	52.27	2.93
59268	3.44	0.26	52.52	2.42
59318	3.61	0.16	50.57	1.44
59325	3.66	0.20	52.20	1.87
59346	3.68	0.13	50.57	1.11
59381	3.53	0.10	50.47	0.84
59417	3.74	0.24	50.93	2.22
59443	3.72	0.29	51.18	2.74
59464	3.56	0.26	50.27	2.40
59477	4.06	0.33	50.02	3.25

TABLE A.16: The list of our anisotropic α and τ_{150} estimations for PSR J1917+1353.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58491	4.85	0.84	25.88	7.63
58807	4.30	1.00	25.75	10.48
58932	5.07	0.65	24.53	4.41
59186	2.26	0.72	23.71	3.90
59219	3.32	1.09	18.84	7.67
59352	4.16	0.96	16.19	5.41
59373	2.86	0.94	15.35	4.10
59387	3.52	0.95	16.76	5.07
59563	4.07	0.92	12.43	3.85
59640	4.35	0.42	18.09	1.57
59689	4.37	0.85	21.54	5.69

TABLE A.18: The list of our anisotropic α and τ_{150} estimations for PSR J2113+4644.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58491	4.53	0.13	98.28	2.36
58520	3.49	0.27	106.22	5.15
58597	3.67	0.42	97.07	8.18
58681	4.56	0.27	101.24	8.94
58702	4.14	0.15	103.80	3.51
59192	5.39	0.40	100.91	8.76
59247	4.17	0.39	112.28	9.02
59268	3.97	0.35	108.28	7.35
59318	4.30	0.13	103.31	2.34
59325	4.31	0.33	109.16	7.08
59346	4.63	0.14	105.03	2.72
59381	4.25	0.10	96.07	1.72
59417	4.85	0.23	106.40	4.74
59443	4.36	0.31	109.22	6.50
59464	4.48	0.39	112.39	9.05
59477	4.30	0.22	108.35	4.86

TABLE A.19: The list of our isotropic α and τ_{150} estimations for PSR J2139+2242.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58463	1.67	0.72	22.44	3.46
58756	1.78	0.18	23.65	0.70
58799	0.98	0.62	26.07	3.00
58918	1.00	0.67	25.85	3.34
59444	2.45	0.85	23.87	5.41
59478	1.25	1.08	28.03	8.38
59513	2.25	0.46	20.28	1.75

TABLE A.20: The list of our anisotropic α and τ_{150} estimations for PSR J2139+2242.

time(MJD)	α	α_{err}	τ_{150} (s)	τ_{150}^{err} (s)
58653	2.22	0.96	35.46	9.79
58799	2.58	1.19	28.83	12.42
58918	1.85	0.73	39.49	6.28
59185	3.10	1.17	28.18	12.98
59213	3.87	0.94	33.71	10.69
59254	1.78	1.02	34.17	9.89
59444	3.50	1.06	34.94	14.67
59513	3.02	0.51	27.66	2.94
59541	2.72	0.64	30.04	4.29
59584	2.33	0.82	45.01	9.30
59605	3.41	1.00	33.58	12.34

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